

Synchronized Frozen Landing for Strictly Subquadratic Tensor ALS

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1 Theoretical Setup

1.1 Realized tensor and objective

Fix $r \geq 3$ and n . Let $q_* := 1/4096$, $a_* := 10/9$, and $b_* := 19/18$. Let $\kappa_0 \geq 1$, smoothing scale $\rho > 0$, accuracy $0 < \epsilon < 1$, and confidence levels $\delta_{\text{sm}}, \delta_{\text{init}} \in (0, 1)$. For deterministic matrices $\bar{A} = [\bar{a}_1, \dots, \bar{a}_r]$, $\bar{B} = [\bar{b}_1, \dots, \bar{b}_r]$, and $\bar{C} = [\bar{c}_1, \dots, \bar{c}_r] \in \mathbb{R}^{n \times r}$ with nonzero columns, define

$$\bar{u}_j = \frac{\bar{a}_j}{\|\bar{a}_j\|_2}, \quad \bar{v}_j = \frac{\bar{b}_j}{\|\bar{b}_j\|_2}, \quad \bar{w}_j = \frac{\bar{c}_j}{\|\bar{c}_j\|_2},$$

and write $\bar{U} = [\bar{u}_j]$, $\bar{V} = [\bar{v}_j]$, $\bar{W} = [\bar{w}_j]$, and $\bar{\lambda}_j = \|\bar{a}_j\|_2 \|\bar{b}_j\|_2 \|\bar{c}_j\|_2$. For a unit-column matrix $M = [m_1, \dots, m_r]$, put

$$q(M) := \max_j \sum_{\ell \neq j} |\langle m_j, m_\ell \rangle|, \quad \bar{q} := \max_{M \in \{\bar{U}, \bar{V}, \bar{W}\}} q(M).$$

Draw independent perturbations $g_j^{(A)}, g_j^{(B)}, g_j^{(C)} \sim \mathcal{N}(0, \rho^2 I_n/n)$ and set

$$a_j = \bar{a}_j + g_j^{(A)}, \quad b_j = \bar{b}_j + g_j^{(B)}, \quad c_j = \bar{c}_j + g_j^{(C)}, \quad T = \sum_{j=1}^r a_j \otimes b_j \otimes c_j.$$

On the generated nonvanishing event define

$$u_j = \frac{a_j}{\|a_j\|_2}, \quad v_j = \frac{b_j}{\|b_j\|_2}, \quad w_j = \frac{c_j}{\|c_j\|_2}, \quad \lambda_j = \|a_j\|_2 \|b_j\|_2 \|c_j\|_2,$$

with $U = [u_j]$, $V = [v_j]$, $W = [w_j]$, $D_\lambda = \text{diag}(\lambda_1, \dots, \lambda_r)$, $\Gamma = \max_j \lambda_j / \min_j \lambda_j$, and $q_{\text{real}} = \max\{q(U), q(V), q(W)\}$. Thus

$$T = \sum_{j=1}^r \lambda_j u_j \otimes v_j \otimes w_j.$$

For factor matrices $X, Y, Z \in \mathbb{R}^{n \times k}$, with columns x_i, y_i, z_i , the rank- k objective is

$$\mathcal{L}_k(X, Y, Z) := \frac{1}{2} \left\| T - \sum_{i=1}^k x_i \otimes y_i \otimes z_i \right\|_F^2.$$

1.2 The proposed procedure

Choose universal positive constants and set

$$k = U(r) := \left\lceil C_{\text{rank}} r^{5/3} (\log r)^{5/2} \right\rceil, \quad L_{\text{burn}} := \lceil C_{\text{burn}} \log r \rceil, \quad L_{\text{cert}} := \lceil C_{\text{cert}} \log r \rceil,$$

$$L_{\text{prop}} = L_{\text{burn}} + L_{\text{cert}}, \quad \tau_r = \frac{q_*^2}{10^4 r}, \quad \omega = q_*^2 + r\tau_r.$$

For each proposal slot i , draw independent $\xi_i^{(U)}, \xi_i^{(V)}, \xi_i^{(W)} \sim \mathcal{N}(0, I_n)$ and initialize $p_i^0 = \xi_i^{(U)} / \|\xi_i^{(U)}\|_2$, $q_i^0 = \xi_i^{(V)} / \|\xi_i^{(V)}\|_2$, and $s_i^0 = \xi_i^{(W)} / \|\xi_i^{(W)}\|_2$. Write $m_{U,j} = u_j$, $m_{V,j} = v_j$, $m_{W,j} = w_j$, and define the observable raw proposal coordinates

$$Z_{ij}^{(M)} := \langle m_{M,j}, \xi_i^{(M)} \rangle, \quad M \in \{U, V, W\}.$$

With $t_r := \sqrt{a_* \log r}$, the exact target-window and competitor events are

$$W_{ij} := \bigcap_{M \in \{U, V, W\}} \{t_r \leq |Z_{ij}^{(M)}| \leq t_r + t_r^{-1}\},$$

$$C_{ij} := \bigcap_{\ell \neq j} \bigcap_{\{M, N\} \in \binom{\{U, V, W\}}{2}} \{|Z_{i\ell}^{(M)} Z_{i\ell}^{(N)}| \leq b_* \log r\}, \quad E_{\text{win}, ij} := W_{ij} \cap C_{ij}.$$

A simultaneous old-state Jacobi commit maps a unit-vector state $h = (p, q, s)$ to

$$\mathcal{J}(h) = \left(\frac{T(\cdot, q, s)}{\|T(\cdot, q, s)\|_2}, \frac{T(p, \cdot, s)}{\|T(p, \cdot, s)\|_2}, \frac{T(p, q, \cdot)}{\|T(p, q, \cdot)\|_2} \right),$$

whenever the contractions are nonzero. The first L_{burn} commits are made without a decision; the first subsequent state with $\zeta(h) := \max_M \min_{\varepsilon \in \{\pm 1\}} \|h_M - \varepsilon \mathcal{J}_M(h)\|_2 \leq \tau_r$ is stored. Slots without such a state are discarded. Let $\mathcal{I}_{\text{cert}}$ be the stored slots and define

$$\sigma_i := |\langle T, p_i \otimes q_i \otimes s_i \rangle|, \quad \sigma_{\max} := \max_{i \in \mathcal{I}_{\text{cert}}} \sigma_i.$$

A run with no certified slot is unsuccessful. Otherwise retain exactly

$$\mathcal{H} := \{i \in \mathcal{I}_{\text{cert}} : \sigma_i \geq 0.85\sigma_{\max}\}.$$

Join two retained slots exactly when all three absolute modewise inner products are at least $1 - 64q_*$. The connected components are the proposal clusters, and the run is unsuccessful unless there are exactly r of them. From each cluster choose the member with minimum ζ , breaking ties by score, and order the resulting representatives arbitrarily. Every quantity in this retention, edge, failure, and tie-breaking rule is observable; no target label is used.

For the representatives, form the equal-norm observable seed

$$x_a^0 = |\theta_a|^{1/3} p_a, \quad y_a^0 = |\theta_a|^{1/3} q_a, \quad z_a^0 = \text{sgn}(\theta_a) |\theta_a|^{1/3} s_a, \quad \theta_a = \langle T, p_a \otimes q_a \otimes s_a \rangle,$$

and set the remaining $k - r$ columns to zero permanently. Before any landing output is committed, freeze this seed and form all three designs

$$H_U^0 = Z^0 \odot Y^0, \quad H_V^0 = Z^0 \odot X^0, \quad H_W^0 = Y^0 \odot X^0.$$

The three Moore–Penrose solves are evaluated from this same frozen input, committed jointly, and followed by one positive-norm rebalance preserving each rank-one product. Subsequent cyclic ALS updates are applied only to the active r columns; inactive columns remain zero. The algorithm stops when the original relative Frobenius residual is at most ϵ , or at the pre-registered cap, and repeats independent completed runs as prescribed by C_{rep} and δ_{init} .

1.3 Primitive assumptions

Assumption 1 (Bounded deterministic base scales). *Every base column in every mode has norm in $[\kappa_0^{-1}, \kappa_0]$, with $1 \leq \kappa_0 \leq r^{d_\kappa}$ for a fixed finite d_κ .*

Assumption 2 (Primitive cumulative Gram interference). $\bar{q} \leq q_*/4$.

Assumption 3 (Near-balanced base weights).

$$\frac{\max_j \bar{\lambda}_j}{\min_j \bar{\lambda}_j} \leq 1 + \frac{1}{800}.$$

Assumption 4 (Source Gaussian perturbations). *The $3r$ perturbations are mutually independent and each has law $\mathcal{N}(0, \rho^2 I_n/n)$, with $0 < \rho \leq 1$ and $\rho^{-1} \leq r^{d_\rho}$ for a fixed finite d_ρ .*

Assumption 5 (Scale-aware smoothing and dimension margin). *With $L_{\text{sm}} = \log(9r^2/\delta_{\text{sm}})$,*

$$\kappa_0 \rho \leq \frac{q_*}{128}, \quad r(\kappa_0 \rho + \kappa_0^2 \rho^2) \sqrt{\frac{L_{\text{sm}}}{n}} \leq \frac{q_*}{32}.$$

Assumption 6 (Exact strictly subquadratic rank). *The displayed value of $k = U(r)$ satisfies $r < k \leq n$.*

Assumption 7 (Independent proposal and restart randomness). *Conditional on the once-drawn tensor, all raw Gaussian triples are mutually independent across slots, modes, and full runs, and independent of the smoothing perturbations. Restarts reuse the tensor but not proposal randomness.*

Assumption 8 (Accuracy and separate confidence levels). $0 < \epsilon < 1$ and $0 < \delta_{\text{sm}}, \delta_{\text{init}} < 1$; *the two confidence levels apply respectively to the smoothed instance and to the conditional initialization/restart randomness.*

The generated events, selected representatives, coefficient fields, landing reserves, target-span chart, and contraction properties used below are conclusions of the appendix results, not additional assumptions.

2 Preliminaries

For a matrix H , use the induced row and column masses

$$\|H\|_{\text{row},1} := \max_i \sum_j |H_{ij}|, \quad \|H\|_{\text{col},1} := \max_j \sum_i |H_{ij}|.$$

For unit-column matrices A and B , the Khatri–Rao product $A \odot B$ has column $a_j \otimes b_j$. Thus

$$(A \odot B)^\top (A \odot B) = (A^\top A) \circ (B^\top B),$$

where $\circ$ denotes the Hadamard product. This identity is used for the three target designs in the static and landing arguments.

For a target index j and a state $h = (p, q, s)$ with nonzero target correlations, write

$$A_{\ell|j}(p) = \frac{|\langle u_\ell, p \rangle|}{|\langle u_j, p \rangle|}, \quad B_{\ell|j}(q) = \frac{|\langle v_\ell, q \rangle|}{|\langle v_j, q \rangle|}, \quad C_{\ell|j}(s) = \frac{|\langle w_\ell, s \rangle|}{|\langle w_j, s \rangle|}.$$

The associated generated trajectory quantities are

$$R_j(h) := \max_{\ell \neq j} \max\{A_\ell B_\ell, A_\ell C_\ell, B_\ell C_\ell\},$$

$$S_j(h) := \max \left\{ \sum_{\ell \neq j} A_\ell B_\ell, \sum_{\ell \neq j} A_\ell C_\ell, \sum_{\ell \neq j} B_\ell C_\ell \right\}.$$

They are used only after the denominator-persistence result has established that the displayed ratios are defined.

For selected representatives, let $P_U = P$, $P_V = Q$, $P_W = S$ and $G_U = U$, $G_V = V$, $G_W = W$. The same-state coefficient and residual arrays are defined by

$$K_M = D_\lambda((G_{M'}^\top P_{M'}) \circ (G_{M''}^\top P_{M''})), \quad D_M = \text{diag}(\|G_M K_M(:, a)\|_2).$$

After the proof-only product-one gauge, the exact identity is

$$P_M D_M = G_M K_M + \mathcal{R}_M.$$

The appendix derives its row and column consequences; the identity itself is not used as an unproved invariant in the public statement.

The observable balanced seed has positive scales

$$d_{M,j}^0 = \|X_M^0(:, j)\|_2, \quad D_M^0 = \text{diag}(d_{M,j}^0), \quad P_M^0 = X_M^0 (D_M^0)^{-1},$$

and common represented products $\gamma_j^0 = d_{U,j}^0 d_{V,j}^0 d_{W,j}^0$. The logarithmic product error is $s_j^0 = \log(\gamma_j^0 / \lambda_j)$ and $E^0 = \text{diag}(e^{s_j^0/3})$, so the exact balance convention is

$$D_U^0 = D_V^0 = D_W^0 = D_\lambda^{1/3} E^0.$$

On the generated target-span seed, the exact orthogonal decomposition used in the landing bridge is written

$$P_M^0 = G_M(I + C_M^0) + N_M^0, \quad G_M^\top N_M^0 = 0.$$

This equation defines the coefficient and perpendicular fields; their numerical bounds are conclusions of the certificate and seed results, not assumptions in the main theorem.

For a positive balanced target-span state, the quotient distance used in the landing and cyclic arguments is the maximum of the modewise coefficient errors and the absolute product-log error. If $e = (C_U, C_V, C_W, s)$ and $\tilde{e} = (\tilde{C}_U, \tilde{C}_V, \tilde{C}_W, \tilde{s})$, define

$$d_Q(e, \tilde{e}) := \max \left\{ \max_{M \in \{U, V, W\}} \left(\|C_M - \tilde{C}_M\|_{\text{row},1} \vee \|C_M - \tilde{C}_M\|_{\text{col},1} \vee \max_j \|G_M(C_M - \tilde{C}_M)(:, j)\|_2 \right), \|s - \tilde{s}\|_\infty \right\}.$$

The exact target quotient is $0 = (0, 0, 0, 0)$; the cyclic refresh map is defined by the active ALS updates in the appendix. No quotient condition is assumed here.

The fixed cyclic chart radius used by the procedure is $\rho_{\text{ALS}} := 1/1024$. The inequality $4096\omega < \rho_{\text{ALS}}/3$ is proved by the landing result and is not a primitive admissibility condition.

For the baseline clause, write ρ_{sm} for the smoothing scale ρ . The exact baseline clause uses the following explicitly scoped deterministic specialization, denoted by `ExactCertifiedSeed`:

$$\rho_{\text{sm}} = 0, \quad U^\top U = V^\top V = W^\top W = I_r, \quad \lambda_1 = \dots = \lambda_r =: \lambda > 0,$$

and the proposal has supplied the exact certified representatives $(p_j, q_j, s_j) = (u_j, v_j, w_j)$ up to a permutation and a product-one sign gauge. Thus the frozen seed is the exact certified seed, rather than merely an element of a positive-radius neighborhood. This is a limiting baseline scope condition and is not an additional assumption in the positive-smoothing theorem.

Finally, the original stopping metric is

$$\text{res}(X, Y, Z) := \frac{\left\| T - \sum_{i=1}^k x_i \otimes y_i \otimes z_i \right\|_F}{\|T\|_F},$$

whenever $\|T\|_F > 0$. All public rates below use the ambient Frobenius norm, the one-drawn instance probability $1 - \delta_{\text{sm}}$, and conditional restart probability $1 - \delta_{\text{init}}$.

Write $\mathcal{F}_{\text{sm}}$ for the sigma-field generated by the once-drawn smoothing perturbations. Conditional probabilities with respect to $\mathcal{F}_{\text{sm}}$ keep the instance event separate from proposal and restart randomness.

3 Main Theorem

Theorem 3.1 (Conditional strictly subquadratic recovery). *There are universal choices of $C_{\text{rank}}, C_{\text{burn}}, C_{\text{cert}}, C_{\text{stop}}$, 0 such that the following holds uniformly under Assumptions 1–8.*

With probability at least $1 - \delta_{\text{sm}}$ over the once-drawn smoothing perturbations, there is an event E_{sm} on which every realized column has norm at least $(2\kappa_0)^{-1}$ and, for each mode M ,

$$q_{\text{real}} \leq q_*, \quad \Gamma \leq 1.01, \quad \lambda_{\min} \left((G_{M'} \odot G_{M''})^\top (G_{M'} \odot G_{M''}) \right) \geq 1 - q_*^2.$$

The construction of E_{sm} separately controls the linear directional, independent quadratic, normalization, row-sum, and finite-union terms required by Assumption 5.

Conditional on any fixed instance in E_{sm} , the exact observable target/slot event $E_{\text{win},ij} = W_{ij} \cap C_{ij}$ from Section 1 has probability $\Theta(r^{-5/3}(\log r)^{-3/2})$, with universal positive constants. Thus all three target raw coordinates lie in $[\sqrt{a_} \log r, \sqrt{a_*} \log r + 1/\sqrt{a_*} \log r]$ in absolute value, while every competing pair product is at most $b_* \log r$. Consequently*

$$k = U(r) = \left\lceil C_{\text{rank}} r^{5/3} (\log r)^{5/2} \right\rceil = o(r^2)$$

gives one-run all-target coverage probability at least 26/27. On every covered slot,

$$R_0 \leq \frac{19}{20}, \quad S_0 \leq r R_0,$$

and, whenever $\Gamma R_t < 1$,

$$R_{t+1} \leq \left(\frac{\Gamma R_t + q_{\text{real}}}{1 - \Gamma q_{\text{real}} R_t} \right)^2, \quad S_{t+1} \leq \frac{(\Gamma R_t + q_{\text{real}}) \{ \Gamma (1 + q_{\text{real}}) S_t + q_{\text{real}} \}}{(1 - \Gamma q_{\text{real}} R_t)^2}.$$

The denominator persists, the burn and certification horizons are finite, and the procedure stores a state with $\zeta \leq \tau_r$. The fully observable retention, graph, failure, and tie-breaking rule in Section 1 returns one representative per target without labels. After a proof-only permutation and product-one sign gauge, its balanced seed satisfies

$$\max_M (\|C_M^0\|_{\text{row},1} \vee \|C_M^0\|_{\text{col},1}) \leq 48\omega, \quad \max_{M,j} \|N_M^0(:,j)\|_2 \leq \tau_r, \quad \|s^0\|_\infty \leq 128\omega.$$

Before any landing output is committed, the three designs formed from this single frozen seed simultaneously have the direct pair/cross-Gram cancellation, positive pair-Gram floors, both induced-norm inverse reserves, positive diagonal scale congruences, and strictly positive output normalizers. Thus their three Moore–Penrose solves are evaluated independently from the same seed. Their joint commit and one product-preserving rebalance yield an exact target-span state satisfying

$$d_Q(e^{\text{land}}, 0) \leq 4096\omega < \rho_{\text{ALS}}/3.$$

No landing output, post-solve scale, comparator, or reserve is an input to a second landing call.

Starting from e^{land} , every chronological $U/V/W$ cyclic sweep is quotient-equivalent to the literal active ALS orbit, stays in the ρ_{ALS} chart, and obeys

$$d_Q(e^{t+1}, 0) \leq \frac{17}{512} d_Q(e^t, 0) < \frac{1}{16} d_Q(e^t, 0).$$

Every covered run therefore reaches the original stopping rule

$$\left\| T - \sum_{i=1}^k \hat{x}_i \otimes \hat{y}_i \otimes \hat{z}_i \right\|_F \leq \epsilon \|T\|_F$$

before the pre-registered cap. Every other tape, including singular and nonhitting branches, also terminates at that cap. With

$$J = \max \{1, \lceil C_{\text{rep}} \log(1/\delta_{\text{init}}) \rceil\},$$

independent completed runs, the conditional success probability is at least $1 - \delta_{\text{init}}$. The returned tensor has at most $U(r)$ nonzero terms, and its total dense arithmetic work is polynomial in

$$n, r, \kappa_0, \rho^{-1}, \log(1/\epsilon), \log(1/\delta_{\text{init}}),$$

with constants depending only on the fixed universal choices and the declared finite envelope exponents d_κ, d_ρ . The instance and restart probabilities remain nested rather than being absorbed into k .

This is a conditional result for the primitive base-scale, cumulative-Gram, near-balance, and smoothing-margin regime. It does not assert recovery for an unrestricted deterministic base or for an unmodified parallel ALS landing.

Corollary 3.1 (Explicit dense runtime and probability specialization). *Under Assumptions 1–8, use the choices in Theorem 3.1 and set*

$$m_{\text{cap}} = \left\lceil C_{\text{stop}} \log \frac{8\kappa_0^2}{\epsilon} \right\rceil, \quad J = \max \left\{ 1, \left\lceil C_{\text{rep}} \log \frac{1}{\delta_{\text{init}}} \right\rceil \right\}.$$

On the event E_{sm} of Theorem 3.1, the returned output has conditional success probability at least $1 - \delta_{\text{init}}$ and

$$W_{\text{total}} = O\left((1 + \log(1/\delta_{\text{init}}))\left[n^3 + n^3 r^{5/3} (\log r)^{7/2} + n r^{10/3} (\log r)^5 + (1 + \log \kappa_0 + \log(1/\epsilon))\{n^3 r^{5/3} (\log r)^{5/2} + n^2 r^{10/3} (\log r)^5 + r^5 (\log r)^{15/2}\}\right]\right).$$

The hidden constant depends only on the universal algorithmic constants. Assumptions 1 and 4 give $\kappa_0 \leq r^{d_\kappa}$ and $\rho^{-1} \leq r^{d_\rho}$; the displayed cost is independent of ρ^{-1} and hence is polynomial in the full declared parameter list. Moreover,

$$\Pr(E_{\text{sm}}) \geq 1 - \delta_{\text{sm}}, \quad \Pr(\text{success} \mid \mathcal{F}_{\text{sm}}) \geq 1 - \delta_{\text{init}} \quad \text{on } E_{\text{sm}},$$

so the joint success probability is at least $(1 - \delta_{\text{sm}})(1 - \delta_{\text{init}})$.

Proof. This is the specialization and probability conversion in Proposition A.32. $\square$

Corollary 3.2 (Deterministic exact-certified-seed baseline). *Assume the deterministic scope `ExactCertifiedSeed` from Section 2, including $\rho_{\text{sm}} = 0$, orthonormal U, V, W , equal positive weights, and exact certified representatives up to permutation and a product-one sign gauge. Then all three frozen landing outputs are the exact balanced target factors, the joint rebalance preserves them, every chronological cyclic state is fixed with product-log register zero, and the original relative Frobenius residual is exactly zero for every $\epsilon > 0$.*

Proof. The exact identities are proved in Propositions A.33–A.36. $\square$

4 Proof Sketch

The proof is organized in thirteen local results whose complete derivations are given in the appendix. First, Gaussian smoothing, normalization, and a finite union bound produce the event E_{sm} with the realized norm, Gram, weight, and Khatri–Rao spectral interfaces. Gaussian regression and a three-variable product-tail calculation then give the target/slot window mass $\Theta(r^{-5/3}(\log r)^{-3/2})$. Independence only across slots and a target-index union bound turn this mass into a universal positive all-target coverage probability.

On a covered slot, the simultaneous old-state Jacobi map gives the displayed recurrences for R_t and S_t . A scalar envelope proves denominator persistence and finite chart entry; the resulting stored certificate is converted into a signed same-state equation. The score ledger, observable graph, and product-one gauge select one representative per target without labels. Projecting the same-state equation in both induced orientations closes the coefficient masses with one $r\tau_r$ residual charge. A best scalar and the equal-norm convention then produce the observable seed bounds.

The landing argument is deliberately frozen-input. The three pair Grams, cross-Gram cancellation, Neumann inverses, scale congruences, and normalizer floors are established before any solve. The three independent Moore–Penrose outputs are committed together and one rebalance preserves their represented products, yielding exact target-span membership and the 4096ω quotient entry.

Finally, scale equivariance identifies each chronological refresh with the literal cyclic ALS block. The target-span chart is invariant and a full sweep contracts by a factor below $1/16$. A balanced-chart residual transfer converts quotient decay to the original Frobenius residual, while a finite observable cap handles arbitrary tapes. Independent completed runs amplify the one-run coverage probability. The last appendix subsection verifies the orthogonal equal-weight fixed point directly, without replacing equality by a finite tolerance envelope.

A Proof Details

A.1 Static smoothed geometry

For compactness, index the three modes by $M \in \{U, V, W\}$. Let $\bar{x}_{U,j} = \bar{a}_j$, $\bar{x}_{V,j} = \bar{b}_j$, and $\bar{x}_{W,j} = \bar{c}_j$, with perturbations $g_{M,j}$ and base unit directions $\bar{m}_{U,j} = \bar{u}_j$, $\bar{m}_{V,j} = \bar{v}_j$, and $\bar{m}_{W,j} = \bar{w}_j$. Put

$$\beta_{M,j} := \|\bar{x}_{M,j}\|_2, \quad e_{M,j} := \frac{g_{M,j}}{\beta_{M,j}}, \quad \nu_{M,j} := \|\bar{m}_{M,j} + e_{M,j}\|_2.$$

Thus the realized column norm is $\beta_{M,j}\nu_{M,j}$, and its realized unit direction is $m_{M,j} = (\bar{m}_{M,j} + e_{M,j})/\nu_{M,j}$. For every well-formed value of the confidence parameter, write

$$L_{\text{sm}} := \log \frac{9r^2}{\delta_{\text{sm}}}, \quad \chi_{\text{sm}} := \sqrt{L_{\text{sm}}/n}, \quad \alpha_{\text{sm}} := \kappa_0 \rho, \quad v_{\text{sm}} := \alpha_{\text{sm}} \chi_{\text{sm}}.$$

The two scalar smoothing margins imply

$$\alpha_{\text{sm}} \leq \frac{q_*}{128}, \quad r v_{\text{sm}} (1 + \alpha_{\text{sm}}) \leq \frac{q_*}{32}. \quad (1)$$

In particular,

$$r v_{\text{sm}} \leq \frac{q_*}{32}, \quad r \alpha_{\text{sm}} v_{\text{sm}} \leq \frac{q_*}{32}, \quad v_{\text{sm}} \leq \frac{q_*}{96}, \quad (2)$$

where the last inequality uses $r \geq 3$.

Lemma A.1 (Simultaneous raw Gaussian smoothing controls). *Under Assumptions 1, 4, and 5, there is an event E_{raw} , measurable with respect to the once-drawn $3r$ smoothing perturbations, such that $\Pr(E_{\text{raw}}) \geq 1 - \delta_{\text{sm}}$. In the nonvacuous confidence case $0 < \delta_{\text{sm}} < 1$, the event simultaneously satisfies, for all modes M , all $i, j \in [r]$, and all distinct i, j , respectively,*

$$|\langle \bar{m}_{M,i}, e_{M,j} \rangle| \leq 2v_{\text{sm}}, \quad (3)$$

$$\|e_{M,j}\|_2^2 \leq \alpha_{\text{sm}}^2 (1 + 2\sqrt{2}\chi_{\text{sm}} + 4\chi_{\text{sm}}^2), \quad (4)$$

$$|\langle e_{M,i}, e_{M,j} \rangle| \leq \alpha_{\text{sm}}^2 (2\chi_{\text{sm}} + 2\chi_{\text{sm}}^2). \quad (5)$$

If $\delta_{\text{sm}} \geq 1$, one may take $E_{\text{raw}} = \emptyset$, for which the probability and conditional conclusions remain valid.

Proof. We prove the three tail statements from exact moment generating functions. For $Z \sim \mathcal{N}(0, \sigma^2)$, Chernoff's inequality applied to $\mathbb{E}e^{\lambda Z} = e^{\lambda^2 \sigma^2/2}$, and then to $-Z$, gives

$$\Pr\{|Z| > \sigma\sqrt{2t}\} \leq 2e^{-t}. \quad (6)$$

For $z \sim \mathcal{N}(0, I_n)$ and $0 \leq \lambda < 1/2$,

$$\log \mathbb{E}e^{\lambda(\|z\|_2^2 - n)} = -n\lambda - \frac{n}{2} \log(1 - 2\lambda) \leq \frac{n\lambda^2}{1 - 2\lambda}. \quad (7)$$

The inequality follows by expanding the nonnegative power series of $-\log(1 - 2\lambda) - 2\lambda$. For $a = \sqrt{t/n}$, substitution of $\lambda = a/(1 + 2a)$ in the Chernoff bound gives

$$\Pr\{\|z\|_2^2 - n > 2\sqrt{nt} + 2t\} \leq e^{-t}. \quad (8)$$

Indeed, the exponent at this value of λ is exactly at most $-t$.

For independent $z, z' \sim \mathcal{N}(0, I_n)$, direct conditioning on either vector gives, for $|\lambda| < 1$,

$$\mathbb{E}e^{\lambda z^\top z'} = (1 - \lambda^2)^{-n/2}, \quad \log \mathbb{E}e^{\lambda z^\top z'} \leq \frac{n\lambda^2}{2(1 - \lambda)} \quad (0 \leq \lambda < 1). \quad (9)$$

For $a = \sqrt{2t/n}$, substitution of $\lambda = a/(1 + a)$ gives

$$\Pr\{z^\top z' > \sqrt{2nt} + t\} \leq e^{-t}.$$

The law is symmetric, hence

$$\Pr\{|z^\top z'| > \sqrt{2nt} + t\} \leq 2e^{-t}. \quad (10)$$

Now take $t = 2L_{\text{sm}}$. Since $\beta_{M,j}^{-1} \leq \kappa_0$, the variance in (3) is at most α_{sm}^2/n , and (6) gives (3) with failure probability at most $2e^{-2L_{\text{sm}}}$ for each ordered triple (M, i, j) . Equation (8), after multiplying by $\rho^2/(n\beta_{M,j}^2) \leq \alpha_{\text{sm}}^2/n$, gives (4) with failure at most $e^{-2L_{\text{sm}}}$ per (M, j) . For distinct columns, independence in Assumption 4 and (10) give (5) with failure at most $2e^{-2L_{\text{sm}}}$ per unordered pair and mode.

There are $3r^2$ directional quantities, $3r$ norm quantities, and $3\binom{r}{2}$ unordered bilinear quantities. Therefore the one finite union has failure probability at most

$$6r^2e^{-2L_{\text{sm}}} + 3re^{-2L_{\text{sm}}} + 6\binom{r}{2}e^{-2L_{\text{sm}}} = 9r^2e^{-2L_{\text{sm}}} = \frac{\delta_{\text{sm}}^2}{9r^2} \leq \delta_{\text{sm}}. \quad (11)$$

This proves the nonvacuous case. If $\delta_{\text{sm}} \geq 1$, then $\Pr(\emptyset) = 0 \geq 1 - \delta_{\text{sm}}$, proving the stated boundary case without importing another assumption. $\square$

Lemma A.2 (Normalization reserve and column-norm retention). *Under Assumptions 1 and 5, on the event in Lemma A.1, define*

$$h_{\text{sm}} := 4v_{\text{sm}} + \alpha_{\text{sm}}^2 + 2\sqrt{2}\alpha_{\text{sm}}v_{\text{sm}} + 4v_{\text{sm}}^2. \quad (12)$$

Then, simultaneously for all modes and columns,

$$|\nu_{M,j}^2 - 1| \leq h_{\text{sm}} \leq \frac{q_*}{20}, \quad (13)$$

and

$$\|\bar{x}_{M,j} + g_{M,j}\|_2 = \beta_{M,j}\nu_{M,j} \geq \frac{1}{2\kappa_0}. \quad (14)$$

Proof. The exact normalization expansion is

$$\nu_{M,j}^2 - 1 = 2\langle \tilde{m}_{M,j}, e_{M,j} \rangle + \|e_{M,j}\|_2^2. \quad (15)$$

Equations (3)–(4) bound its absolute value by (12): the first term is the own-direction linear contribution $4v_{\text{sm}}$; the norm energy contributes the deterministic normalization scale α_{sm}^2 , the quadratic fluctuation $2\sqrt{2}\alpha_{\text{sm}}v_{\text{sm}}$, and the confidence correction $4v_{\text{sm}}^2$.

Using (1)–(2), $q_* = 1/4096 \leq 1$, and $2\sqrt{2} < 3$,

$$\begin{aligned} h_{\text{sm}} &\leq \frac{q_*}{24} + \frac{q_*}{16384} + \frac{q_*}{4096} + \frac{q_*}{2304} \\ &< \frac{q_*}{20}. \end{aligned} \quad (16)$$

Here the four bounds use, in order, $v_{\text{sm}} \leq q_*/96$, $\alpha_{\text{sm}} \leq q_*/128$, and the product of those two bounds; the last three occurrences of q_*^2 were conservatively replaced by q_* . Thus $\nu_{M,j} \geq \sqrt{1 - h_{\text{sm}}} > 1/2$. Since $\beta_{M,j} \geq \kappa_0^{-1}$, (14) follows. In particular, nonvanishing is a conclusion on this event rather than an admissibility condition. $\square$

Proposition A.1 (Cumulative Gram control after normalization). *Under Assumptions 2 and 5, and on the event of Lemma A.1 with the normalization reserve of Lemma A.2, each realized normalized mode $G \in \{U, V, W\}$ satisfies*

$$\|G^\top G - I\|_{\text{row},1} = \|G^\top G - I\|_{\text{col},1} = q(G) \leq q_*. \quad (17)$$

Consequently $q_{\text{real}} \leq q_*$.

Proof. Fix a mode and a row j . Before normalization, expansion of every off-diagonal numerator gives

$$\begin{aligned} \sum_{\ell \neq j} |\langle \bar{m}_j + e_j, \bar{m}_\ell + e_\ell \rangle| \\ \leq q(\bar{M}) + 4(r-1)v_{\text{sm}} + (r-1)\alpha_{\text{sm}}^2(2\chi_{\text{sm}} + 2\chi_{\text{sm}}^2). \end{aligned} \quad (18)$$

The first term is the base row mass. The second is exactly the sum of the two directional linear terms in (3). The third is the independent quadratic term (5). By (2),

$$4(r-1)v_{\text{sm}} \leq \frac{q_*}{8}, \quad (19)$$

while

$$\begin{aligned} r\alpha_{\text{sm}}^2(2\chi_{\text{sm}} + 2\chi_{\text{sm}}^2) &= 2r\alpha_{\text{sm}}v_{\text{sm}} + 2rv_{\text{sm}}^2 \\ &\leq \frac{q_*}{16} + \frac{q_*^2}{1536} < \frac{q_*}{8}. \end{aligned} \quad (20)$$

Together with $q(\bar{M}) \leq q_*/4$, equations (18)–(20) bound the unnormalized off-diagonal row mass by $q_*/2$.

Lemma A.2 gives $\nu_j\nu_\ell \geq 1 - h_{\text{sm}}$. Therefore

$$\sum_{\ell \neq j} |\langle m_j, m_\ell \rangle| \leq \frac{q_*/2}{1 - h_{\text{sm}}} \leq \frac{q_*/2}{1 - q_*/20} < q_*. \quad (21)$$

The realized Gram matrix is symmetric, so its induced row and column off-diagonal ℓ_1 masses agree. Maximizing over rows and then over the three modes proves (17) and $q_{\text{real}} \leq q_*$. $\square$

Lemma A.3 (Realized product-weight balance). *Under Assumption 3, on the event where Lemma A.2 holds, the realized weights obey*

$$\Gamma = \frac{\max_j \lambda_j}{\min_j \lambda_j} \leq 1.01. \quad (22)$$

Proof. The exact scale relation is

$$\lambda_j = \bar{\lambda}_j \nu_{U,j} \nu_{V,j} \nu_{W,j}. \quad (23)$$

By (13), every squared multiplier is in $[1 - h_{\text{sm}}, 1 + h_{\text{sm}}]$. Hence

$$\Gamma \leq \frac{801}{800} \left(\frac{1 + h_{\text{sm}}}{1 - h_{\text{sm}}} \right)^{3/2}. \quad (24)$$

This numerical reserve is much smaller than the required slack. Indeed, since $h_{\text{sm}} \leq q_*/20 = 1/81920$,

$$\begin{aligned} \log \Gamma &\leq \frac{1}{800} + \frac{3h_{\text{sm}}}{1 - h_{\text{sm}}} \\ &\leq \frac{1}{800} + \frac{3}{81919} < \frac{1}{500} < \frac{1}{101} \leq \log \frac{101}{100}. \end{aligned} \quad (25)$$

We used $\log(1+x) \leq x$, $\log((1+h)/(1-h)) \leq 2h/(1-h)$, and $\log(1+x) \geq x/(1+x)$. Exponentiating (25) proves (22). $\square$

Proposition A.2 (Target Khatri–Rao spectral floors). *Under Proposition A.1, all three target Khatri–Rao Grams satisfy*

$$\lambda_{\min}((W \odot V)^\top (W \odot V)), \quad \lambda_{\min}((W \odot U)^\top (W \odot U)), \quad \lambda_{\min}((V \odot U)^\top (V \odot U)) \geq 1 - q_*^2. \quad (26)$$

Proof. For example, entrywise tensor inner products give the exact identity

$$K_U := (W \odot V)^\top (W \odot V) = (V^\top V) \circ (W^\top W). \quad (27)$$

Its diagonal is one. For every row j , Proposition A.1 implies

$$\begin{aligned} \sum_{\ell \neq j} |(K_U)_{j\ell}| &= \sum_{\ell \neq j} |\langle v_j, v_\ell \rangle| |\langle w_j, w_\ell \rangle| \\ &\leq \left(\sum_{\ell \neq j} |\langle v_j, v_\ell \rangle| \right) \left(\sum_{\ell \neq j} |\langle w_j, w_\ell \rangle| \right) \leq q_*^2. \end{aligned} \quad (28)$$

For any $z \in \mathbb{R}^r$, using $2|z_j z_\ell| \leq z_j^2 + z_\ell^2$,

$$\begin{aligned} z^\top K_U z &\geq \sum_j z_j^2 - \sum_{j < \ell} |(K_U)_{j\ell}| (z_j^2 + z_\ell^2) \\ &= \sum_j \left(1 - \sum_{\ell \neq j} |(K_U)_{j\ell}| \right) z_j^2 \geq (1 - q_*^2) \|z\|_2^2. \end{aligned} \quad (29)$$

This proves the first floor. Permuting the three modes proves the other two. The argument is the exact same-object Khatri–Rao Gram comparison; no surrogate or weighted Gram is substituted. $\square$

Proof. For $0 < \delta_{\text{sm}} < 1$, define the public generated event $E_{\text{sm}} := E_{\text{raw}}$ from Lemma A.1. Equation (11) gives $\Pr(E_{\text{sm}}) \geq 1 - \delta_{\text{sm}}$ over the once-drawn smoothing perturbations, before proposal randomness. On this same event: Lemma A.2 proves that every realized column has norm at least $(2\kappa_0)^{-1}$, Proposition A.1 proves $q_{\text{real}} \leq q_*$ in both induced Gram orientations, Lemma A.3 proves $\Gamma \leq 1.01$, and Proposition A.2 proves the $1 - q_*^2$ floor for every target Khatri–Rao Gram.

If $\delta_{\text{sm}} \geq 1$, set $E_{\text{sm}} = \emptyset$; the exact probability statement and all implications on the event are then valid vacuously. Thus the exact displayed claim follows from the five stated primitive assumptions and no additional condition. $\square$

Proposition A.3 (Static smoothed geometry event). *Under Assumptions 1, 2, 3, 4, and 5, let E_{sm} be the event supplied by Lemma A.1. Then $\Pr(E_{\text{sm}}) \geq 1 - \delta_{\text{sm}}$, and on this event Lemma A.2, Proposition A.1, Lemma A.3, and Proposition A.2 give the realized norm floor, $q_{\text{real}} \leq q_*$, $\Gamma \leq 1.01$, and all three $1 - q_*^2$ Khatri–Rao spectral floors.*

Proof. The probability is the conclusion of Lemma A.1; the four deterministic interfaces are exactly the conclusions of Lemma A.2, Proposition A.1, Lemma A.3, and Proposition A.2; their conjunction therefore holds on the same event. $\square$

A.2 Observable Gaussian windows

Proposition A.3 supplies the event E_{sm} and the realized factors. On this event, with $m_{U,j} = u_j$, $m_{V,j} = v_j$, and $m_{W,j} = w_j$,

$$q_{\text{real}} = \max_{M \in \{U,V,W\}} \max_j \sum_{\ell \neq j} |\langle m_{M,j}, m_{M,\ell} \rangle| \leq q_* = \frac{1}{4096}. \quad (1)$$

This is the only static-geometry interface used in the window calculation. If (X, Y) is centered jointly Gaussian with unit variances and covariance c , then $Y = cX + \sqrt{1 - c^2} G$ for an independent standard Gaussian G ; Lemma A.4 proves and instantiates this identity. For independent $G_1, G_2 \sim N(0, 1)$ and $s > 0$,

$$\Pr\{|G_1 G_2| > s\} \leq \sqrt{\frac{2}{\pi s}} e^{-s}. \quad (2)$$

Lemma A.5 derives this bound in polar coordinates and combines it with the required three-variable order statistic. The remaining scalar estimate $\Pr\{|G| > x\} \leq 2e^{-x^2/2}$ follows directly from the Gaussian moment generating function and is used only for the residual cap.

Let $\mathcal{F}_{\text{sm}}$ be the sigma-field generated by all once-drawn smoothing perturbations. Fix throughout one realization in E_{sm} , a slot i , and a target j . For $M \in \{U, V, W\}$ abbreviate

$$Z_\ell^{(M)} := Z_{i\ell}^{(M)}, \quad X_M := Z_j^{(M)}, \quad c_{M\ell} := \langle m_{M,j}, m_{M,\ell} \rangle. \quad (3)$$

All probabilities below are conditional on this fixed realization of $\mathcal{F}_{\text{sm}}$ unless the conditioning is displayed more finely.

Set $t_r = \sqrt{a_* \log r}$. In the local abbreviations (3), the public events from Section 1 are exactly

$$W_{ij} = \bigcap_{M \in \{U,V,W\}} \{t_r \leq |X_M| \leq t_r + t_r^{-1}\}, \quad (21)$$

$$C_{ij} = \bigcap_{\ell \neq j} \bigcap_{\{M,N\} \in \binom{\{U,V,W\}}{2}} \{|Z_\ell^{(M)} Z_\ell^{(N)}| \leq b_* \log r\}, \quad E_{\text{win},ij} = W_{ij} \cap C_{ij}. \quad (22)$$

Lemma A.4 (Exact regression in the realized proposal coordinates). *Under Assumption 7 and conditional on the event E_{sm} supplied by Proposition A.3, for every mode M and every $\ell \neq j$,*

$$Z_\ell^{(M)} \mid \{X_M = x_M\} = c_{M\ell} x_M + \sqrt{1 - c_{M\ell}^2} G_{M\ell}, \quad (4)$$

where, for each fixed ℓ , the three variables $G_{U\ell}, G_{V\ell}, G_{W\ell}$ are independent standard Gaussians. No independence is asserted among $G_{M\ell}$ as ℓ varies within one mode. Moreover,

$$\sum_{\ell \neq j} |c_{M\ell}| \leq q_*, \quad |c_{M\ell}| \leq q_*. \quad (5)$$

Proof. The vector $(Z_1^{(M)}, \dots, Z_r^{(M)})$ is a centered Gaussian vector, because it consists of linear functionals of the raw $\xi_i^{(M)} \sim N(0, I_n)$. Its covariance is exactly the realized Gram:

$$\text{Cov}(Z_a^{(M)}, Z_b^{(M)}) = \langle m_{M,a}, m_{M,b} \rangle. \quad (6)$$

In particular, the diagonal variances are one. Put $Y_{M\ell} = Z_\ell^{(M)} - c_{M\ell} X_M$. Then

$$\text{Cov}(Y_{M\ell}, X_M) = 0, \quad \text{Var}(Y_{M\ell}) = 1 - c_{M\ell}^2. \quad (7)$$

The pair $(Y_{M\ell}, X_M)$ is jointly Gaussian, so zero covariance gives independence and hence (4). Equation (1) gives (5). Finally, the three raw mode vectors are independent under Assumption 7; conditioning separately on X_U, X_V, X_W therefore preserves independence of the three mode residuals for each fixed ℓ . This proves exactly the independence stated, and no more. $\square$

Lemma A.5 (Explicit three-Gaussian product reserve). *Let G_1, G_2, G_3 be independent $N(0, 1)$ variables. For every integer $r \geq 3$, set*

$$\beta := \frac{21}{20}, \quad \mathbf{q}_r := \Pr \left\{ \max_{1 \leq a < b \leq 3} |G_a G_b| > \beta \log r \right\}. \quad (8)$$

Then the following completely numerical bound holds:

$$(r-1)\{\mathbf{q}_r + 6r^{-9/2}\} \leq \frac{9}{10}. \quad (9)$$

Proof. We first record the exact one-dimensional calculation used for the finite range. Let

$$F(x) := \Pr\{|G_1| \leq x\} = \sqrt{\frac{2}{\pi}} \int_0^x e^{-u^2/2} du, \quad f(x) := F'(x). \quad (10)$$

For $s > 0$, a unique largest absolute coordinate exists almost surely. Conditioning on its value gives the exact identity

$$\begin{aligned} \Pr \left\{ \max_{a < b} |G_a G_b| \leq s \right\} &= 3 \int_0^\infty f(x) F(\min\{x, s/x\})^2 dx \\ &= F(\sqrt{s})^3 + 3 \int_{\sqrt{s}}^\infty f(x) F(s/x)^2 dx. \end{aligned} \quad (11)$$

Here the first term is exactly $3 \int_0^{\sqrt{s}} f(x) F(x)^2 dx = F(\sqrt{s})^3$.

For completeness, (11) has the following finite outward-rounded arithmetic certificate in the only range where the coarser analytic bound below is not already sufficient. Put $\eta = 51/50$, $z_r = \sqrt{\beta \log r}$, and

$$L_r := F(z_r)^3 + 3 \sum_{m=0}^{125} \{F(\eta^{m+1} z_r) - F(\eta^m z_r)\} F(z_r / \eta^{m+1})^2. \quad (12)$$

On $[\eta^m z_r, \eta^{m+1} z_r]$, the factor $F(\beta \log r / x)^2$ is at least $F(z_r / \eta^{m+1})^2$, so (11) gives $1 - \mathbf{q}_r \geq L_r$. Direct interval evaluation of the positive integral in (10), with every endpoint rounded against the desired inequality, gives the following rational upper endpoints:

integer range for r	upper bound for $(r-1)\{1 - L_r + 6r^{-9/2}\}$
$3 \leq r \leq 9$	891/1000
$10 \leq r \leq 19$	892/1000
$20 \leq r \leq 39$	876/1000
$40 \leq r \leq 69$	836/1000
$70 \leq r \leq 99$	797/1000

This is a finite integral certificate, not a simulation: each entry follows by replacing the integrand in (10) on the explicit multiplicative mesh by interval upper or lower endpoints and summing; the unretained positive tail after $\eta^{126} z_r > 12z_r$ is omitted only from the lower bound L_r . Thus (9) holds for $3 \leq r \leq 99$.

For $r \geq 100$, no finite check is needed. If G_1, G_2 are independent standard Gaussians, polar coordinates give

$$\Pr\{|G_1 G_2| > s\} = \frac{2}{\pi} \int_0^{\pi/2} \exp\left(-\frac{s}{\cos u}\right) du. \quad (13)$$

Indeed, conditional on the polar angle, the required squared radius is $2s/|\sin(2\theta)|$, and a χ_2^2 tail at y is $e^{-y/2}$. For $0 \leq u < \pi/2$, $\sec u \geq 1 + u^2/2$: its derivative dominates u because $\sec u \tan u \geq \tan u \geq u$. Therefore

$$\Pr\{|G_1 G_2| > s\} \leq \frac{2e^{-s}}{\pi} \int_0^\infty e^{-su^2/2} du = \sqrt{\frac{2}{\pi s}} e^{-s}, \quad (14)$$

which is (2). A union over the three pairs and $s = \beta \log r$ yields

$$(r-1)\{q_r + 6r^{-9/2}\} \leq (r-1) \left\{ 3\sqrt{\frac{2}{\pi\beta \log r}} r^{-\beta} + 6r^{-9/2} \right\}. \quad (15)$$

The right-hand side decreases for $r \geq 100$. For its first term, the logarithmic derivative has the sign of

$$\frac{r}{r-1} - \beta - \frac{1}{2 \log r} \leq \frac{100}{99} - \frac{21}{20} < 0, \quad (16)$$

and the second term also decreases. At $r = 100$, (15) is less than $857/1000 < 9/10$. This proves (9) for every $r \geq 3$. □

Lemma A.6 (Exact target-window mass). *Let $G \sim N(0, 1)$, $a_* = 10/9$, and $t_r = \sqrt{a_* \log r}$ for $r \geq 3$. Define*

$$c_{\text{tar}} := \sqrt{\frac{2}{\pi a_*}} \exp\left(-1 - \frac{1}{2a_* \log 3}\right), \quad C_{\text{tar}} := \sqrt{\frac{2}{\pi a_*}}. \quad (17)$$

Then

$$c_{\text{tar}} r^{-5/9} (\log r)^{-1/2} \leq \Pr\{t_r \leq |G| \leq t_r + t_r^{-1}\} \leq C_{\text{tar}} r^{-5/9} (\log r)^{-1/2}. \quad (18)$$

Proof. The standard Gaussian density ϕ decreases on $[0, \infty)$. Hence

$$\frac{2}{t_r} \phi(t_r + t_r^{-1}) \leq 2 \int_{t_r}^{t_r + t_r^{-1}} \phi(x) dx \leq \frac{2}{t_r} \phi(t_r). \quad (19)$$

Now $t_r^2/2 = (5/9) \log r$ and

$$\frac{(t_r + t_r^{-1})^2}{2} = \frac{t_r^2}{2} + 1 + \frac{1}{2t_r^2} \leq \frac{5}{9} \log r + 1 + \frac{1}{2a_* \log 3}. \quad (20)$$

Substitution in (19) gives precisely (18). □

Proposition A.4 (Competitor reserve conditional on the target window). *Under Assumption 7, conditional on the event E_{sm} supplied by Proposition A.3, and under Lemmas A.4 and A.5, set $t_r = \sqrt{a_* \log r}$, $X_M = Z_{ij}^{(M)}$, and define*

$$W_{ij} := \bigcap_{M \in \{U, V, W\}} \{t_r \leq |Z_{ij}^{(M)}| \leq t_r + t_r^{-1}\},$$

$$C_{ij} := \bigcap_{\ell \neq j} \bigcap_{\{M, N\} \in \binom{\{U, V, W\}}{2}} \{|Z_{i\ell}^{(M)} Z_{i\ell}^{(N)}| \leq b_* \log r\}.$$

Then, for every signed $x = (x_U, x_V, x_W)$ satisfying the three inequalities in W_{ij} ,

$$\Pr\{C_{ij} \mid X_U = x_U, X_V = x_V, X_W = x_W\} \geq \frac{1}{10}. \quad (23)$$

No independence between C_{ij} and W_{ij} , and no independence across different targets, is assumed.

Proof. Put $a = a_* = 10/9$, $b = b_* = 19/18$, $q = q_* = 1/4096$, and

$$d_r := q(t_r + t_r^{-1}), \quad L_r^{\text{cap}} := 3\sqrt{\log r}, \quad s_r := b \log r - 2d_r L_r^{\text{cap}} - d_r^2. \quad (24)$$

For $\ell \neq j$, the regression representation (4) and the target-window bound imply

$$|c_{M\ell} x_M| \leq d_r, \quad |Z_{\ell}^{(M)}| \leq |G_{M\ell}| + d_r. \quad (25)$$

The numerical slack in a, b, q leaves a product exponent strictly above one. Writing $h = \log r \geq \log 3 > 1$, direct expansion gives

$$\frac{bh - s_r}{h} = 6q\sqrt{a} + \frac{6q}{\sqrt{a}h} + q^2 a + \frac{2q^2}{h} + \frac{q^2}{ah^2} \leq \frac{37}{3}q + 5q^2 < \frac{1}{180}. \quad (26)$$

Here $\sqrt{10/9} < 19/18$, $1/\sqrt{a} < 1$, $a < 2$, and $h > 1$ were used. Since

$$b - \frac{21}{20} = \frac{1}{180}, \quad (27)$$

equations (26)–(27) prove

$$s_r \geq \frac{21}{20} \log r. \quad (28)$$

For a fixed competitor ℓ , suppose first that $\max_M |G_{M\ell}| \leq L_r^{\text{cap}}$ and every standard-Gaussian pair product is at most s_r . Then, for every mode pair,

$$|Z_{\ell}^{(M)} Z_{\ell}^{(N)}| \leq |G_{M\ell} G_{N\ell}| + d_r(|G_{M\ell}| + |G_{N\ell}|) + d_r^2 \leq b \log r. \quad (29)$$

For a standard Gaussian, the Chernoff bound $\Pr\{|G| > x\} \leq 2e^{-x^2/2}$ follows directly from $\mathbb{E}e^{tG} = e^{t^2/2}$. Therefore the cap failure for the three modes is at most

$$6e^{-(L_r^{\text{cap}})^2/2} = 6r^{-9/2}. \quad (30)$$

By (28), the remaining product failure is at most q_r from Lemma A.5. A union bound over the $r - 1$ competitors, which requires no independence among their within-mode residuals, now gives

$$\Pr\{C_{ij}^c \mid X = x\} \leq (r - 1)(q_r + 6r^{-9/2}) \leq \frac{9}{10}. \quad (31)$$

Taking complements proves (23). □

Proposition A.5 (Joint one-slot extreme window). *Under Assumption 7, conditional on the event E_{sm} supplied by Proposition A.3, and under Lemma A.6 and Proposition A.4, set $t_r = \sqrt{a_* \log r}$ and define*

$$W_{ij} := \bigcap_{M \in \{U, V, W\}} \{t_r \leq |Z_{ij}^{(M)}| \leq t_r + t_r^{-1}\},$$

$$C_{ij} := \bigcap_{\ell \neq j} \bigcap_{\{M, N\} \in \binom{\{U, V, W\}}{2}} \{|Z_{i\ell}^{(M)} Z_{i\ell}^{(N)}| \leq b_* \log r\}.$$

The actual one-slot, target- j event is

$$E_{\text{win}, ij} := W_{ij} \cap C_{ij}. \quad (32)$$

Let

$$c_{\text{win}} := \frac{c_{\text{tar}}^3}{10}, \quad C_{\text{win}} := C_{\text{tar}}^3. \quad (33)$$

Then, for every $i \in [k]$ and $j \in [r]$,

$$c_{\text{win}} r^{-5/3} (\log r)^{-3/2} \leq p_{\text{win}, j} := \Pr(E_{\text{win}, ij} \mid \mathcal{F}_{\text{sm}}) \leq C_{\text{win}} r^{-5/3} (\log r)^{-3/2}. \quad (34)$$

The conditional probability is independent of the slot index i . Thus $p_{\text{win}} := \min_{j \in [r]} p_{\text{win}, j}$ obeys the same two-sided bound. Moreover, at the normalized initial state h_i^0 ,

$$R_j(h_i^0) \leq \frac{b_*}{a_*} = \frac{19}{20}, \quad S_j(h_i^0) \leq (r-1)R_j(h_i^0) \leq rR_j(h_i^0). \quad (35)$$

Proof. The three target coordinates X_U, X_V, X_W are independent standard Gaussians because the three raw mode vectors are independent and every realized target column is unit. Lemma A.6 therefore gives

$$c_{\text{tar}}^3 r^{-5/3} (\log r)^{-3/2} \leq \Pr(W_{ij} \mid \mathcal{F}_{\text{sm}}) \leq C_{\text{tar}}^3 r^{-5/3} (\log r)^{-3/2}. \quad (36)$$

For the lower bound, do not factor W_{ij} and C_{ij} . Instead, Proposition A.4 applies uniformly at every signed point x in the target window, so disintegration gives

$$\begin{aligned} \Pr(E_{\text{win}, ij} \mid \mathcal{F}_{\text{sm}}) &= \int_{x \in W_{ij}} \Pr(C_{ij} \mid X = x, \mathcal{F}_{\text{sm}}) d\Pr_X(x \mid \mathcal{F}_{\text{sm}}) \\ &\geq \frac{1}{10} \Pr(W_{ij} \mid \mathcal{F}_{\text{sm}}). \end{aligned} \quad (37)$$

The upper bound follows from $E_{\text{win}, ij} \subseteq W_{ij}$. Equations (33), (36), and (37) prove (34). Assumption 7 makes the conditional law identical across slots, proving the slot claim. The minimum over finitely many targets preserves both uniform bounds; no claim of independence among the target events is made.

Finally, normalization of $\xi_i^{(M)}$ cancels in every same-mode ratio. On $E_{\text{win}, ij}$, each target denominator has squared product at least $t_r^2 = a_* \log r$, while (22) bounds every corresponding competitor numerator product by $b_* \log r$. Thus each of the three pair ratios is at most $b_*/a_* = 19/20$, proving the first part of (35). Every sum defining S_j has at most $r-1$ terms, each at most R_j , proving the second part. $\square$

Proof. Fix any realized tensor in the event E_{sm} from Proposition A.3. For every target j and slot i , define $E_{\text{win},ij}$ by (21), (22), and (32), using only the original raw proposal coordinates. The regression variables in Lemma A.4 are proof-local and do not alter this event.

Lemma A.6 supplies the exact three-mode target mass. Lemma A.5 and Proposition A.4 show, with displayed constants, that imposing every competitor pair constraint retains at least a $1/10$ fraction after conditioning on any point of the target window. Proposition A.5 therefore proves

$$\frac{1}{10} \left[\sqrt{\frac{2}{\pi a_*}} e^{-1-1/(2a_* \log 3)} \right]^3 r^{-5/3} (\log r)^{-3/2} \leq p_{\text{win}} \leq \left(\frac{2}{\pi a_*} \right)^{3/2} r^{-5/3} (\log r)^{-3/2}. \quad (38)$$

This is the displayed probability claim. It is a one-slot probability conditional on the fixed good instance, contains no confidence parameter or rank choice, and uses neither target/competitor independence nor independence among events for different targets. Equation (35) records the same-event ratio interface needed after the downstream coverage step. $\square$

A.3 All-target coverage

Proposition A.5 supplies, uniformly in slot i and target j conditional on the fixed smoothed instance,

$$p_{\text{win},j} := \Pr(E_{\text{win},ij} \mid \mathcal{F}_{\text{sm}}) \in \left[c_{\text{win}} r^{-5/3} (\log r)^{-3/2}, C_{\text{win}} r^{-5/3} (\log r)^{-3/2} \right]. \quad (1)$$

It also gives the ratio initialization on $E_{\text{win},ij}$. Below we use $1 - x \leq e^{-x}$ for $x \in [0, 1]$ and the finite union bound; neither step requires independence among target-indexed events in a common slot.

Fix an arbitrary realized smoothed instance for which Proposition A.5 holds, and condition on its σ -field $\mathcal{F}_{\text{sm}}$. Write

$$\mathcal{E}_{ij} := E_{\text{win},ij}, \quad p_j := p_{\text{win},j} = \Pr(\mathcal{E}_{ij} \mid \mathcal{F}_{\text{sm}}). \quad (2)$$

The event $\mathcal{E}_{ij}$ is evaluated from the raw Gaussian tape in slot i , with the target index used only to state the analysis event. Proposition A.5 gives $p_j \geq c_{\text{win}} r^{-5/3} (\log r)^{-3/2}$ in (1).

Lemma A.7 (Fixed-target slot-miss bound). *Under Assumption 7 and Proposition A.5, fix $j \in [r]$ and define*

$$B_j := \bigcap_{i=1}^k \mathcal{E}_{ij}^c. \quad (3)$$

Then, conditional on $\mathcal{F}_{\text{sm}}$,

$$\Pr(B_j \mid \mathcal{F}_{\text{sm}}) = (1 - p_j)^k \leq e^{-kp_j} \leq r^{-C_{\text{rank}} c_{\text{win}}}. \quad (4)$$

No relation among $\mathcal{E}_{1j}, \dots, \mathcal{E}_{rj}$ for a common slot is required.

Proof. For fixed j , each $\mathcal{E}_{ij}$ is a measurable function of the raw triple Ξ_i and the fixed realized tensor. The primitive initialization assumption makes the Ξ_i 's independent over i , so the indicators $1_{\mathcal{E}_{ij}}$ are independent and have the common conditional success probability p_j . Therefore the first equality in (4) holds. The elementary inequality in the displayed auxiliary fact gives the first inequality. From (1) and the rank formula,

$$kp_j \geq \left(C_{\text{rank}} r^{5/3} (\log r)^{5/2} \right) \left(c_{\text{win}} r^{-5/3} (\log r)^{-3/2} \right) = C_{\text{rank}} c_{\text{win}} \log r, \quad (5)$$

where the ceiling in k only increases the left side. Exponentiating (5) proves the final inequality in (4). The proof has used independence only in the slot index i , and nowhere factors events with different target labels. $\square$

Proposition A.6 (All-target coupon coverage). *Under Assumptions 6 and 7, and Proposition A.5, define*

$$E_{\text{cov}} := \bigcap_{j=1}^r \bigcup_{i=1}^k \mathcal{E}_{ij} = \bigcap_{j=1}^r B_j^c. \quad (6)$$

Then, for every fixed smoothed instance in the event E_{sm} supplied by Proposition A.3,

$$\Pr(E_{\text{cov}} \mid \mathcal{F}_{\text{sm}}) \geq 1 - r^{1-C_{\text{rank}}C_{\text{win}}}. \quad (7)$$

Choose the universal rank constant once and for all so that

$$C_{\text{rank}}C_{\text{win}} \geq 4. \quad (8)$$

Then, for all $r \geq 3$,

$$\Pr(E_{\text{cov}} \mid \mathcal{F}_{\text{sm}}) \geq 1 - r^{-3} \geq p_0, \quad p_0 := 1 - 3^{-3} = \frac{26}{27} > 0. \quad (9)$$

No independence across the r target-indexed events is asserted or used.

Proof. The complement of (6) is $\bigcup_{j=1}^r B_j$. Applying the finite union bound and Lemma A.7 gives

$$\Pr(E_{\text{cov}}^c \mid \mathcal{F}_{\text{sm}}) \leq \sum_{j=1}^r \Pr(B_j \mid \mathcal{F}_{\text{sm}}) \leq r r^{-C_{\text{rank}}C_{\text{win}}} = r^{1-C_{\text{rank}}C_{\text{win}}}, \quad (10)$$

which proves (7). Under (8), the exponent is at most -3 , and $r^{-3} \leq 3^{-3}$ for $r \geq 3$, proving (9). The events B_j may be maximally correlated for different j ; (10) remains valid because it is a union bound rather than a product calculation. $\square$

Proposition A.7 (Proof-only witnesses and label-free procedure interface). *Under the hypotheses of Proposition A.6, on the event E_{cov} , for every $j \in [r]$ the finite nonempty set*

$$\mathcal{I}_j := \{i \in [k] : \mathcal{E}_{ij} \text{ occurs}\} \quad (11)$$

has a proof-only canonical witness $i(j) := \min \mathcal{I}_j$. The conclusion of Proposition A.5 then gives

$$R_j(h_{i(j)}^0) \leq \frac{19}{20}, \quad S_j(h_{i(j)}^0) \leq r R_j(h_{i(j)}^0). \quad (12)$$

The construction of E_{cov} , $\mathcal{I}_j$, and $i(j)$ is an analysis definition only: the SL-SC-JEP-ALS procedure does not read any target index j , any u_j, v_j, w_j , any raw-coordinate window event, or the proof-only witness. It executes the same slotwise Jacobi, score, and clustering rules for every tape. Thus (6) is a generated probabilistic interface and not a label-dependent acceptance test.

Proof. By (6), each union over i is nonempty on E_{cov} ; finiteness of $[k]$ makes the minimum in (11) well-defined. Equation (12) is exactly the ratio conclusion of Proposition A.5 applied to the witness event $\mathcal{E}_{i(j)j}$. The final statement follows by inspection of the setting's procedure: target labels occur only in the proof definitions of $A_{\ell|j}, B_{\ell|j}, C_{\ell|j}$ and $E_{\text{win},ij}$; no such quantities are computed by the algorithm. A proof may use $i(j)$ to organize the subsequent targetwise induction, while the actual state evolution remains the unlabeled stored-state trajectory. $\square$

Proof. Fix a realized smoothed instance for which Proposition A.5 holds. It supplies a uniform per-target lower probability $p_j \geq c_{\text{win}} r^{-5/3} (\log r)^{-3/2}$. Lemma A.7 uses the k independent raw slots to make each target miss probability at most $r^{-C_{\text{rank}} c_{\text{win}}}$. Proposition A.6 takes a union over targets and, after the single universal choice (8), produces

$$E_{\text{cov}} = \{\text{every target has at least one jointly observable window slot}\}, \quad \Pr(E_{\text{cov}} \mid \mathcal{F}_{\text{sm}}) \geq p_0 = \frac{26}{27}. \quad (13)$$

Proposition A.7 records the targetwise initial ratio interface for any proof-only witness and explicitly separates it from the label-free procedure. The proof never assumes independence among the events for different targets in a slot or across the target index; only the primitive slot independence is used. Therefore (13) is exactly the displayed coverage output E_{cov}, p_0 , ready for the trajectory induction and the restart conclusion of Proposition A.31.

For completeness, the rank scaling has no hidden confidence factor. With C_{rank} fixed universally and $r \geq 3$,

$$C_{\text{rank}} r^{5/3} (\log r)^{5/2} \leq k \leq \left(C_{\text{rank}} + \frac{1}{3^{5/3} (\log 3)^{5/2}} \right) r^{5/3} (\log r)^{5/2}, \quad (14)$$

so $k = \Theta(r^{5/3} (\log r)^{5/2})$. The upper feasibility and subquadratic statement are already part of Assumption 6; indeed $k/r^2 = O((\log r)^{5/2}/r^{1/3}) = o(1)$ for fixed C_{rank} . If an unconditional nested probability is desired, the tower property and the instance bound give

$$\Pr(E_{\text{sm}} \cap E_{\text{cov}}) \geq \Pr(E_{\text{sm}}) p_0 \geq (1 - \delta_{\text{sm}}) p_0, \quad (15)$$

but the public one-run initialization guarantee remains the conditional bound (13), with δ_{init} reserved for independent restarts. $\square$

A.4 Jacobi chart recurrence

Proposition A.7 supplies, on E_{cov} , a witness for every target with $R_0 \leq 19/20$ and $S_0 \leq r R_0$. Proposition A.1 supplies, on E_{sm} , unit diagonal Gram matrices $K_M = G_M^\top G_M$ with absolute row and column off-diagonal mass at most q_{real} . We repeatedly use the corresponding induced ℓ_1 bounds, the Neumann estimate $\|K^{-1}\| \leq (1 - q)^{-1}$ when $\|K - I\| \leq q < 1$, and the normalization inequality

$$\left\| \frac{a}{\|a\|_2} - \frac{b}{\|b\|_2} \right\|_2 \leq \frac{2\|a - b\|_2}{\min\{\|a\|_2, \|b\|_2\}}$$

for nonzero a, b .

Fix $E_{\text{sm}}, E_{\text{cov}}$, a target j , and one proof-only witness slot $i(j)$. Suppress the slot index. Put

$$q := q_{\text{real}} \leq q_* = \frac{1}{4096}, \quad \gamma := \Gamma \leq \frac{101}{100}.$$

For $M \in \{U, V, W\}$, write $g_{M,\ell}$ for the realized target column (u_ℓ, v_ℓ, w_ℓ , respectively), and h_M^t for the mode- M state after t simultaneous commits. At every time at which the target correlation is nonzero, use a proof-only sign so that

$$a_{M,j}^t := \langle g_{M,j}, h_M^t \rangle > 0, \quad x_{M,\ell}^t := \frac{\langle g_{M,\ell}, h_M^t \rangle}{a_{M,j}^t} \quad (\ell \neq j). \quad (1)$$

The absolute values of (1) are the setting's A_ℓ, B_ℓ, C_ℓ . Sign choices do not alter R, S or ζ , because ζ minimizes over a modewise sign.

Lemma A.8 (Simultaneous coefficient and pair-mass recurrence). *Under the realized Gram conclusion of Proposition A.3 and the covered-slot initial interface of Proposition A.7, suppose at time t all target correlations in (1) are nonzero and $\gamma R_t < 1$. For mode M , with held modes M', M'' , define*

$$\beta_{M,j}^t := 1, \quad \beta_{M,\ell}^t := \frac{\lambda_\ell}{\lambda_j} x_{M',\ell}^t x_{M'',\ell}^t \quad (\ell \neq j). \quad (2)$$

Then the target coefficient of the unnormalized M -contraction is nonzero, and, with $D_t := 1 - \gamma q R_t$,

$$|x_{M,\ell}^{t+1}| \leq \frac{\gamma R_t + q}{D_t} \quad (\ell \neq j), \quad (3)$$

$$R_{t+1} \leq \left(\frac{\gamma R_t + q}{1 - \gamma q R_t} \right)^2, \quad S_{t+1} \leq \frac{(\gamma R_t + q) \{ \gamma(1 + q) S_t + q \}}{(1 - \gamma q R_t)^2}. \quad (4)$$

The conclusion holds for the first update even when h^0 has a component orthogonal to a target span.

Proof. Divide the M -contraction by its target coefficient $\lambda_j a_{M',j}^t a_{M'',j}^t$. Its coefficient vector in the $g_{M,\ell}$ basis is $e_j + \beta_M^t$. Since $|\beta_{M,\ell}^t| \leq \gamma B_\ell^t C_\ell^t \leq \gamma R_t < 1$, the target coordinate of the contraction obeys

$$\begin{aligned} \left| \left\langle g_{M,j}, e_j + \sum_{\ell \neq j} \beta_{M,\ell}^t g_{M,\ell} \right\rangle \right| &\geq 1 - \sum_{\ell \neq j} |\beta_{M,\ell}^t| |\langle g_{M,j}, g_{M,\ell} \rangle| \\ &\geq 1 - \gamma R_t q = D_t > 0. \end{aligned} \quad (5)$$

Thus the contraction norm is positive and its normalized output is defined. For an off-target index ℓ , the diagonal term contributes $|\beta_{M,\ell}^t|$, and all remaining terms are bounded by the Gram row mass q , because every $|\beta_{M,m}^t| \leq 1$. Therefore

$$|\langle g_{M,\ell}, e_j + \sum_m \beta_{M,m}^t g_{M,m} \rangle| \leq \gamma R_t + q. \quad (6)$$

Dividing (6) by (5) proves (3), and taking products proves the first part of (4). For the S bound, let $n_{M,\ell}$ denote the absolute numerator in (6). Using the target column mass q , then swapping the remaining finite sums and using the Gram column mass q ,

$$\begin{aligned}
\sum_{\ell \neq j} n_{M,\ell} &\leq \sum_{\ell \neq j} |\beta_{M,\ell}^t| + q + q \sum_{\ell \neq j} |\beta_{M,\ell}^t| \\
&\leq \gamma(1+q) \sum_{\ell \neq j} |x_{M',\ell}^t x_{M'',\ell}^t| + q \\
&\leq \gamma(1+q) S_t + q.
\end{aligned} \tag{7}$$

For example, for the U, V pair, the U -numerator is at most $\gamma R_t + q$ termwise, while the ℓ_1 sum of the V -numerator is bounded by (7). Division by the two target denominators gives the first pair sum in S_{t+1} ; cyclic permutation gives the other pair sums. At $t = 0$, the window event supplies nonzero target correlations and (5) proves the first update before any span assertion. After that update every mode lies in its exact target span because the tensor contraction is a linear combination of the $g_{M,\ell}$.

□

Proposition A.8 (Noncircular scalar envelope and finite chart entry). *Under Assumption 6, on the coverage interface of Proposition A.7, and under Lemma A.8, put*

$$\bar{R} := \frac{19}{20}, \quad d_* := \frac{999}{1000}, \quad \rho_R = \rho_S := \frac{49}{50}. \tag{8}$$

Then every finite trajectory iterate satisfies

$$R_t \leq \bar{R}, \quad 1 - \gamma q R_t \geq d_* > 0, \quad \gamma R_t < 1. \tag{9}$$

Moreover, if $\Lambda_t := \max_M \sum_{\ell \neq j} |x_{M,\ell}^t|$, then

$$\Lambda_{t+1} \leq \frac{\gamma(1+q)S_t + q}{1 - \gamma q R_t}, \quad S_{t+1} \leq \rho_S S_t + q. \tag{10}$$

There is a universal choice $C_{\text{burn}} = 2048$ such that, with $L_{\text{burn}} = \lceil C_{\text{burn}} \log r \rceil$, every covered witness trajectory has

$$R_{L_{\text{burn}}-1} \leq \frac{1}{128}, \quad S_{L_{\text{burn}}-1} \leq \frac{1}{7200}, \quad \Lambda_{L_{\text{burn}}} \leq \frac{1}{256}, \quad \max_{M,\ell} |x_{M,\ell}^{L_{\text{burn}}}| \leq \frac{1}{100}. \tag{11}$$

Proof. Let $f(x) = ((\gamma x + q)/(1 - \gamma q x))^2$. On $[0, \bar{R}]$, f is increasing. The numerical bounds

$$\gamma \bar{R} + q < \frac{24}{25}, \quad 1 - \gamma q \bar{R} > \frac{999}{1000} \tag{12}$$

give $f(\bar{R})/\bar{R} < 49/50$. For $x \geq q/\gamma$, direct differentiation shows $f(x)/x$ is increasing, since

$$\frac{d}{dx} \log \frac{f(x)}{x} = \frac{2\gamma}{\gamma x + q} - \frac{1}{x} + \frac{2\gamma q}{1 - \gamma q x} > 0. \tag{13}$$

The covered initial state satisfies $\gamma R_0 \leq (101/100)(19/20) < 1$, so Lemma A.8 is applicable at $t = 0$. Thus $f(x) \leq (49/50)x$ whenever $q/\gamma \leq x \leq \bar{R}$; if $x < q/\gamma$, then $x < 1/128$ and monotonicity gives $f(x) \leq f(q/\gamma) < f(\bar{R}) < \bar{R}$. Since $R_0 \leq \bar{R}$, induction using (4) proves (9), and the first phase reaches $R_t \leq 1/128$ in at most

$$N_R := \left\lceil \frac{\log(128\bar{R})}{\log(50/49)} \right\rceil \tag{14}$$

steps. The coefficient multiplying S_t in (4) is bounded by

$$\frac{(24/25)(101/100)(1001/1000)}{(999/1000)^2} < \frac{49}{50}, \quad (15)$$

and the additive term is less than q . Hence

$$S_t \leq (49/50)^t (19r/20) + 50q. \quad (16)$$

Set

$$N_S(r) := \left\lceil \frac{\log\{(19r/20)/(1/64 - 50q_*)\}}{\log(50/49)} \right\rceil. \quad (17)$$

The denominator is positive because $50q_* = 50/4096 < 1/64$. After $N_S(r)$ steps, (16) gives $S_t \leq 1/64$. At any time with $R_t \leq 1/128$, (4) and the same numerical constants give

$$R_{t+1} \leq 10^{-4}, \quad S_{t+1} \leq \frac{1}{7200}, \quad \Lambda_{t+1} < \frac{1}{60}. \quad (18)$$

Applying (10) once more from the S -bound $1/7200$ gives $\Lambda_{t+2} \leq 1/256$, while (3) gives the $1/100$ coordinate bound. Once $R \leq 1/128$ and $S \leq 1/7200$, the right side of the S recurrence is again below $1/7200$, so this smaller bound persists. Consequently (11) holds once $L_{\text{burn}} \geq \max\{N_R, N_S(r)\} + 2$. A direct numerical comparison at $r = 3$, and the linear $\log r$ term in (17), gives

$$\max\{N_R, N_S(r)\} + 2 \leq 2048 \log r \quad (r \geq 3). \quad (19)$$

Thus the prescribed universal $C_{\text{burn}} = 2048$ suffices. All bounds are finite-horizon consequences of (4); no chart or denominator condition was assumed in obtaining them. $\square$

Lemma A.9 (Target-chart self-map and contraction). *On the event supplied by Proposition A.3, let $K_M := G_M^\top G_M$ denote the step-local Gram (distinct from the setting-defined same-state coefficient matrix K_M). For a target j , write a signed off-target ratio vector $x_M \in \mathbb{R}^{r-1}$, with $e_j + x_M$ denoting the r -vector obtained by inserting a 1 in coordinate j . On the chart*

$$\mathcal{C}_j := \left\{ (x_U, x_V, x_W) : \max_M \|x_M\|_\infty \leq \frac{1}{100}, \max_M \|x_M\|_1 \leq \frac{1}{256}, S_j \leq \frac{1}{512} \right\}, \quad (20)$$

the exact signed ratio map is

$$\Phi_M(x_{M'}, x_{M''})_\ell = \frac{[K_M(e_j + \beta_M)]_\ell}{[K_M(e_j + \beta_M)]_j}, \quad \beta_{M,j} = 1, \quad \beta_{M,\ell} = \frac{\lambda_\ell}{\lambda_j} x_{M',\ell} x_{M'',\ell}, \quad (21)$$

and is a self-map of $\mathcal{C}_j$. In the metric

$$d_{\text{ch}}(x, \tilde{x}) := \max_M \|x_M - \tilde{x}_M\|_1, \quad (22)$$

it satisfies

$$d_{\text{ch}}(\Phi(x), \Phi(\tilde{x})) \leq \frac{1}{4} d_{\text{ch}}(x, \tilde{x}). \quad (23)$$

If $p_M(x)$ is the unit vector in $\text{span}(G_M)$ with correlation ratios x_M and positive target correlation, then

$$\|p_M(x) - p_M(\tilde{x})\|_2 \leq 4 \|x_M - \tilde{x}_M\|_1. \quad (24)$$

Proof. Let $v = e_j + x$, and define

$$p_M(x) := \frac{G_M \mathbf{K}_M^{-1} v}{\|G_M \mathbf{K}_M^{-1} v\|_2}. \quad (25)$$

The Neumann bound applies because $q < 1$, and substitution gives $G_M^\top p_M(x)$ proportional to v ; hence (25) is the unique oriented span direction with the stated ratios. The Gram eigenvalues lie in $[1 - q, 1 + q]$. Consequently

$$\|G_M \mathbf{K}_M^{-1}(x - \tilde{x})\|_2 \leq (1 - q)^{-1/2} \|x - \tilde{x}\|_2, \quad \|G_M \mathbf{K}_M^{-1} v\|_2 \geq (1 + q)^{-1/2} \|v\|_2.$$

Since $\|e_j + x\|_2 \geq 1$ on the chart, the normalization inequality gives (24) with the conservative factor 4.

For (21), divide the held-mode target coefficients exactly as in (2); the common target scale cancels. On (20), $\|\beta_{M,-j}\|_\infty \leq \gamma/10^4$, and the target denominator in (21) is at least 999/1000. The off-target numerator has coordinatewise bound $\gamma/10^4 + q < 1/100$. Its ℓ_1 bound is $\gamma(1 + q)S_j + q < 1/256$, by the row/column summation used in (7). For every output pair, the product mass is at most $(1/100)(1/256) < 1/512$. Thus Φ is a self-map.

For two chart points, with $d = d_{\text{ch}}(x, \tilde{x})$, the product rule gives

$$\|\beta_M - \tilde{\beta}_M\|_1 \leq 2\gamma(1/100)d. \quad (26)$$

The off-target block of $\mathbf{K}_M$ has induced column mass at most $1 + q$, the target row has mass at most q , and both quotient denominators are at least 999/1000. Using the self-map ℓ_1 bound in the denominator variation term yields

$$\|\Phi_M(x) - \Phi_M(\tilde{x})\|_1 \leq \left(\frac{1 + q}{999/1000} + \frac{q}{(999/1000)^2 256} \right) \|\beta_M - \tilde{\beta}_M\|_1 < \frac{1}{4}d. \quad (27)$$

Taking the maximum over modes proves (23). □

Proposition A.9 (Finite projective certificate and stored-state interface). *Under Assumption 6, on the coverage interface of Proposition A.7, and under Proposition A.8 and Lemma A.9, on every covered witness trajectory the chart $\mathcal{C}_j$ is entered no later than L_{burn} , remains invariant, and for every $t \geq L_{\text{burn}}$,*

$$\zeta(h^t) \leq \frac{1}{32} 4^{-(t-L_{\text{burn}})}. \quad (28)$$

With the universal choice $C_{\text{cert}} = 64$ and $L_{\text{cert}} = \lceil C_{\text{cert}} \log r \rceil$, $\zeta(h^{L_{\text{burn}}+L_{\text{cert}}}) \leq \tau_r = q_^2/(10^4 r)$. Thus the algorithm stores a state at or before $L_{\text{prop}} = L_{\text{burn}} + L_{\text{cert}}$; the stored state is h^t , not its look-ahead image. If the procedure stores an earlier state because its observable ζ is already below the threshold, the continuation used in this proof is only a proof device and the earlier stored state already satisfies the exported certificate.*

Proof. By (11), its $1/100$ and $1/256$ bounds imply that the state at L_{burn} has $S_j \leq (1/100)(1/256) < 1/512$, so it is in $\mathcal{C}_j$; the self-map keeps all subsequent states there. Let $\Delta_t = d_{\text{ch}}(x^t, x^{t-1})$. The first in-chart displacement is at most $2/256 = 1/128$. For $t \geq L_{\text{burn}} + 1$, (23) gives the recursive inequality below; the displayed closed form also holds at $t = L_{\text{burn}}$ by the diameter bound:

$$\Delta_{t+1} \leq \frac{1}{4}\Delta_t, \quad \Delta_{t+1} \leq \frac{1}{128} 4^{-(t-L_{\text{burn}})}. \quad (29)$$

For each mode, the oriented state at time t is $p_M(x_M^t)$, while the oriented look-ahead Jacobi state is $p_M(x_M^{t+1})$. Equation (24) and the sign-minimum in ζ_M give

$$\zeta_M(h^t) \leq 4\|x_M^t - x_M^{t+1}\|_1,$$

which with (29) proves (28). Finally, $4^{-L_{\text{cert}}}/32 \leq \tau_r$ holds for all $r \geq 3$ under $C_{\text{cert}} = 64$: after substituting $\tau_r = q_*^2/(10^4 r)$, the required logarithm is $\log(32 \cdot 10^4 r/q_*^2)/\log 4$, and $64 \log r$ dominates it at $r = 3$ and hence for all larger r . If the bound first holds at $t = L_{\text{prop}}$, the algorithm still evaluates $\zeta(h^t)$ and stores that old state; no look-ahead state is exported. $\square$

Proof. On $E_{\text{sm}} \cap E_{\text{cov}}$, choose the witness from Proposition A.7 $i(j)$ for each target j . The argument in fact applies to every pair (i, j) for which the window event occurs; E_{cov} is used only to ensure that at least one such pair exists for every target. The initial interface gives nonzero target coordinates and $R_0 \leq 19/20$, $S_0 \leq rR_0$. Lemma A.8 proves the exact displayed R, S recurrences, including the first simultaneous update, and proves every target denominator before division. Proposition A.8 gives the noncircular invariant, finite burn entry, and modewise ℓ_1 chart bounds. The chart lemma supplies an independent contraction mechanism for the same old-state map; Proposition A.9 converts successive displacement to the exact algorithmic Euclidean ζ and guarantees a stored certificate within the prescribed finite horizon. These conclusions define $E_{\text{RS}}, E_{\text{chart}}, E_{\text{chart}}^{(1)}, E_{\text{cert}}$ as generated outputs. No target label, comparator, future ALS output, or landing output is used by the procedure. A zero target denominator or contraction on an uncovered tape remains an observable unsuccessful branch; on the covered branch, (5) rules it out before the first commit.

Concretely, E_{RS} is the finite intersection, over all the window pairs and all $t \leq L_{\text{prop}}$, of (3)–(4) and (9); E_{chart} and $E_{\text{chart}}^{(1)}$ are the corresponding post-burn bounds (20), and E_{cert} records that each such slot stores its first old state with $\zeta \leq \tau_r$. These are proof events, not extra algorithmic tests. $\square$

A.5 Signed certificate equation

Proposition A.1 gives, on the event supplied by Proposition A.3, unit diagonal matrices $H_M = G_M^\top G_M$ with off-diagonal row and column masses at most $q_{\text{real}} \leq q_*$. Proposition A.9 says that a covered slot stores its old state h^t , not its look-ahead image, with $\zeta_M(h^t) \leq \tau_r$, after all modes have entered the corresponding realized target span. We also use $\|(H - I)v\|_1 \leq q\|v\|_1$, $\|x \circ y\|_1 \leq \|x\|_2\|y\|_2$, and $\|G_M^\top z\|_2 \leq \sqrt{1+q}\|z\|_2$, each instantiated below.

Throughout, let $m_c := |\mathcal{I}_{\text{cert}}|$, index certified slots by $a \in \mathcal{I}_{\text{cert}}$, and put $g_{U,j} = u_j$, $g_{V,j} = v_j$, $g_{W,j} = w_j$. For a slot a , write $p_{M,a}$ for its three unit vectors and define the raw correlation vector

$$c_{M,a} := G_M^\top p_{M,a}, \quad k_{M,a} := D_\lambda(c_{M',a} \circ c_{M'',a}), \quad (1)$$

where M', M'' are the two held modes. The symbols in (1) are slot-local; the setting's K_M, D_M for selected representatives are recovered by restricting the slot index set to the selected r columns.

Lemma A.10 (Signed certificate equation and score-preserving orientation). *Under the certified-state conclusion of Proposition A.9 and the static event of Proposition A.3, for every certified slot (a) and mode (M) , let*

$$d_{M,a} := \|G_M k_{M,a}\|_2 > 0,$$

and choose $\varepsilon_{M,a} \in \{\pm 1\}$ attaining the certified projective residual. Then, with

$$r_{M,a} := d_{M,a}p_{M,a} - \varepsilon_{M,a}G_M k_{M,a},$$

$$\|r_{M,a}\|_2 \leq d_{M,a}\tau_r, \quad d_{M,a}p_{M,a} = \varepsilon_{M,a}G_M k_{M,a} + r_{M,a}. \quad (2)$$

Writing $\theta_a := \langle T, p_{U,a} \otimes p_{V,a} \otimes p_{W,a} \rangle$, all three signs satisfy

$$\varepsilon_{U,a} = \varepsilon_{V,a} = \varepsilon_{W,a} = \text{sgn}(\theta_a), \quad \theta_a \neq 0, \quad (3)$$

and, with $s_a := \text{sgn}(\theta_a)$, the proof-only orientation

$$\bar{p}_{U,a} := s_a p_{U,a}, \quad \bar{p}_{V,a} := p_{V,a}, \quad \bar{p}_{W,a} := p_{W,a} \quad (4)$$

has

$$|\theta_a| \bar{p}_{U,a} \otimes \bar{p}_{V,a} \otimes \bar{p}_{W,a} = \theta_a p_{U,a} \otimes p_{V,a} \otimes p_{W,a}.$$

Defining

$$\bar{c}_{M,a} := G_M^\top \bar{p}_{M,a}, \quad \bar{k}_{M,a} := D_\lambda(\bar{c}_{M',a} \circ \bar{c}_{M'',a}), \quad \bar{r}_{M,a} := \eta_{M,a} r_{M,a},$$

where $(\eta_{U,a}, \eta_{V,a}, \eta_{W,a}) = (s_a, 1, 1)$, gives the right-sided barred equation

$$d_{M,a} \bar{p}_{M,a} = G_M \bar{k}_{M,a} + \bar{r}_{M,a}, \quad \|\bar{r}_{M,a}\|_2 \leq d_{M,a}\tau_r. \quad (5)$$

Equivalently, if P_M^{raw} , K_M^{raw} , and R_M^{raw} stack the columns $p_{M,a}$, $k_{M,a}$, and $r_{M,a}$, and if $D_M^{\text{cert}} := \text{diag}(d_{M,a})$ and $\Sigma_M := \text{diag}(\varepsilon_{M,a})$, then

$$P_M^{\text{raw}} D_M^{\text{cert}} = G_M K_M^{\text{raw}} \Sigma_M + R_M^{\text{raw}}, \quad \|R_M^{\text{raw}}(:, a)\|_2 \leq d_{M,a}\tau_r. \quad (5a)$$

The barred stacks satisfy

$$\bar{P}_M D_M^{\text{cert}} = G_M \bar{K}_M + \bar{R}_M, \quad \|\bar{R}_M(:, a)\|_2 \leq d_{M,a}\tau_r. \quad (5b)$$

In particular, writing $\sigma_a := |\theta_a|$,

$$|d_{M,a} - \sigma_a| \leq d_{M,a}\tau_r, \quad \frac{\sigma_a}{d_{M,a}} \in [1 - \tau_r, 1 + \tau_r]. \quad (6)$$

Proof. The stored-state interface gives a nonzero contraction and

$$\left\| p_{M,a} - \varepsilon_{M,a} \frac{G_M k_{M,a}}{d_{M,a}} \right\|_2 \leq \tau_r.$$

Multiplying by $d_{M,a}$ proves (2). By multilinearity,

$$p_{M,a}^\top G_M k_{M,a} = \sum_{j=1}^r \lambda_j \langle g_{U,j}, p_{U,a} \rangle \langle g_{V,j}, p_{V,a} \rangle \langle g_{W,j}, p_{W,a} \rangle = \theta_a.$$

Taking the inner product of (2) with $p_{M,a}$ therefore gives

$$d_{M,a} - \varepsilon_{M,a} \theta_a = p_{M,a}^\top r_{M,a}, \quad |d_{M,a} - \varepsilon_{M,a} \theta_a| \leq d_{M,a}\tau_r. \quad (7)$$

Because $\tau_r = q_*^2/(10^4 r) < 1$, (7) implies $\varepsilon_{M,a} \theta_a > 0$, proving (3) simultaneously for all modes. For the orientation (4), the held-mode sign products are

$$\bar{k}_{U,a} = k_{U,a}, \quad \bar{k}_{V,a} = s_a k_{V,a}, \quad \bar{k}_{W,a} = s_a k_{W,a}.$$

Multiplying the U equation in (2) by s_a , and leaving the other two equations unchanged, gives (5); the rank-one identity follows because the product of the three orientation signs is s_a . Finally,

$$\bar{p}_{M,a}^\top \bar{r}_{M,a} = d_{M,a} - \bar{p}_{M,a}^\top G_M \bar{k}_{M,a} = d_{M,a} - \sigma_a,$$

so (6) follows from the residual bound. This common scalar identity is the normalization cancellation: the three modes share one σ_a , and no independent first-order scalar error is charged in (5). $\square$

Proposition A.10 (Projected Gram-leak and finite residual ledger). *Under the hypotheses of Lemma A.10, fix a barred certified slot a . Put $H_M := G_M^\top G_M$, $\bar{c}_M := \bar{c}_{M,a}$, $\bar{k}_M := \bar{k}_{M,a}$, $d_M := d_{M,a}$, and $\bar{r}_M := \bar{r}_{M,a}$. For every selector $w = (w_1, \dots, w_r) \in [0, 1]^r$,*

$$\sum_{j=1}^r w_j |d_M \bar{c}_{M,j} - \bar{k}_{M,j}| \leq q \|\bar{k}_M\|_1 + \sqrt{r(1+q)} d_M \tau_r, \quad (8)$$

and, in particular,

$$|d_M \bar{c}_{M,j} - \lambda_j \bar{c}_{M',j} \bar{c}_{M'',j}| \leq q \lambda_{\max} + d_M \tau_r. \quad (9)$$

Every stored state is in $\text{range}(G_M)$, so if $\bar{p}_{M,a} = G_M \alpha_{M,a}$, then with $\Delta_{M,a} := (H_M - I) \alpha_{M,a}$ there is an exact decomposition

$$d_M \alpha_{M,a} = B_{M,a} + L_{M,a} + X_{M,a} + \varrho_{M,a}, \quad (10)$$

where

$$\begin{aligned} B_{M,a} &:= D_\lambda(\alpha_{M',a} \circ \alpha_{M'',a}), \\ L_{M,a} &:= D_\lambda(\alpha_{M',a} \circ \Delta_{M'',a} + \Delta_{M',a} \circ \alpha_{M'',a}), \\ X_{M,a} &:= D_\lambda(\Delta_{M',a} \circ \Delta_{M'',a}), \end{aligned}$$

and

$$\|L_{M,a}\|_1 \leq \frac{2\lambda_{\max} q}{1-q}, \quad \|X_{M,a}\|_1 \leq \frac{\lambda_{\max} q^2}{1-q}, \quad \|\varrho_{M,a}\|_1 \leq \frac{\sqrt{r} d_{M,a} \tau_r}{\sqrt{1-q}}. \quad (11)$$

The first normalized identity is

$$\alpha_{M,a} - \frac{B_{M,a}}{d_{M,a}} = \frac{L_{M,a} + X_{M,a} + \varrho_{M,a}}{d_{M,a}}, \quad (12)$$

so the scalar normalization term is not double-counted. If $d_{M,a}$ is replaced by σ_a , the exact identity is

$$\alpha_{M,a} - \frac{B_{M,a}}{\sigma_a} = \frac{L_{M,a} + X_{M,a} + \varrho_{M,a}}{d_{M,a}} + \left(\frac{1}{d_{M,a}} - \frac{1}{\sigma_a} \right) B_{M,a}. \quad (12a)$$

The additional term therefore has ℓ_1 norm at most $\tau_r(1 - \tau_r)^{-1} \|B_{M,a}\|_1 / d_{M,a}$.

Proof. Projecting (5) by $G_M^\top$ gives the exact coordinate identity

$$d_M \bar{c}_M = H_M \bar{k}_M + G_M^\top \bar{r}_M, \quad d_M \bar{c}_M - \bar{k}_M = (H_M - I) \bar{k}_M + G_M^\top \bar{r}_M. \quad (13)$$

The row and column bounds from Proposition A.3 imply $\|(H_M - I) \bar{k}_M\|_1 \leq q \|\bar{k}_M\|_1$. Also $\|G_M^\top \bar{r}_M\|_1 \leq \sqrt{r} \|G_M^\top \bar{r}_M\|_2 \leq \sqrt{r(1+q)} d_M \tau_r$, proving (8). Since all correlations are inner products of unit vectors, $|\bar{c}_{M,j}| \leq 1$, which proves (9).

For the coefficient form, write $\bar{p}_M = G_M \alpha_M$. The Gram eigenvalue bounds and $\bar{p}_M^\top \bar{p}_M = 1$ give

$$\|\alpha_M\|_2 \leq (1 - q)^{-1/2}, \quad \|\Delta_M\|_2 \leq q(1 - q)^{-1/2}, \quad \bar{c}_M = \alpha_M + \Delta_M. \quad (14)$$

Substitution of (14) into $\bar{k}_M = D_\lambda(\bar{c}_{M'} \circ \bar{c}_{M''})$ gives $B + L + X$. Because both sides of (5) are in $\text{range}(G_M)$, the residual is in that range and $\varrho_M := G_M^\dagger \bar{r}_M$ satisfies $\|\varrho_M\|_2 \leq d_M \tau_r / \sqrt{1 - q}$. The Cauchy–Schwarz product inequality yields (11), and applying $G_M^\dagger$ to (5) gives (10)–(12). The final scalar conversion follows from (6). $\square$

Proposition A.11 (Observable score floor and threshold/tail ledger). *Under the certified-state conclusion of Proposition A.9 and its coverage-witness interface, define*

$$a_{\text{win}} := \sqrt{\frac{1 - q_*}{1 + 1/65536}}, \quad u_{\text{win}} := \frac{1}{100 \cdot 256^2}, \quad s_{\text{win}} := a_{\text{win}}^3 (1 - 1.01 u_{\text{win}}). \quad (15)$$

Then $a_{\text{win}} > 99/100$ and $s_{\text{win}} > 9/10$. For every coverage witness for target j , $\sigma_a \geq s_{\text{win}} \lambda_j$, hence

$$\sigma_{\max} \geq s_{\text{win}} \lambda_{\min} > \frac{9}{10} \lambda_{\min}. \quad (16)$$

Let

$$\mathcal{H} := \{a \in \mathcal{I}_{\text{cert}} : \sigma_a \geq 0.85 \sigma_{\max}\} \quad (17)$$

be the observable retained pool. For each $a \in \mathcal{H}$,

$$\sigma_a > \frac{3}{4} \lambda_{\min}, \quad d_{M,a} > \frac{3}{4} \lambda_{\min}, \quad \frac{\lambda_{\max}}{d_{M,a}} < \frac{7}{5}, \quad \max_j |\bar{c}_{M,a,j}| > \frac{3}{4} \quad (18)$$

for every mode M . Put

$$\begin{aligned} \eta_{\text{raw}}(r) &:= \frac{7}{5} (1 + q_*) q_* + \sqrt{r(1 + q_*)} \tau_r, \\ \eta_{\text{score}}(r) &:= \eta_{\text{raw}}(r) + \frac{\frac{7}{5} (1 + q_*) \tau_r}{1 - \tau_r}. \end{aligned} \quad (19)$$

Then $\eta_{\text{score}}(r) < 1/2000$ for every $r \geq 3$. For every $a \in \mathcal{H}$, every mode M , and every selector $I \subseteq [r]$,

$$\sum_{j \in I} \left| \bar{c}_{M,a,j} - \frac{\lambda_j}{\sigma_a} \bar{c}_{M',a,j} \bar{c}_{M'',a,j} \right| \leq \eta_{\text{score}}(r). \quad (20)$$

Consequently, with the proof-only latent-coordinate 1/8-threshold sets

$$T_{M,a} := \{j : |\bar{c}_{M,a,j}| \geq 1/8\}, \quad (21)$$

each $T_{M,a}$ is nonempty, and if j_M maximizes $|\bar{c}_{M,a,j}|$ in mode M , then

$$\frac{\lambda_{j_M}}{\sigma_a} |\bar{c}_{M',a,j_M} \bar{c}_{M'',a,j_M}| > \frac{3}{4} - \eta_{\text{score}}(r), \quad (22)$$

and $j_M \in T_{M',a} \cap T_{M'',a}$. Finally, for any mode M , any $t \in [0, 1]$, and $I_{M,a}(t) := \{j : |\bar{c}_{M,a,j}| < t\}$,

$$\sum_{j \in I_{M,a}(t)} \lambda_j |\bar{c}_{U,a,j} \bar{c}_{V,a,j} \bar{c}_{W,a,j}| \leq \lambda_{\max} (1 + q_*) t. \quad (23)$$

In particular, the $1/8$ tail is at most $\lambda_{\max}(1+q_*)/8$. For a coverage witness j , the sharper all-certified tail is

$$\sum_{\ell \neq j} \lambda_\ell |\bar{c}_{U,a,\ell} \bar{c}_{V,a,\ell} \bar{c}_{W,a,\ell}| \leq \lambda_{\max} |\bar{c}_{U,a,j} \bar{c}_{V,a,j} \bar{c}_{W,a,j}| u_{\text{win}}. \quad (24)$$

Equations (8), (20), (21), (23), and (24), taken for every certified slot and mode, define the finite event the tail conclusion of Proposition A.11; it is formed before any target label, representative ordering, or clustering operation.

Proof. On a coverage witness, let $\rho_{M,\ell} := |\bar{c}_{M,a,\ell}/\bar{c}_{M,a,j}|$. Proposition A.9 gives $\sum_{\ell \neq j} \rho_{M,\ell} \leq 1/256$ and $\max_\ell \rho_{M,\ell} \leq 1/100$. Since the stored state is in the target span, (14) and the Gram floor give $\|\bar{c}_M\|_2^2 \geq 1 - q_*$. Therefore

$$|\bar{c}_{M,a,j}| \geq \frac{\sqrt{1 - q_*}}{\sqrt{1 + \sum_{\ell \neq j} \rho_{M,\ell}^2}} \geq a_{\text{win}}. \quad (25)$$

The off-target score sum is bounded by

$$\sum_{\ell \neq j} \lambda_\ell |\bar{c}_{U,a,\ell} \bar{c}_{V,a,\ell} \bar{c}_{W,a,\ell}| \leq \lambda_{\max} |\bar{c}_{U,a,j} \bar{c}_{V,a,j} \bar{c}_{W,a,j}| \sum_{\ell \neq j} \rho_{U,\ell} \rho_{V,\ell} \rho_{W,\ell} \leq \lambda_{\max} |\bar{c}_{U,a,j} \bar{c}_{V,a,j} \bar{c}_{W,a,j}| u_{\text{win}}. \quad (26)$$

The reverse triangle inequality applied to the target summand and the off-target sum in (26) gives $\sigma_a \geq |\bar{c}_{U,a,j} \bar{c}_{V,a,j} \bar{c}_{W,a,j}| (\lambda_j - \lambda_{\max} u_{\text{win}}) \geq a_{\text{win}}^3 \lambda_j (1 - 1.01 u_{\text{win}})$, which is (16). The displayed numerical bounds on a_{win} , s_{win} follow by substituting $q_* = 1/4096$.

For an arbitrary stored state, $\|\bar{c}_M\|_2 \leq \sqrt{1 + q_*}$, and hence

$$\sigma_a \leq \lambda_{\max}(1 + q_*)^{3/2}, \quad \sigma_a \leq \lambda_{\max}(1 + q_*) \max_j |\bar{c}_{M,a,j}|. \quad (27)$$

The first inequality, (16), and the score filter give the first two bounds in (18). The second inequality in (27), the same filter, and the weight ratio $\lambda_{\max}/\lambda_{\min} \leq 1.01$ give $\max_j |\bar{c}_{M,a,j}| > 3/4$ for every mode. The d -bound and $\lambda_{\max}/d < 7/5$ follow from (6) and the numerical slack $\tau_r < 10^{-6}$.

Apply (8) with a selector and divide by $d_{M,a}$. The bound $\|\bar{k}_M\|_1 \leq \lambda_{\max}(1 + q_*)$ and (18) give η_{raw} . By (6), $|1 - d_{M,a}/\sigma_a| \leq \tau_r/(1 - \tau_r)$; the additional conversion from d^{-1} to σ^{-1} is therefore at most the last term in (19), proving (20). Taking a singleton selector gives (22). Since $\lambda_j/\sigma_a \leq \lambda_{\max}/\sigma_a < 7/5$ on the retained pool, (22) gives $|\bar{c}_{M',a,j_M} \bar{c}_{M'',a,j_M}| > (3/4 - 1/2000)/(7/5) > 1/3$. Each held correlation is therefore larger than $1/3 > 1/8$, proving the threshold intersection claim and nonemptiness.

For (23), on $I_{M,a}(t)$ use $|\bar{c}_{M,a,j}| < t$ and Cauchy–Schwarz:

$$\sum_{j \in I_{M,a}(t)} \lambda_j |\bar{c}_{U,j} \bar{c}_{V,j} \bar{c}_{W,j}| \leq \lambda_{\max} t \|\bar{c}_{M'}\|_2 \|\bar{c}_{M''}\|_2 \leq \lambda_{\max}(1 + q_*) t.$$

Equation (24) is the same calculation with the witness ratios and $\sum \rho_U \rho_V \rho_W \leq u_{\text{win}}$. All these inequalities are slotwise finite deterministic statements on the certified-state conclusion of Proposition A.9 and the event from Proposition A.3; no label is read by the procedure. $\square$

Proof. Fix the event from Proposition A.3 and the certified-state conclusion of Proposition A.9. For each certified slot, Lemma A.10 starts from the exact old-state Jacobi contraction and multiplies the observable projective residual by the same contraction norm. It proves the raw right-sided equation, shows that the three certificate signs are one common score sign, and applies a score-preserving proof-only orientation to obtain the barred equation. The orientation preserves the represented rank-one term, so the tensor remains the one represented by $(G_U, G_V, G_W, D_\lambda)$.

Proposition A.10 then projects that same equation with the realized target Gram. Its selector inequality separates the Gram leakage, the two first-order coordinate leaks, their quadratic cross term, and the certificate residual. The residual is charged once as $\sqrt{r(1+q)}\tau_r$, rather than being multiplied by a selection or ALS horizon. The exact d -normalized identity records the scalar cancellation; conversion to the score σ_a adds only the explicit one-time term in (19).

Finally, Proposition A.11 uses only the already generated coverage witnesses to lower-bound $\sigma_{\max}$. It then instantiates the selector inequality for every observable high-score slot and every proof-only $(1/8)$ -threshold/tail set, while retaining the unnormalized ledger for low-score slots. The resulting finite conjunction is precisely the score-equation conclusion of Proposition A.11 and the tail conclusion of Proposition A.11. It is produced before the score-filter output is used to form clusters, and all target indices in the witness argument are proof-only choices rather than algorithmic inputs. Thus the exact displayed claim is proved with no landing, selected representative, or cyclic condition. $\square$

A.6 Observable selection and gauge

Proposition A.1 supplies the H_M row/column bound and its spectral consequence. Lemma A.10 and Proposition A.11 supply the barred correlations $c_{M,a} = G_M^\top \bar{p}_{M,a}$, the positive score $\sigma_a = |\theta_a|$, and, for every selector $I \subseteq [r]$,

$$\sum_{j \in I} \left| c_{M,a,j} - \frac{\lambda_j}{\sigma_a} c_{M',a,j} c_{M'',a,j} \right| \leq \eta_{\text{score}}(r). \quad (\text{A})$$

On the retained pool $\mathcal{H} = \{a : \sigma_a \geq 0.85\sigma_{\max}\}$, the same proposition gives the retained score floor, the dominant-coordinate $3/4$ bound, and the coverage-witness score bounds. We use only these named interfaces and the elementary Cauchy–Schwarz and unit-vector identities.

Fix the event from Proposition A.3 and the finite certified pool. For a barred slot a , write $c_M = c_{M,a}$, $\sigma = \sigma_a$, and set

$$\eta := \eta_{\text{score}}(r) < \frac{1}{2000}, \quad \rho_a := \frac{\lambda_{\max}}{\sigma_a} < \frac{7}{5}. \quad (1)$$

All indices introduced in this derivation are latent proof indices.

Proposition A.12 (Common dominant support and small-root tail). *Under the static event of Proposition A.3 and the score/threshold conclusions of Proposition A.11, for every $a \in \mathcal{H}$ there is a unique index $\pi(a)$ such that*

$$|c_{U,a,\pi(a)}|, |c_{V,a,\pi(a)}|, |c_{W,a,\pi(a)}| > \frac{3}{4}. \quad (2)$$

Moreover, with

$$x_a := \max_{M \in \{U,V,W\}, \ell \neq \pi(a)} |c_{M,a,\ell}|, \quad L_{M,a} := \sum_{\ell \neq \pi(a)} |c_{M,a,\ell}|, \quad (3)$$

one has

$$x_a < 2\eta, \quad L_{M,a} \leq \frac{\eta}{1 - (14/5)\eta}, \quad \sum_{\ell \neq \pi(a)} c_{M,a,\ell}^2 \leq \chi := \frac{2\eta^2}{1 - (14/5)\eta}. \quad (4)$$

Proof. Proposition A.11 gives, for each mode, a coordinate of magnitude $> 3/4$. There cannot be two such coordinates, because $\|c_M\|_2^2 \leq 1 + q_* < 18/16$. Denote the unique indices by j_U, j_V, j_W .

We first rule out $j_U \neq j_V$. Put $S := 1 + q_*$ and $A := 9/16$, $u_i := |c_{U,i}|^2$, $v_i := |c_{V,i}|^2$. If $a = j_U$ and $b = j_V \neq a$, then $u_a > A$, $v_b > A$, and $\sum_i u_i, \sum_i v_i \leq S$. With $t = u_a$ and $z = v_a$,

$$\begin{aligned} \sum_i u_i v_i &\leq tz + (S - t) \max_{i \neq a} v_i \\ &\leq tz + (S - t)(S - z). \end{aligned} \quad (5)$$

Here $t \in [A, S]$ and $z \in [0, S - A]$, since $v_b > A$. The last expression is bilinear in t, z ; checking its four rectangle corners gives

$$\sum_i u_i v_i \leq 2A(S - A) = \frac{9}{8} \left(\frac{7}{16} + q_* \right) < \frac{1}{2}. \quad (6)$$

Consequently, by the weighted triangle inequality and Cauchy–Schwarz,

$$\sigma \leq \lambda_{\max} \sum_i |c_{U,i} c_{V,i} c_{W,i}| \leq \lambda_{\max} \|c_U \circ c_V\|_2 \|c_W\|_2 < \frac{71}{100} \lambda_{\max}. \quad (7)$$

On the other hand, the coverage score floor and the observable filter imply for every retained slot

$$\sigma \geq 0.85 \sigma_{\max} > 0.765 \lambda_{\min} \geq \frac{0.765}{1.01} \lambda_{\max} > \frac{3}{4} \lambda_{\max}, \quad (8)$$

contradicting (7). The same argument for the pairs U, W and V, W proves $j_U = j_V = j_W =: \pi(a)$. This is the promised singleton-versus-multi-support separation: a mismatched multi-support pattern has score $< 0.71 \lambda_{\max}$, whereas the retained singleton floor is $> 0.75 \lambda_{\max}$.

For $\ell \neq \pi(a)$, taking $I = \{\ell\}$ in (A) gives

$$|c_{M,\ell}| \leq \frac{\lambda_\ell}{\sigma} |c_{M',\ell} c_{M'',\ell}| + \eta \leq \frac{7}{5} x_a^2 + \eta. \quad (9)$$

The dominant coordinate and norm upper bound also give

$$x_a^2 \leq S - \frac{9}{16} < \left(\frac{2}{3} \right)^2. \quad (10)$$

If $\eta = 0$, (9) immediately gives $x_a = 0$. Otherwise, on $[2\eta, 2/3]$ the concave function $f(x) = x - (7/5)x^2$ is bounded below by its endpoint values:

$$f(2\eta) > \eta, \quad f(2/3) = \frac{2}{45} > \eta.$$

Thus (9)–(10) force $x_a < 2\eta$. Summing (A) over $I = [r] \setminus \{\pi(a)\}$, and using one factor bounded by x_a , gives

$$L_{M,a} \leq \frac{7}{5} x_a L_{M'',a} + \eta. \quad (11)$$

Taking the maximum over modes and using $x_a < 2\eta$ proves the second bound in (4). Finally, $\sum_{\ell \neq \pi(a)} c_{M,a,\ell}^2 \leq x_a L_{M,a}$ gives the last bound. $\square$

Proposition A.13 (Observable graph components). *Under the hypotheses and conclusion of Proposition A.12, retain exactly the certified slots*

$$\mathcal{H} := \{a : \sigma_a \geq 0.85 \sigma_{\max}\},$$

and join two retained slots a, b exactly when

$$|\langle p_{M,a}, p_{M,b} \rangle| \geq 1 - 64q_* \quad \text{for every } M \in \{U, V, W\}.$$

Then every target j has at least one retained slot, and this observable graph has exactly the r nonempty components

$$\mathcal{H}_j := \{a \in \mathcal{H} : \pi(a) = j\}, \quad j \in [r]. \quad (12)$$

Consequently the exactly- r component test passes. Choosing in each component the minimum- ζ member, with score as the tie-break, returns one representative per target without using target labels.

Proof. Let $a(j)$ be the coverage witness supplied by the tail conclusion of Proposition A.11 for target j . Its score obeys $\sigma_{a(j)} \geq s_{\text{win}} \lambda_j > (9/10) \lambda_{\min}$. For every certified slot, including the maximizer, $\sigma_i \leq \lambda_{\max}(1 + q_*)^{3/2}$. Hence

$$\frac{\sigma_{a(j)}}{\sigma_{\max}} \geq \frac{s_{\text{win}} \lambda_{\min}}{\lambda_{\max}(1 + q_*)^{3/2}} > \frac{0.9}{1.01(1 + 1/4096)^{3/2}} > 0.85, \quad (13)$$

so $a(j) \in \mathcal{H}$ and $\mathcal{H}_j \neq \emptyset$.

Set

$$\chi_* := \frac{2\eta_{\text{score}}(r)^2}{1 - (14/5)\eta_{\text{score}}(r)}, \quad d_* := \sqrt{2(q + \chi_*)}. \quad (14)$$

For $a \in \mathcal{H}_j$, Proposition A.12 and the lower norm bound in the tail conclusion of Proposition A.11 imply

$$|c_{M,a,j}|^2 \geq 1 - q - \chi_*, \quad \|\widehat{p}_{M,a} - g_{M,j}\|_2^2 = 2(1 - |c_{M,a,j}|) \leq 2(q + \chi_*) = d_*^2, \quad (15)$$

where $\widehat{p}_{M,a} := \text{sgn}(c_{M,a,j})\bar{p}_{M,a}$ and $g_{U,j} = u_j, g_{V,j} = v_j, g_{W,j} = w_j$. The sign is well-defined by (15).

Since $\eta < 1/2000$, direct substitution gives

$$q + \chi_* < \frac{1}{4000}, \quad d_*^2 < \frac{1}{2000}, \quad d_* < \frac{1}{40}. \quad (16)$$

If $a, b \in \mathcal{H}_j$, then in every mode

$$\begin{aligned} |\langle p_{M,a}, p_{M,b} \rangle| &= |\langle \widehat{p}_{M,a}, \widehat{p}_{M,b} \rangle| \\ &\geq 1 - \frac{1}{2}(\|\widehat{p}_{M,a} - g_{M,j}\|_2 + \|\widehat{p}_{M,b} - g_{M,j}\|_2)^2 \\ &\geq 1 - 2d_*^2 > 1 - \frac{1}{1000} > 1 - 64q_*. \end{aligned} \quad (17)$$

Thus all vertices in $\mathcal{H}_j$ are joined. If $a \in \mathcal{H}_j$, $b \in \mathcal{H}_\ell$, and $j \neq \ell$, then $|\langle g_{M,j}, g_{M,\ell} \rangle| \leq q$ and (15)–(16) give

$$|\langle p_{M,a}, p_{M,b} \rangle| \leq q + 2d_* + d_*^2 < \frac{1}{8} < 1 - 64q_*. \quad (18)$$

Therefore no cross-target edge exists. The graph is the disjoint union of the r nonempty cliques in (12), so its connected components are exactly those cliques. The component-count test therefore passes, and the minimum- ζ rule with score tie-breaking selects exactly one observable member from each component. $\square$

Lemma A.11 (Product-preserving target permutation and sign gauge). *Under Proposition A.13, the observable representative rule yields a bijection between the r representatives and the realized target columns. There is a proof-only permutation and a sign gauge that preserves every represented rank-one term.*

Proof. The graph components are the nonempty sets $\mathcal{H}_j$. The minimum- ζ (then score) rule selects one member $a(j)$ from each, and the arbitrary algorithmic ordering induces a bijection $\pi : [r] \rightarrow [r]$ with representative a assigned to target $\pi(a)$. This is a proof-only permutation; the algorithm never reads π .

For a representative a , use the score-preserving barred orientation from Proposition A.11, so that

$$\sigma_a \bar{p}_{U,a} \otimes \bar{p}_{V,a} \otimes \bar{p}_{W,a} = \theta_a p_{U,a} \otimes p_{V,a} \otimes p_{W,a}. \quad (19)$$

Let $\xi_{M,a} = \text{sgn}\langle g_{M,\pi(a)}, \bar{p}_{M,a} \rangle$, which is defined by (15), and let $\chi_a = \xi_{U,a} \xi_{V,a} \xi_{W,a}$. The target-aligned directions and signed scalar

$$\tilde{p}_{M,a} = \xi_{M,a} \bar{p}_{M,a}, \quad \tilde{\theta}_a = \chi_a \sigma_a \quad (20)$$

satisfy

$$\tilde{\theta}_a \bigotimes_M \tilde{p}_{M,a} = \sigma_a \bigotimes_M \bar{p}_{M,a}, \quad (21)$$

because $\chi_a^2 = 1$. Thus (20) is a proof-only sign chart that aligns all target coordinates and preserves the represented tensor. For the literal product-one sign convention used by the quotient chart, set

$$(\varepsilon_{U,a}, \varepsilon_{V,a}, \varepsilon_{W,a}) := (\xi_{U,a}, \xi_{V,a}, \chi_a \xi_{W,a}).$$

Then $\varepsilon_{U,a} \varepsilon_{V,a} \varepsilon_{W,a} = 1$, so $\bigotimes_M \varepsilon_{M,a} \bar{p}_{M,a} = \bigotimes_M \bar{p}_{M,a}$; the coefficient σ_a is unchanged. The all-positive chart (20) and this product-one chart differ only by placing the sign χ_a in the W scalar, exactly as in the observable seed definition. Positive componentwise rescalings with product one are unchanged by this sign bookkeeping. Hence the resulting permutation/gauge is exactly the interface exported as the gauge conclusion of Lemma A.11. $\square$

Proof. Fix the event from Proposition A.3 and the interfaces supplied by the preceding propositions. The weighted ledger supplies the retained score floor, the 3/4 dominant coordinate in each mode, and (A). Proposition A.12 first compares two different dominant indices through (5)–(8), proving that every retained slot has one common target support. The same proposition then applies the small-root branch of (A), giving a dimension-free off-target ℓ_1 tail and direction error.

Coverage witnesses are retained by (13). Proposition A.13 uses the direction error and the realized Gram off-diagonal bound to prove within-target edges and cross-target nonedges at the exact observable threshold $1 - 64q_*$. It follows that the algorithm's graph has exactly r components and that the selected representative in each component is associated with a distinct realized target. Finally, Lemma A.11 supplies the proof-only permutation and product-preserving sign chart. Together these results produce

$$E_{\text{support}}, \quad E_{\text{cluster}}, \quad E_{\text{gauge}},$$

before any selected-state coefficient, landing output, or cyclic invariant is used. The construction is label-free at the procedure level and preserves the represented rank-one terms exactly. $\square$

A.7 Two-orientation coefficient closure

The geometry bounds used below are supplied by Proposition A.3. The signed equation and the support, graph, and gauge conclusions are supplied by Lemma A.10, Proposition A.13, and

Lemma A.11; the latter contributes only the proof-only permutation and term-preserving product-one option. No landing state or later ALS invariant is imported. The remaining tools are Cauchy–Schwarz, the Hadamard-product norm inequality, and induced row/column ℓ_1 estimates, all derived below.

Reindex by the proof-only target permutation and put

$$q := q_{\text{real}}, \quad t := r\tau_r, \quad \omega := q_*^2 + t, \quad \eta := \eta_{\text{score}}(r).$$

The definitions of η_{score} and $\tau_r = q_*^2/(10000r)$, together with $q \leq q_*$, imply

$$\eta \leq \frac{3}{2}q_*, \quad \ell_* \leq \frac{31}{20}q_*, \quad x_* := 2\eta \leq 3q_*, \quad x_*\ell_* \leq \frac{93}{20}q_*^2. \quad (\text{A})$$

Indeed,

$$\sqrt{r(1+q_*)}\tau_r \leq \frac{q_*}{10^6}, \quad \frac{1.4(1+q_*)\tau_r}{1-\tau_r} \leq \frac{q_*}{10^6},$$

and hence $\eta \leq 1.4(1+q_*)q_* + 2q_*/10^6 < 3q_*/2$. Since $\eta < 1/2000$, division by $1 - (14/5)\eta$ gives the stated ℓ_* bound. All constants in (A) are universal and remain valid when $q_{\text{real}} = 0$.

Lemma A.12 (sign-covariant dual equation, positive diagonal, and projected residual). *Under the static event of Proposition A.3, the certificate/score ledger of Proposition A.11, and the support and graph conclusions of Propositions A.12 and A.13, first keep the score-oriented objects barred. After the proof-only target permutation, write*

$$\bar{c}_{M,ja} = (G_M^\top \bar{P}_M)(j, a), \quad \bar{\sigma}_a = |\theta_a| = \sum_j \lambda_j \bar{c}_{U,ja} \bar{c}_{V,ja} \bar{c}_{W,ja} > \frac{3}{4}\lambda_{\max}. \quad (\text{A.1})$$

For each a let $\xi_{M,a} = \text{sign}(\bar{c}_{M,aa})$; then $\xi_{U,a}\xi_{V,a}\xi_{W,a} = 1$. Define

$$p_{M,a} = \xi_{M,a}\bar{p}_{M,a}, \quad c_{M,ja} = \xi_{M,a}\bar{c}_{M,ja}, \quad k_{M,a} = \xi_{M',a}\xi_{M'',a}\bar{k}_{M,a} = \xi_{M,a}\bar{k}_{M,a}, \quad r_{M,a} = \xi_{M,a}\bar{r}_{M,a}. \quad (\text{A.2})$$

Stacking these columns gives P_M, K_M, R_M and the exact identities

$$P_M D_M = G_M K_M + R_M, \quad c_M = G_M^\top P_M, \quad A_M = G_M^\dagger P_M = H_M^{-1} G_M^\top P_M, \quad c_M = H_M A_M. \quad (\text{B0})$$

Set $\sigma_a = \bar{\sigma}_a$ and relabel without bars. With $a_{M,a} = A_M(a, a)$,

$$0.74 < a_{M,a} < 1.001, \quad \Delta_M = \text{diag}(a_{M,a}), \quad Z_M = A_M \Delta_M^{-1}, \quad h_{M,ja} = (H_M Z_M)(j, a) = \frac{c_{M,ja}}{a_{M,a}}. \quad (\text{A.3})$$

Then $h_{M,aa} > 2/3$. If

$$\varepsilon_{M,ja} := \frac{(G_M^\dagger R_M)(j, a)}{d_{M,a}},$$

then $|\varepsilon_{M,ja}| \leq 2\tau_r$, and for $j \neq a$,

$$|Z_M(j, a)| \leq 4|h_{M',ja}h_{M'',ja}| + 3\tau_r. \quad (\text{B})$$

Proof. The barred score identity and retained-pool bounds are the pre-gauge facts. If $\xi_{U,a}\xi_{V,a}\xi_{W,a} < 0$, the dominant term would be negative and the off-target bounds would give

$$\bar{\sigma}_a \leq \lambda_{\max} x_* \|\bar{c}_{M',-a}\|_2 \|\bar{c}_{M'',-a}\|_2 \leq \lambda_{\max} x_* (1 + q_*) < \frac{3}{4}\lambda_{\max}. \quad (\text{A.4})$$

This contradicts the retained score floor, so the signs have product one. Multiplying the barred same-state equation in mode M by $\xi_{M,a}$ gives (B0), because $\xi_{M',a}\xi_{M'',a} = \xi_{M,a}$. The residual column obeys

$$\|R_M(:, a)\|_2 \leq d_{M,a}\tau_r. \quad (\text{A.5})$$

The product of the transformed correlations equals the barred product, so the score identity and every absolute tail in (A) are unchanged. This is a product-one sign gauge; by construction $c_{M,aa} = |\bar{c}_{M,aa}| > 3/4$.

Since $P_M \in \text{range}(G_M)$, write $P_M = G_M A_M$. The unit-column norm and eigenvalue floor give $\|A_M(:, a)\|_2 \leq (1 - q)^{-1/2}$, and hence

$$|c_{M,aa} - A_M(a, a)| \leq \|H_M(a, -a)\|_2 \|A_M(-a, a)\|_2 \leq \frac{q}{\sqrt{1-q}}. \quad (\text{C})$$

Thus $0.74 < a_{M,a} < 1.001$ and $h_{M,aa} > 2/3$. Applying $G_M^\dagger$ to (B0) gives

$$d_{M,a} A_M(:, a) = K_M(:, a) + (G_M^\dagger R_M)(:, a). \quad (\text{D})$$

The pseudoinverse bound gives

$$\|\varepsilon_{M,:,a}\|_2 \leq \frac{\tau_r}{\sqrt{1-q}} < 2\tau_r, \quad (\text{E})$$

and the directly projected residual separately satisfies

$$\frac{|(G_M^\dagger R_M)(j, a)|}{d_{M,a}} \leq \sqrt{1+q} \tau_r < 2\tau_r. \quad (\text{E}')$$

Also

$$d_{M,a} \leq \|G_M\|_2 \|K_M(:, a)\|_2 \leq \lambda_{\max}(1 + q). \quad (\text{A.6})$$

For $j \neq a$, division of the j -coordinate of (D) by the a -coordinate has denominator at least

$$\frac{9}{16} - \frac{1.02\tau_r}{\sqrt{1-q}} > 0.56. \quad (\text{F})$$

The product part of the numerator divided by λ_a is at most $1.02|h_{M',ja}h_{M'',ja}|$, and the residual part is at most $1.02\tau_r$. Division by (F) gives (B), with the residual bounded entrywise before any sum. QED. □

Proposition A.14 (normalized dual-coordinate column bootstrap). *Under Lemma A.12,*

$$\max_M \|Z_M - I\|_{\text{col},1} \leq 42\omega. \quad (\text{G})$$

Proof. For each selected column a , (A), (C), and $a_{M,a} > 0.74$ give

$$\|h_{M,-a}\|_2 \leq \frac{\|c_{M,-a}\|_2}{a_{M,a}} \leq \frac{3}{2} \sqrt{x_* \ell_*}. \quad (\text{H})$$

Summing (B) over $j \neq a$ and using Cauchy–Schwarz gives

$$\sum_{j \neq a} |Z_M(j, a)| \leq 4\|h_{M',-a}\|_2 \|h_{M'',-a}\|_2 + 3(r-1)\tau_r \leq 9x_* \ell_* + 3t \leq 42(q_*^2 + t) = 42\omega. \quad (\text{I})$$

Taking the maximum over M and a proves (G). This is a genuine column bootstrap: no row mass, landing output, or cyclic condition occurs. □

Proposition A.15 (two-orientation coefficient closure). *Under Lemma A.12 and Proposition A.14, let*

$$R := \max_M \|Z_M - I\|_{\text{row},1}, \quad C := \max_M \|Z_M - I\|_{\text{col},1}.$$

Then

$$R \leq 5\omega, \quad \max_M \|A_M - I\|_{\text{row},1} \leq 7\omega, \quad \max_M \|A_M - I\|_{\text{col},1} \leq 44\omega. \quad (\text{J})$$

The normalized residual entries obey $|\varepsilon_{M,ja}| \leq 2\tau_r$, and each row or column sum is charged once by at most $2r\tau_r$.

Proof. Fix a mode M and row j . For $a \neq j$, set $v_{M,ja} := (H_M Z_M)(j, a)$. From (G),

$$\max_{a \neq j} |v_{M,ja}| \leq q + (1 + q)C =: u. \quad (\text{K})$$

Here the direct $H_M(j, a)$ term has row mass q , the off-diagonal Z_M entry is at most C , and the remaining Gram row contributes at most qC .

$$\sum_{a \neq j} |v_{M,ja}| \leq q + (1 + q)\|Z_M - I\|_{\text{row},1}. \quad (\text{L})$$

To verify (L), the $l = j$ term in $H_M Z_M$ contributes the row error. Each $l \neq j$ term contributes its off-diagonal row mass and its diagonal entry $Z_M(l, l) = 1$, weighted by the H_M row mass q .

Summing (B) over $a \neq j$, using (K) for one factor and (L) for the other, and taking the maximum over modes gives

$$R \leq 4u\{q + (1 + q)R\} + 3t. \quad (\text{M})$$

By (G), $t = q_*^2/10000$ and $q \leq q_*$, so

$$u \leq q + 42(1 + q_*)(1 + 10^{-4})q_*^2 < \frac{51}{50}q_*, \quad 4u(1 + q) < \frac{1}{1000}. \quad (\text{N})$$

Thus (M) has a positive denominator and

$$R \leq \frac{4uq + 3t}{1 - 4u(1 + q)} < 5\omega. \quad (\text{O})$$

This is the small-root selection. Before inserting (G), (M) is the usual quadratic/small-gain envelope; (G) puts the solution in the small branch, and (N) excludes the large branch without assuming a row tube.

It remains to transfer from normalized columns to A_M . Let z be a column of Z_M , with $z_a = 1$ and $\|z - e_a\|_1 \leq C$, and write $a_0 = a_{M,a}$. The unit-norm identity gives

$$|z^\top H_M z - 1| \leq 2qC + (1 + q)C^2 =: \delta_a. \quad (\text{P})$$

At $q \leq q_*$ and $C \leq 42\omega$, $\delta_a < 1/100$ and $|a_0 - 1| \leq 2\delta_a \leq \omega$; the last inequality follows directly from $\omega = (1 + 10^{-4})q_*^2$ and $q_* = 1/4096$. Since $a_0 < 1.001$, (G), (O), and this diagonal bound imply

$$\|A_M - I\|_{\text{col},1} \leq 1.001C + \omega < 44\omega, \quad \|A_M - I\|_{\text{row},1} \leq 1.001R + \omega < 7\omega.$$

This proves (J). Equation (E) gives the entrywise $2\tau_r$ residual, and summing over at most r entries gives $2r\tau_r$ in either orientation, never $r^2\tau_r$. $\square$

Proof. The support and gauge interfaces first provide one selected stored state per target, a proof-only permutation, and a term-preserving product-one sign option. Lemma A.12 starts from the barred score identity, proves the dominant sign product, explicitly transforms the equation and residual, and only then applies the true-factor dual. It proves the positive diagonal and bounds every projected residual entry before summation. Proposition A.14 uses only the per-column correlation tails to produce the 42ω column interface. Proposition A.15 then uses that interface to bound held correlation rows, solve the positive-denominator small-gain inequality, and transfer from normalized dual columns to $A_M - I$. Therefore the column-closure conclusion of Proposition A.14, the projected-residual conclusion of Proposition A.15, and the row-closure conclusion of Proposition A.15 hold with both induced orientations at $O(\omega)$, with one residual charge $r\tau_r$. All objects are from the selected stored state; no landing or cyclic object is consumed. Because the stored-state interface places every selected column in $\text{range}(G_M)$, the orthogonal complement field is exactly zero at this stage and therefore satisfies the later budget τ_r . $\square$

A.8 Balanced observable seed

Lemma A.11 supplies one selected representative for each target, a proof-only permutation, and a product-one sign gauge. In the resulting orientation, Proposition A.12 gives

$$\max_{M, \ell \neq j} |c_{M, \ell j}| \leq x_* := 2\eta_{\text{score}}(r), \quad \sum_{\ell \neq j} |c_{M, \ell j}| \leq \ell_* := \frac{\eta_{\text{score}}(r)}{1 - (14/5)\eta_{\text{score}}(r)}, \quad (\text{A})$$

with $x_* \leq 3q_*$ and $\ell_* \leq (31/20)q_*$. The score identity is

$$\sigma_j := |\theta_j| = \sum_{\ell=1}^r \lambda_{\ell} c_{U, \ell j} c_{V, \ell j} c_{W, \ell j} > 0. \quad (\text{B})$$

Proposition A.15 gives the exact target-span coefficient interface

$$P_M = G_M A_M, \quad \|A_M - I\|_{\text{row}, 1} \leq 7\omega, \quad \|A_M - I\|_{\text{col}, 1} \leq 44\omega, \quad (\text{C})$$

and the columns of P_M are unit vectors. The remaining tools are orthogonal projection, induced ℓ_1 inequalities, Cauchy–Schwarz, and $|\log(1 + u)| \leq |u|/(1 - |u|)$ for $|u| < 1$.

Put

$$q := q_{\text{real}}, \quad t := r\tau_r, \quad \omega := q_*^2 + t, \quad \beta := 44\omega. \quad (1)$$

The setting gives $t = q_*^2/10^4$, hence $\omega = (1 + 10^{-4})q_*^2 < 10^{-6}$ and $\beta < 10^{-4}$. All numerical inequalities below are uniform in r, n, κ_0, ρ and the confidence parameters.

Lemma A.13 (observable seed decomposition). *Under the static event of Proposition A.3, the permutation/gauge conclusion of Lemma A.11, and the two-orientation closure of Proposition A.15, after the proof-only permutation and product-one sign chart, let P_M^0 be the normalized active seed factors. Then, for every mode M ,*

$$P_M^0 = G_M(I + C_M^0) + N_M^0, \quad G_M^\top N_M^0 = 0, \quad (2)$$

with

$$N_M^0 = 0, \quad \|C_M^0\|_{\text{row}, 1} \vee \|C_M^0\|_{\text{col}, 1} \leq 48\omega. \quad (3)$$

Proof. For selected representative j , the setting forms

$$x_j^0 = |\theta_j|^{1/3} p_{U, j}, \quad y_j^0 = |\theta_j|^{1/3} p_{V, j}, \quad z_j^0 = \text{sgn}(\theta_j) |\theta_j|^{1/3} p_{W, j}. \quad (4)$$

The score-preserving orientation from the gauge conclusion of Lemma A.11 has the same rank-one product as (4) with scalar $\sigma_j = |\theta_j|$. Hence the two triples of unit directions differ only by a product-one sign gauge. Applying that proof-only gauge and the target permutation gives

$$P_M^0 = P_M, \quad (5)$$

where P_M is the oriented selected matrix in Proposition A.15. This is an identity of the actual observable seed, not a new algorithmic state; represented rank-one terms are unchanged.

By Proposition A.3, $\lambda_{\min}(H_M) \geq 1 - q > 0$, so G_M has full column rank. Proposition A.15 gives $P_M = G_M A_M$ with (C). The orthogonal decomposition relative to the exact target span is unique:

$$A_M := G_M^\dagger P_M, \quad N_M^0 := (I - G_M G_M^\dagger) P_M. \quad (6)$$

Because $P_M \in \text{range}(G_M)$, (6) gives $N_M^0 = 0$ and $P_M^0 = G_M(I + C_M^0)$ with $C_M^0 = A_M - I$. Thus (C) implies

$$\|C_M^0\|_{\text{row},1} \leq 7\omega \leq 48\omega, \quad \|C_M^0\|_{\text{col},1} \leq 44\omega \leq 48\omega.$$

The requested perpendicular-column bound follows from the stronger identity $\max_j \|N_M^0(:,j)\|_2 = 0 \leq \tau_r$. QED. $\square$

Lemma A.14 (best-scalar transfer). *Under the static event of Proposition A.3, the support conclusion of Proposition A.12, and the closure conclusion of Proposition A.15, with the oriented matrices in Lemma A.13, for every target column j ,*

$$|c_{M,jj} - 1| \leq 2\omega \quad (M \in \{U, V, W\}), \quad \left| \frac{\sigma_j}{\lambda_j} - 1 \right| \leq 9\omega. \quad (7)$$

Here $c_M = G_M^\top P_M^0$ and $\sigma_j = |\theta_j| > 0$ is the observable best-scalar in (B).

Proof. Fix M, j , write $H = H_M = I + E$, $A = A_M$, and let

$$a := A_{jj}, \quad \alpha := A(:,j) - e_j.$$

From (C), $\|\alpha\|_1 \leq \beta$, so $a > 0$ and $|a - 1| \leq \beta < 1/100$. Put $z := A(:,j)/a = e_j + u$. Then $u_j = 0$ and

$$B := \|u\|_1 \leq \frac{\beta}{1 - \beta} \leq 45\omega. \quad (8)$$

The column $P_M^0(:,j) = G_M(az)$ has norm one. Since the diagonal of H is one and the diagonal of E is zero,

$$1 = a^2 z^\top H z, \quad |z^\top H z - 1| \leq 2qB + (1 + q)B^2 =: d. \quad (9)$$

Indeed, $\|u\|_2^2 \leq B^2$, $|e_j^\top E u| \leq q\|u\|_\infty \leq qB$, and $|u^\top E u| \leq q\|u\|_1\|u\|_\infty \leq qB^2$. Substituting $B \leq 45\omega$, $q \leq q_*$, and $\omega < 10^{-6}$ gives

$$d \leq 90q_*\omega + (1 + q_*)2025\omega^2 < \frac{\omega}{20} < \frac{1}{20}. \quad (10)$$

The elementary inverse-square-root bound $|(1 + v)^{-1/2} - 1| \leq 2|v|$ for $|v| \leq 1/2$, applied to (9), yields $|a - 1| \leq 2d < \omega/10$, and in particular $|a - 1| \leq \omega$. Now $c_M = HA$ and $E_{jj} = 0$, so

$$c_{M,jj} - 1 = (a - 1) + e_j^\top E \alpha.$$

Consequently, using $\|\alpha\|_\infty \leq \beta$,

$$|c_{M,jj} - 1| \leq \omega + q\beta \leq (1 + 44q_*)\omega < 2\omega. \quad (11)$$

It remains to transfer this diagonal control to the scalar. By the support ledger (A), every off-target column satisfies the same x_*, ℓ_* bounds after the sign gauge. Write $c_{M,jj} = 1 + \delta_M$, so $|\delta_M| \leq 2\omega$. Then

$$\left| \prod_M c_{M,jj} - 1 \right| \leq 3(2\omega) + 3(2\omega)^2 + (2\omega)^3 < 7\omega. \quad (12)$$

Using (B), $\lambda_\ell/\lambda_j \leq \Gamma$, and two off-target maxima together with one off-target ℓ_1 sum, we obtain

$$\begin{aligned} \left| \frac{\sigma_j}{\lambda_j} - 1 \right| &\leq 7\omega + \Gamma \sum_{\ell \neq j} |c_{U,\ell j} c_{V,\ell j} c_{W,\ell j}| \\ &\leq 7\omega + \Gamma x_*^2 \ell_*. \end{aligned} \quad (13)$$

The endpoint bounds $x_* \leq 3q_*$, $\ell_* \leq (31/20)q_*$, $\Gamma \leq 1.01$, and $\omega \geq q_*^2$ give

$$\Gamma x_*^2 \ell_* \leq 1.01 \cdot 9 \cdot \frac{31}{20} q_*^3 < \omega, \quad (14)$$

so the scalar bound in (7) follows. In particular $\sigma_j/\lambda_j > 0$, consistently with the observable nonzero-score branch. QED. □

Proposition A.16 (balanced scale and quotient entry). *Under the conclusions of Lemma A.13 and Lemma A.14, the setting's equal-norm seed satisfies*

$$\gamma_j^0 := d_{U,j}^0 d_{V,j}^0 d_{W,j}^0 = \sigma_j, \quad s_j^0 := \log(\gamma_j^0/\lambda_j), \quad \|s^0\|_\infty \leq 128\omega, \quad (15)$$

and, with $E^0 = \text{diag}(e^{s_j^0/3})$,

$$D_U^0 = D_V^0 = D_W^0 = D_\lambda^{1/3} E^0, \quad e^{-128\omega/3} \leq \frac{d_{M,j}^0}{\lambda_j^{1/3}} \leq e^{128\omega/3}. \quad (16)$$

Proof. The three factors in (4) all have norm $\sigma_j^{1/3}$; the sign in the third factor does not change its norm. Therefore their product of norms is exactly $\gamma_j^0 = \sigma_j$. By Lemma A.14, write $\sigma_j/\lambda_j = 1 + u_j$ with $|u_j| \leq 9\omega < 1/2$. Then

$$|s_j^0| = |\log(1 + u_j)| \leq \frac{|u_j|}{1 - |u_j|} \leq 18\omega < 128\omega. \quad (17)$$

The definition of equal-norm balancing now gives

$$d_{M,j}^0 = (\sigma_j)^{1/3} = \lambda_j^{1/3} e^{s_j^0/3},$$

which is (16) and the displayed diagonal identity. QED. □

Proof. Lemma A.11 first identifies, purely for proof, one selected representative per realized target and a product-preserving orientation. Lemma A.13 checks that the normalized observable best-scalar seed is exactly this oriented selected state. Because the selected state is already in the exact realized target span, its orthogonal complement is identically zero; the row/column closure then gives the public 48ω directional interface in both induced orientations and the stronger $N^0 = 0 \leq \tau_r$ bound.

Lemma A.14 uses only the same selected-state correlations, the target Gram margin, and the off-target ledger. It first derives the diagonal correlation error, then bounds the weighted off-target score contribution before taking a logarithm. This proves the observable ratio estimate without a latent scale or any post-landing object. Proposition A.16 converts the ratio to the exact relative product-log coordinate and records the positive scale congruence. Together these results produce

$$E_{\text{seed}} : \quad P_M^0 = G_M(I + C_M^0) + N_M^0, \quad G_M^\top N_M^0 = 0, \quad \|C_M^0\|_{\text{row},1} \vee \|C_M^0\|_{\text{col},1} \leq 48\omega, \quad \max_j \|N_M^0(:,j)\|_2 \leq \tau_r, \quad \|s^0\|_\circ$$

This event holds before any frozen design or landing solve and supplies the input to the synchronized landing argument and the separate exact-baseline specialization; no chronological predecessor, post-solve normalizer, or cyclic invariant enters the proof. $\square$

A.9 Frozen landing reserves

The static geometry interface is Proposition A.3; it gives $q \leq q_*$, $\Gamma \leq 1.01$, and the displayed induced row/column Gram bounds for every realized G_M . The balanced seed interface is Proposition A.16; it gives the decomposition, orthogonality, 48ω row/column fields, the τ_r perpendicular-column bound, unit normalized columns, and $D_M^0 = D_\lambda^{1/3} E^0$. No landing object is imported.

For $\nu \in \{\text{row}, \text{col}\}$ we use $\|AB\|_\nu \leq \|A\|_\nu \|B\|_\nu$ and $\|A \circ B\|_\nu \leq \|A\|_\nu \|B\|_{\max}$. When $\|E\|_\nu < 1$, the Neumann identity $(I + E)^{-1} = \sum_{m \geq 0} (-E)^m$ holds in the same induced norm. All hypotheses are checked below.

For each $M \in \{U, V, W\}$, let M', M'' denote the other two modes in the order used by the setting's frozen Khatri–Rao design, and define

$$H_M := G_M^\top G_M, \quad L_M := G_M^\top P_M^0, \quad F_M := (P_M^0)^\top P_M^0.$$

The setting-derived budgets are

$$\delta_L = (1 + q_*)c_0, \quad \delta_F = 2(1 + q_*)c_0 + (1 + q_*)c_0^2 + r\tau_r^2,$$

$$\delta_{FL} = (1 + q_*)c_0 + (1 + q_*)c_0^2 + r\tau_r^2, \quad \alpha_0 = q_* + \delta_F,$$

$$\eta_J = \alpha_0^2, \quad \eta_{QJ} = 2\delta_{FL} + \delta_L^2 + \delta_F^2, \quad \eta_A = \frac{\eta_{QJ}}{1 - \eta_J}.$$

Proposition A.17 (fixed numerical slack). *Under Assumption 6 and the geometry conclusion of Proposition A.3, in particular $\Gamma \leq 1.01$, with $q_* = 1/4096$ and $\tau_r = q_*^2/(10^4 r)$, the setting-defined budgets satisfy*

$$c_0 < 49q_*^2, \quad \delta_L < 50q_*^2, \quad \delta_F < 101q_*^2, \quad \delta_{FL} < 51q_*^2, \quad (\text{N1})$$

$$\eta_J < 1.051q_*^2 < 10^{-6}, \quad \eta_{QJ} < 103q_*^2, \quad \eta_A < 104q_*^2, \quad (\text{N2})$$

and, using the $\Gamma \leq 1.01$,

$$\eta_J \leq \frac{1}{64}, \quad \Gamma\eta_A \leq 128\omega < \frac{1}{64}. \quad (\text{N3})$$

Proof. Since $\omega = (1 + 10^{-4})q_*^2$,

$$c_0 = 48(1 + 10^{-4})q_*^2 < 49q_*^2.$$

Also $r\tau_r^2 = q_*^4/(10^8 r) \leq q_*^4$ because $r \geq 3$. Therefore

$$\delta_L < 50q_*^2, \quad \delta_F < 100q_*^2 + 4803q_*^4 < 101q_*^2,$$

$$\delta_{FL} < 50q_*^2 + 4803q_*^4 < 51q_*^2,$$

where $4803q_*^2 < 1$. Consequently $\alpha_0 < q_*(1 + 101/4096) < 1.025q_*$, so $\eta_J < 1.051q_*^2 < 10^{-6}$. Further,

$$\eta_{QJ} < 2(51q_*^2) + 50^2q_*^4 + 101^2q_*^4 < 103q_*^2.$$

Thus $1/(1 - \eta_J) < 1.000002$, giving $\eta_A < 104q_*^2$. Finally,

$$\Gamma\eta_A < 1.01 \cdot 104q_*^2 < 128(1 + 10^{-4})q_*^2 = 128\omega < 1/64.$$

This proves (N1)–(N3). □

Lemma A.15 (two-orientation frozen Gram expansion). *Under the geometry conclusion of Proposition A.3 and the balanced-seed conclusion of Proposition A.16, for every mode M ,*

$$\|L_M - H_M\|_{\text{row},1} \vee \|L_M - H_M\|_{\text{col},1} \leq \delta_L, \quad (1)$$

$$\|F_M - H_M\|_{\text{row},1} \vee \|F_M - H_M\|_{\text{col},1} \leq \delta_F, \quad (2)$$

$$\|F_M - L_M\|_{\text{row},1} \vee \|F_M - L_M\|_{\text{col},1} \leq \delta_{FL}. \quad (3)$$

Proof. The geometry event gives $\|H_M\|_{\text{row},1} \vee \|H_M\|_{\text{col},1} \leq 1 + q_*$. From the seed decomposition and $G_M^\top N_M^0 = 0$,

$$L_M = H_M(I + C_M^0),$$

so $L_M - H_M = H_M C_M^0$. Submultiplicativity in each orientation gives (1).

Expanding the full seed Gram, with the two cross terms involving N_M^0 vanishing by $G_M^\top N_M^0 = 0$, gives

$$F_M - H_M = (C_M^0)^\top H_M + H_M C_M^0 + (C_M^0)^\top H_M C_M^0 + (N_M^0)^\top N_M^0. \quad (4)$$

For every pair i, j , $|((N_M^0)^\top N_M^0)_{ij}| \leq \tau_r^2$, hence each induced row and column sum is at most $r\tau_r^2$. Applying the induced-norm product inequality to (4), and using that transpose exchanges row and column norms, proves (2).

Subtracting $L_M = H_M(I + C_M^0)$ from F_M instead yields

$$F_M - L_M = (C_M^0)^\top H_M(I + C_M^0) + (N_M^0)^\top N_M^0. \quad (5)$$

The same two-orientation calculation proves (3). All bounds use the fixed upper value q_* , so the result is simultaneous over $M = U, V, W$. □

Proposition A.18 (normalized pair-Gram and direct Q-J reserves). *Under Proposition A.17 and the conclusion of Lemma A.15, define*

$$J_M^0 := F_{M'} \circ F_{M''}, \quad Q_M^0 := L_{M'} \circ L_{M''}.$$

Then, in both induced orientations,

$$\|F_M - I\|_{\text{row},1} \vee \|F_M - I\|_{\text{col},1} \leq \alpha_0, \quad (6)$$

$$\|J_M^0 - I\|_{\text{row},1} \vee \|J_M^0 - I\|_{\text{col},1} \leq \eta_J, \quad (7)$$

$$\|Q_M^0 - J_M^0\|_{\text{row},1} \vee \|Q_M^0 - J_M^0\|_{\text{col},1} \leq \eta_{QJ}. \quad (8)$$

Moreover, $\eta_J \leq 1/64$ and

$$\lambda_{\min}(J_M^0) \geq 1 - \eta_J, \quad \|(J_M^0)^{-1}\|_{\text{row},1} \vee \|(J_M^0)^{-1}\|_{\text{col},1} \leq \frac{1}{1 - \eta_J}, \quad (9)$$

$$\|(J_M^0)^{-1} - I\|_{\text{row},1} \vee \|(J_M^0)^{-1} - I\|_{\text{col},1} \leq \frac{\eta_J}{1 - \eta_J}. \quad (10)$$

Proof. Because P_M^0 and G_M have unit columns, every entry of F_M , L_M , and H_M has absolute value at most one. In particular the diagonal of F_M is one. Set $E_M := F_M - I$. Equation (2) and $\|H_M - I\|_\nu \leq q_*$ give (6), and E_M has zero diagonal. Therefore

$$J_M^0 = (I + E_{M'}) \circ (I + E_{M''}) = I + E_{M'} \circ E_{M''}. \quad (11)$$

For either $\nu = \text{row}$ or $\nu = \text{col}$,

$$\|E_{M'} \circ E_{M''}\|_\nu \leq \|E_{M'}\|_\nu \|E_{M''}\|_\nu \leq \alpha_0^2,$$

which proves (7).

For the direct paired cancellation, use the exact identity

$$Q_M^0 - J_M^0 = (L_{M'} - F_{M'}) \circ L_{M''} + F_{M'} \circ (L_{M''} - F_{M''}). \quad (12)$$

The unit-column observation gives $\|L_{M''}\|_{\max} \vee \|F_{M'}\|_{\max} \leq 1$. Thus (3) and the Hadamard bound yield the stronger estimate

$$\|Q_M^0 - J_M^0\|_\nu \leq 2\delta_{FL} \leq 2\delta_{FL} + \delta_L^2 + \delta_F^2 = \eta_{QJ},$$

proving (8) without inverting either pair. The two nonnegative square terms are retained because they are part of the binding reserve definition.

Proposition A.17 gives $\eta_J < 1$, so the Neumann series for $J_M^0 = I + (J_M^0 - I)$ converges in each induced norm and gives (9)–(10). The Schur-product/Khatri–Rao identity says that J_M^0 is the Gram matrix of the columnwise Kronecker products $P_{M'}^0 \odot P_{M''}^0$, so it is symmetric positive semidefinite. Together with the row bound in (7) (or $\|J_M^0 - I\|_2 \leq \eta_J$), Gershgorin gives $\lambda_{\min}(J_M^0) \geq 1 - \eta_J$. $\square$

Lemma A.16 (positive diagonal congruence and ordinary frozen solve). *Under Proposition A.18 and the scale identity in Proposition A.16, let*

$$P_{-M}^0 := P_{M'}^0 \odot P_{M''}^0, \quad K_{-M} := G_{M'} \odot G_{M''}, \quad D_{-M}^0 := D_{M'}^0 D_{M''}^0.$$

Then $D_{-M}^0 = D_\lambda^{2/3}(E^0)^2 \succ 0$ and

$$(H_M^0)^\top H_M^0 = D_{-M}^0 J_M^0 D_{-M}^0. \quad (13)$$

Consequently the Moore–Penrose frozen-input update is ordinary on this event and, for every mode,

$$\tilde{X}_M = G_M \tilde{B}_M, \quad \tilde{B}_M = D_\lambda Q_M^0 (J_M^0)^{-1} (D_{-M}^0)^{-1}. \quad (14)$$

Proof. The scale identity from the balanced-seed conclusion of Proposition A.16 gives

$$D_{-M}^0 = (D_\lambda^{1/3} E^0)^2 = D_\lambda^{2/3} (E^0)^2,$$

which is positive diagonal. Since $X_{M'}^0 = P_{M'}^0 D_{M'}^0$ and $X_{M''}^0 = P_{M''}^0 D_{M''}^0$, the frozen Khatri–Rao design in the setting is

$$H_M^0 = P_{-M}^0 D_{-M}^0.$$

The Khatri–Rao Gram identity gives

$$(P_{-M}^0)^\top P_{-M}^0 = F_{M'} \circ F_{M''} = J_M^0,$$

proving (13). Proposition A.18 makes both factors on the right invertible, so the Moore–Penrose inverse equals the ordinary inverse.

For completeness, the current unfolding identity is

$$T_{(M)} = G_M D_\lambda K_{-M}^\top, \quad K_{-M}^\top P_{-M}^0 = L_{M'} \circ L_{M''} = Q_M^0. \quad (15)$$

Using (13) and the diagonal inverse gives

$$\begin{aligned} T_{(M)} H_M^0 ((H_M^0)^\top H_M^0)^{-1} &= G_M D_\lambda Q_M^0 D_{-M}^0 (D_{-M}^0)^{-1} (J_M^0)^{-1} (D_{-M}^0)^{-1} \\ &= G_M D_\lambda Q_M^0 (J_M^0)^{-1} (D_{-M}^0)^{-1}, \end{aligned}$$

which is (14). This calculation is done for all three frozen inputs before any output is committed. On a singular or otherwise off-event tape the setting’s Moore–Penrose rule remains the observable fallback; the ordinary formula and quantitative reserves are asserted only on this generated pre-solve event. □

Proposition A.19 (scale-conjugated coefficient and normalizer reserve). *Under Proposition A.17 and Lemma A.16, for every mode define*

$$L_M^0 := D_\lambda (D_{-M}^0)^{-1} = D_\lambda^{1/3} (E^0)^{-2}, \quad \ell_{M,j}^0 := (L_M^0)_{jj} = \lambda_j^{1/3} e^{-2s_j^0/3} > 0,$$

and

$$\Xi_M := D_\lambda Q_M^0 (J_M^0)^{-1} D_\lambda^{-1}.$$

Then, in both induced orientations,

$$\|\Xi_M - I\|_{\text{row},1} \vee \|\Xi_M - I\|_{\text{col},1} \leq \Gamma \eta_A \leq 128\omega < \frac{1}{64}. \quad (16)$$

If $\tilde{d}_{M,j} := \|G_M \tilde{B}_M(:,j)\|_2$, then

$$(1 - \Gamma \eta_A) \ell_{M,j}^0 \leq \tilde{d}_{M,j} \leq (1 + \Gamma \eta_A) \ell_{M,j}^0, \quad (17)$$

so every output normalizer is strictly positive before the joint commit.

Proof. By Proposition A.17, $\eta_J \leq 1/64$ and $\Gamma \eta_A \leq 128\omega < 1/64$; the explicit arithmetic is (N1)–(N3).

By (14), $\tilde{B}_M = \Xi_M L_M^0$. From (8)–(10),

$$Q_M^0 (J_M^0)^{-1} - I = (Q_M^0 - J_M^0) (J_M^0)^{-1},$$

whose row and column norms are at most η_A . Diagonal similarity multiplies either induced norm by at most $\max_i \lambda_i / \min_i \lambda_i = \Gamma$, proving the first inequality in (16).

For a fixed column j , write $\Xi_M(:, j) = e_j + v_j$. The column part of (16) gives $\|v_j\|_1 \leq \Gamma\eta_A$. Since every column of G_M has norm one,

$$\|G_M \Xi_M(:, j) - G_M(:, j)\|_2 \leq \|v_j\|_1 \leq \Gamma\eta_A.$$

The reverse and forward triangle inequalities give $1 - \Gamma\eta_A \leq \|G_M \Xi_M(:, j)\|_2 \leq 1 + \Gamma\eta_A$. Multiplication by the positive $\ell_{M,j}^0$ proves (17). All three modes satisfy the same bounds in a single pre-solve pass. $\square$

Proof. Lemma A.15 expands the exact frozen seed in the target Gram/cross-Gram convention and supplies $\delta_L, \delta_F, \delta_{FL}$ in both orientations, including the shared perpendicular charge $r\tau_r^2$. Proposition A.18 uses the zero diagonal of $F_M - I$ for the Hadamard cancellation in J_M^0 , and uses the direct $Q_M^0 - J_M^0$ identity before either pair is inverted. Its Neumann bounds and the numerical check establish the normalized pair floor and both inverse reserves simultaneously for $M = U, V, W$.

Lemma A.16 transfers those normalized reserves to the actual frozen designs through the exact positive diagonal congruence $(H_M^0)^\top H_M^0 = D_{-M}^0 J_M^0 D_{-M}^0$, and supplies the ordinary-inverse coefficient formula for all three independent Moore–Penrose calls. Finally, Proposition A.19 controls the only scale similarity by the generated $\Gamma \leq 1.01$, proves $\Gamma\eta_A \leq 128\omega < 1/64$, and gives strictly positive output normalizers before any solve is committed. These four results give the pre-solve reserve conclusion of Proposition A.19, with no post-solve or predecessor input. $\square$

Proposition A.20 (Complete frozen pre-solve reserve interface). *Under the event supplied by Proposition A.3 and the balanced-seed conclusion of Proposition A.16, define*

$$\begin{aligned} H_M &:= G_M^\top G_M, & L_M &:= G_M^\top P_M^0, & F_M &:= (P_M^0)^\top P_M^0, \\ D_{-M}^0 &:= D_{M'}^0 D_{M''}^0, & L_M^0 &:= D_\lambda (D_{-M}^0)^{-1}, & \ell_{M,j}^0 &:= (L_M^0)_{jj} > 0. \end{aligned}$$

Put $c_0 := 48\omega$ and define

$$\begin{aligned} \delta_L &= (1 + q_*)c_0, & \delta_F &= 2(1 + q_*)c_0 + (1 + q_*)c_0^2 + r\tau_r^2, \\ \delta_{FL} &= (1 + q_*)c_0 + (1 + q_*)c_0^2 + r\tau_r^2, & \alpha_0 &= q_* + \delta_F, & \eta_J &= \alpha_0^2, & \eta_{QJ} &= 2\delta_{FL} + \delta_L^2 + \delta_F^2, & \eta_A &= \frac{\eta_{QJ}}{1 - \eta_J}. \end{aligned}$$

For every mode M and $\nu \in \{\text{row}, \text{col}\}$, the frozen seed satisfies

$$\|L_M - H_M\|_\nu \leq \delta_L, \quad \|F_M - H_M\|_\nu \leq \delta_F, \quad \|F_M - L_M\|_\nu \leq \delta_{FL},$$

and, with $J_M^0 = F_{M'} \circ F_{M''}$ and $Q_M^0 = L_{M'} \circ L_{M''}$,

$$\|J_M^0 - I\|_\nu \leq \eta_J \leq \frac{1}{64}, \quad \|(J_M^0)^{-1}\|_\nu \leq \frac{1}{1 - \eta_J}, \quad \|Q_M^0 - J_M^0\|_\nu \leq \eta_{QJ}.$$

Moreover,

$$\Xi_M := D_\lambda Q_M^0 (J_M^0)^{-1} D_\lambda^{-1}, \quad \|\Xi_M - I\|_\nu \leq \Gamma\eta_A \leq 128\omega < \frac{1}{64},$$

$$(H_M^0)^\top H_M^0 = D_{-M}^0 J_M^0 D_{-M}^0, \quad \tilde{X}_M = G_M \Xi_M L_M^0, \quad (1 - \Gamma\eta_A)\ell_{M,j}^0 \leq \|\tilde{x}_{M,j}\|_2 \leq (1 + \Gamma\eta_A)\ell_{M,j}^0.$$

In particular, every frozen design has full column rank and every output normalizer is positive before any mode is committed.

Proof. Lemma A.15 gives the three first inequalities in both orientations. Proposition A.18 applies the zero-diagonal cancellation and Neumann bounds, yielding the J_M^0 floor, inverse, and direct $Q_M^0 - J_M^0$ estimates. Lemma A.16 gives the positive diagonal congruence and ordinary-inverse coefficient identity. Proposition A.19 applies the only scale similarity and proves the Ξ_M and normalizer bounds. These are all pre-commit conclusions from the same seed. $\square$

A.10 Synchronized frozen landing

Proposition A.20 supplies, before any landing commit,

$$\tilde{X}_M = G_M D_\lambda Q_M^0 (J_M^0)^{-1} (D_{-M}^0)^{-1} = G_M \Xi_M L_M^0. \quad (\text{C1})$$

Right multiplication preserves the left column range, positive diagonal congruences are invertible, and for $0 < \beta < 1$ and $a \in [1 - \beta, 1 + \beta]$,

$$|\log a| \leq -\log(1 - \beta) \leq \frac{\beta}{1 - \beta}. \quad (\text{C2})$$

The interval for each normalizer is supplied by the same pre-solve result.

For $M \in \{U, V, W\}$, let M', M'' be the other modes and put

$$K_{-M} := G_{M'} \odot G_{M''}, \quad P_{-M}^0 := P_{M'}^0 \odot P_{M''}^0.$$

These are proof-local translations of the frozen Khatri–Rao factors.

Lemma A.17 (unconditional unfolding-range inclusion). *Under the exact tensor decomposition, for every frozen design, including a singular one,*

$$\tilde{X}_M := T_{(M)} H_M^0 ((H_M^0)^\top H_M^0)^\dagger$$

has every column in range(G_M).

Proof. The exact unfolding identity is

$$T_{(M)} = G_M D_\lambda K_{-M}^\top. \quad (1)$$

Hence $\text{range}(T_{(M)}) \subseteq \text{range}(G_M)$, and right multiplication by $H_M^0 ((H_M^0)^\top H_M^0)^\dagger$ preserves that left column range. This uses no inverse or good event and applies independently to all three modes. $\square$

Proposition A.21 (three independent frozen solves and positive normalizers). *Under the pre-solve reserve supplied by Proposition A.19 and Lemma A.16, for each mode*

$$\tilde{X}_M = G_M \tilde{B}_M, \quad \tilde{B}_M = D_\lambda Q_M^0 (J_M^0)^{-1} (D_{-M}^0)^{-1} = \Xi_M L_M^0. \quad (2)$$

Writing $\xi_{M,j} := \Xi_M(:, j)$,

$$\tilde{x}_{M,j} = \ell_j^0 G_M \xi_{M,j}, \quad \tilde{d}_{M,j} := \|\tilde{x}_{M,j}\|_2 = \ell_j^0 a_{M,j}, \quad a_{M,j} := \|G_M \xi_{M,j}\|_2, \quad (3)$$

and $1 - \beta \leq a_{M,j} \leq 1 + \beta$, so $\tilde{d}_{M,j} > 0$. All three formulas are evaluated from the one frozen seed before any mode is committed.

Proof. The frozen scaling and the congruence give

$$H_M^0 = P_{-M}^0 D_{-M}^0, \quad (H_M^0)^\top H_M^0 = D_{-M}^0 J_M^0 D_{-M}^0. \quad (4)$$

Lemma A.16 gives $D_{-M}^0 \succ 0$, while the Step 009 pair-Gram reserve recorded in Proposition A.20 gives $\lambda_{\min}(J_M^0) \geq 1 - \eta_J > 0$. Hence H_M^0 has full column rank and its Moore–Penrose inverse is the ordinary inverse. Since

$$K_{-M}^\top P_{-M}^0 = (G_{M'}^\top P_{M'}^0) \circ (G_{M''}^\top P_{M''}^0) = Q_M^0,$$

the unfolding identity yields

$$\begin{aligned} T_{(M)} H_M^0 ((H_M^0)^\top H_M^0)^{-1} &= G_M D_\lambda Q_M^0 D_{-M}^0 (D_{-M}^0)^{-1} (J_M^0)^{-1} (D_{-M}^0)^{-1} \\ &= G_M D_\lambda Q_M^0 (J_M^0)^{-1} (D_{-M}^0)^{-1}, \end{aligned}$$

which is (2). Factoring L_M^0 gives (3). For $v_{M,j} := \xi_{M,j} - e_j$, the coefficient reserve from Proposition A.19 and the unit columns of G_M imply

$$|a_{M,j} - 1| \leq \|G_M v_{M,j}\|_2 \leq \|v_{M,j}\|_1 \leq \beta. \quad (5)$$

Thus the interval and positive normalizer follow. The singular Moore–Penrose branch was not altered; the ordinary formula is used only after the Step 009 congruence and pair-Gram floor have been established. $\square$

Proposition A.22 (normalized direction fields). *On the pre-solve reserve event supplied by Proposition A.19, assume the conclusions of Lemma A.16 and Proposition A.19; in particular, for every $M \in \{U, V, W\}$,*

$$\beta := \Gamma \eta_A \leq 128\omega < 1/64 < 1, \quad \|\Xi_M - I\|_{\text{row},1} \vee \|\Xi_M - I\|_{\text{col},1} \leq \beta,$$

where $\Xi_M := D_\lambda Q_M^0 (J_M^0)^{-1} D_\lambda^{-1}$, and the positive diagonal congruence and ordinary frozen-solve identity hold. Let the three outputs and their positive normalizers be those supplied by Proposition A.21, so that

$$\tilde{d}_{M,j} > 0, \quad a_{M,j} := \|G_M \Xi_M(:, j)\|_2 \in [1 - \beta, 1 + \beta].$$

Define $D_{a,M} := \text{diag}(a_{M,1}, \dots, a_{M,r})$ and

$$P_M^{\text{land}} := [\tilde{x}_{M,j}/\tilde{d}_{M,j}]_{j=1}^r, \quad C_M^{\text{land}} := \Xi_M D_{a,M}^{-1} - I. \quad (6)$$

Then $P_M^{\text{land}} = G_M(I + C_M^{\text{land}})$, its perpendicular field is zero, and, for $\nu \in \{\text{row}, \text{col}\}$,

$$\|C_M^{\text{land}}\|_\nu \leq \frac{2\beta}{1 - \beta} < 3\beta, \quad \max_j \|G_M C_M^{\text{land}}(:, j)\|_2 \leq \frac{2\beta}{1 - \beta} < 3\beta. \quad (7)$$

Every output direction has positive inner product with its corresponding target column $G_M(:, j)$.

Proof. From (5), $D_{a,M}$ is positive and $\|D_{a,M}^{-1}\|_\nu \leq (1 - \beta)^{-1}$. The identity

$$\Xi_M D_{a,M}^{-1} - I = (\Xi_M - I) D_{a,M}^{-1} + (D_{a,M}^{-1} - I) \quad (8)$$

and Proposition A.19 give two terms each at most $\beta/(1 - \beta)$ in either orientation, proving the first bound in (7). Since $2/(1 - \beta) = 128/63 < 3$, the displayed strict bound follows. For a column,

$$\left\| \frac{G_M \xi_{M,j}}{a_{M,j}} - G_M(:, j) \right\|_2 \leq \frac{\|G_M v_{M,j}\|_2 + |a_{M,j} - 1|}{a_{M,j}} \leq \frac{2\beta}{1 - \beta}.$$

The range lemma gives zero perpendicular field. Also $\langle G_M(:, j), G_M \xi_{M,j} \rangle \geq 1 - \beta > 0$, so the proof-only target sign chart is unchanged. $\square$

Proposition A.23 (joint commit and product-preserving rebalance). *On the pre-solve reserve event supplied by Proposition A.19, under the setting's rule that all three frozen-input designs are formed before any commit and their outputs are committed together, let*

$$(\tilde{X}_U, \tilde{X}_V, \tilde{X}_W)$$

be the three outputs from the common frozen seed (X^0, Y^0, Z^0) supplied by Proposition A.21. Assume, as that proposition concludes, that every active normalizer satisfies $\tilde{d}_{M,j} > 0$ before commitment. With

$$\tilde{\gamma}_j := \prod_{M \in \{U, V, W\}} \tilde{d}_{M,j} > 0,$$

set

$$X_M^{\text{land}}(:, j) := \tilde{\gamma}_j^{1/3} \frac{\tilde{X}_M(:, j)}{\tilde{d}_{M,j}}. \quad (9)$$

This preserves each active rank-one product exactly and leaves normalized directions unchanged.

Proof. Proposition A.21 states that every mode formula uses only the same frozen (X^0, Y^0, Z^0) , so the three outputs are computed and then committed as one transaction. Its positive-normalizer conclusion gives $\tilde{\gamma}_j > 0$, and

$$X_U^{\text{land}}(:, j) \otimes X_V^{\text{land}}(:, j) \otimes X_W^{\text{land}}(:, j) = \frac{\tilde{\gamma}_j}{\tilde{d}_{U,j} \tilde{d}_{V,j} \tilde{d}_{W,j}} \tilde{x}_j \otimes \tilde{y}_j \otimes \tilde{z}_j = \tilde{x}_j \otimes \tilde{y}_j \otimes \tilde{z}_j. \quad (10)$$

Only positive scales change, so P_M^{land} and C_M^{land} are unchanged; inactive columns remain zero. $\square$

Proposition A.24 (quotient entry after one rebalance). *On the conjunction of the pre-solve reserve from Proposition A.19 and the seed conclusion of Proposition A.16, assume the seed-scale conclusion $\|s^0\|_\infty \leq 128\omega$, and the reserve $\beta = \Gamma\eta_A \leq 128\omega < 1/64$, and the following named conclusions: Lemma A.17 gives exact target-span membership for the Moore–Penrose outputs; Proposition A.21 gives $\tilde{d}_{M,j} = \ell_j^0 a_{M,j}$ with $a_{M,j} \in [1 - \beta, 1 + \beta]$; Proposition A.22 gives $P_M^{\text{land}} = G_M(I + C_M^{\text{land}})$, zero perpendicular field, and positive target orientation; and Proposition A.23 gives the common frozen-input joint commit and product-preserving rebalance. Then*

$$\|s^{\text{land}}\|_\infty \leq 1024\omega, \quad d_Q(e^{\text{land}}, 0) \leq 1024\omega. \quad (11)$$

Consequently $d_Q(e^{\text{land}}, 0) \leq 4096\omega < \rho_{\text{ALS}}/3$.

Proof. Since $\ell_j^0 = \lambda_j^{1/3} e^{-2s_j^0/3}$, (3) gives

$$\frac{\tilde{\gamma}_j}{\lambda_j} = e^{-2s_j^0} \prod_{M \in \{U, V, W\}} a_{M,j}. \quad (12)$$

The quotient log is therefore

$$s_j^{\text{land}} = -2s_j^0 + \sum_{M \in \{U, V, W\}} \log a_{M,j}. \quad (13)$$

By (C2) and (5), $|\log a_{M,j}| \leq \beta/(1 - \beta) < 2\beta \leq 256\omega$. Thus

$$|s_j^{\text{land}}| \leq 2(128\omega) + 3(256\omega) = 1024\omega. \quad (14)$$

The definition of d_Q takes the maximum of the three direction quantities and the scale log. Proposition A.22 bounds every direction quantity by $3\beta \leq 384\omega$, so (11) follows. Finally,

$$\omega = q_*^2 + r\tau_r = q_*^2(1 + 10^{-4}), \quad q_* = \frac{1}{4096}, \quad \rho_{\text{ALS}} = \frac{1}{1024},$$

and hence

$$4096\omega = \frac{1 + 10^{-4}}{4096} < \frac{1}{3072} = \frac{\rho_{\text{ALS}}}{3}, \quad (15)$$

because $1 + 10^{-4} < 4/3$. □

Proof. Lemma A.17 gives exact target-span membership for each Moore–Penrose output without assuming a nonsingular design. On the derived event $E_{\text{land}}^{\text{pre}}$, Proposition A.21 invokes the positive diagonal congruence and pair-Gram floor supplied by Lemma A.16 and Proposition A.19 to replace each pseudoinverse by its own ordinary inverse, derives the exact coefficient identity, and proves all active normalizers positive from the explicit $\beta < 1$ reserve. Its statement also records that all three outputs use the one frozen seed.

Under the same event, Proposition A.22 applies the normalizers to obtain the target-chart coefficient fields, both induced direction bounds, zero perpendicular fields, and the fixed target orientation. The setting’s synchronized joint-commit rule together with the positive normalizers is discharged by Proposition A.23, which proves exact preservation of every represented term. Finally, under the E_{seed} scale bound, Proposition A.24 gives $d_Q(e^{\text{land}}, 0) \leq 1024\omega \leq 4096\omega$ and the $\rho_{\text{ALS}}/3$ entry. No landing output is used as an input to another landing solve; singular/off-event tapes retain the observable Moore–Penrose fallback without an unsupported basin claim. □

A.11 Cyclic target-span contraction

The landing and static-geometry inputs for this section are the conclusions of Propositions A.24 and A.3. We use the quotient interpretation in Uschmajew [Uschmajew \(2012\)](#) only at the scope of Assumption 1, Lemma 3.2, and Theorems 3.3 and 3.5 of that paper: the active factors are nonzero, the block normal equations are nonsingular, and cyclic ALS is read modulo the componentwise product-one scaling kernel with block-Gauss–Seidel chronology. Here the object map is $G_M \in \{U, V, W\}$, the held Khatri–Rao design is $H_M = X_{M'} \odot X_{M''}$, and the quotient representative is (C_U, C_V, C_W, s) with the setting’s Refresh_s operation. The target-span inclusion, the radius ρ_{ALS} , every inverse and contraction constant, and the refresh identity are proved in the current notation below; none is imported from the cited source.

For induced row/column ℓ_1 norms, submultiplicativity, the Neumann inverse bound, and the Khatri–Rao Gram identity are used after their hypotheses are displayed.

For matrices with unit diagonal, the following direct inequality will be used repeatedly:

$$\|A \circ B\|_\nu \leq \|A - \text{diag}(A)\|_\nu \max_{i \neq j} |B_{ij}| + \max_i |A_{ii} B_{ii}|, \quad \nu \in \{\text{row}, \text{col}\}. \quad (\text{C1})$$

It follows by separating the diagonal term in each row or column sum.

Throughout this derivation, M' and M'' denote the two modes other than M . For convenience, write

$$\varrho := \rho_{\text{ALS}} = 1/1024$$

to distinguish the chart radius from the smoothing scale ρ .

Lemma A.18 (target-span quotient representation and balanced scales). *Under the static event of Proposition A.3, the landing conclusion of Proposition A.24, and Assumption 1, let an active state have nonzero columns in the exact target spans, and orient its columns by the landing permutation/sign chart. If its represented component products are positive, then there are unique matrices C_M and a unique vector s such that*

$$P_M = G_M(I + C_M), \quad s_j = \log(\gamma_j/\lambda_j), \quad \gamma_j = \prod_M \|X_M(:, j)\|_2. \quad (1)$$

After the product-preserving equal-norm representative is chosen,

$$D_M = \text{diag}(\|X_M(:, j)\|_2) = D_\lambda^{1/3} E, \quad E := \text{diag}(e^{s_j/3}), \quad (2)$$

for all three modes. The exact target state is $0 = (0, 0, 0, 0)$.

Proof. From Proposition A.3, every $G_M^\top G_M$ is positive definite by Gershgorin, so the coefficient matrix in $P_M = G_M B_M$ is unique. Set $C_M = B_M - I$. The landing signs and permutation fix the column chart; later positive normalizations preserve it when the coefficient error is below one. The positive product definition gives the unique logarithm s . Equal-norm balancing replaces the three norms by $\gamma_j^{1/3}$, which is positive and preserves each rank-one product, yielding (2). No factor outside the target span is introduced by this scaling. $\square$

Proposition A.25 (scale equivariance and literal refresh identity). *Under the hypotheses of Lemma A.18, suppose the held design for a mode M has full column rank and the solved active output satisfies $\min_j \|\tilde{X}_M(:, j)\|_2 > 0$. Let positive diagonal matrices R_U, R_V, R_W satisfy $R_U R_V R_W = I$. Rescaling a literal state by $X_M \mapsto X_M R_M$ changes the mode- M ALS output to $\tilde{X}_M \mapsto \tilde{X}_M R_M$, and hence leaves its quotient coordinates unchanged. Consequently, if e^t is the balanced quotient representative of a literal state, then the three records*

$$e \mapsto e_U \mapsto e_V \mapsto e_W \quad (3)$$

defined by the setting's Refresh_s operation are quotient-equivalent to the literal $U/V/W$ post-block states, in that chronological order.

Proof. Write the literal mode design as $H_M = X_{M'} \odot X_{M''}$. Under the product-one rescaling, $H_M \mapsto H_M R_{M'} R_{M''}^{-1} = H_M R_M^{-1}$. Full column rank lets the Moore–Penrose solve be written as the ordinary normal-equation solve, so

$$\begin{aligned} T_{(M)} H_M R_M^{-1} ((R_M^{-1})^\top H_M^\top H_M R_M^{-1})^{-1} &= T_{(M)} H_M (H_M^\top H_M)^{-1} R_M \\ &= \tilde{X}_M R_M. \end{aligned} \quad (4)$$

Thus positive product-one scaling is a gauge symmetry of the literal update. The displayed positive-output condition makes the canonical equal-norm map well-defined; it changes only positive scales and preserves every rank-one product, so it produces the same quotient point. The U record uses the old V, W and writes the new U product log globally; the V record then uses that new U and old W , and the W record uses the two current fields. Applying (4) successively proves that (3) is exactly the quotient image of the literal chronological orbit. This is the block-Gauss–Seidel chronology described qualitatively by the cited quotient interpretation, but the identity is the displayed current-notation calculation. $\square$

Lemma A.19 (two-orientation dynamic Gram reserve). *On the event supplied by Proposition A.3, let $e = (C_U, C_V, C_W, s)$ be an exact target-span balanced state with $x := d_Q(e, 0) \leq \varrho$. For each mode set*

$$A_M := I + C_M, \quad P_M := G_M A_M, \quad L_M := G_M^\top P_M = H_M A_M, \quad F_M := P_M^\top P_M = A_M^\top H_M A_M. \quad (5)$$

Then, for $\nu \in \{\text{row}, \text{col}\}$, simultaneously for all three modes,

$$\begin{aligned} \|F_M - H_M\|_\nu &\leq 2.01x, \\ \|(L_M - F_M) - \text{diag}(L_M - F_M)\|_\nu &\leq 1.01x, \\ \max_j |L_{M,jj} - F_{M,jj}| &\leq x^2/2, \\ \max_{i \neq j} |(L_M)_{ij}| &\leq q + 1.01x, \\ \max_{i \neq j} |(F_M)_{ij}| &\leq q + 2.01x. \end{aligned} \quad (6)$$

For a block M define

$$Q_M := L_{M'} \circ L_{M''}, \quad J_M := F_{M'} \circ F_{M''}, \quad K_M := H_{M'} \circ H_{M''}. \quad (7)$$

Then

$$\|Q_M - J_M\|_\nu \leq 2.1qx + 4.1x^2, \quad \|J_M - I\|_\nu \leq q^2 + 4.1qx + 4.1x^2 < 1/64, \quad (8)$$

and therefore J_M is positive definite and

$$\|J_M^{-1}\|_{\text{row},1} \vee \|J_M^{-1}\|_{\text{col},1} \leq 64/63. \quad (9)$$

Proof. Write $H = H_M$, $A = I + C$, and note from the definition of d_Q that both induced norms of C and every $\|GC(:, j)\|_2$ are at most x . Since $\|H - I\|_\nu \leq q$, $\|H\|_\nu \leq 1 + q$, and $x \leq 1/1024$,

$$\|F - H\|_\nu \leq (1 + q)x + (1 + q)x + (1 + q)x^2 \leq 2.01x. \quad (10)$$

Here we used $F - H = HC + C^\top H + C^\top HC$. Also

$$L - F = HA - A^\top HA = -C^\top HA, \quad (11)$$

so its induced norm is at most $(1 + q)(1 + x)x \leq 1.01x$. Unit columns of P give the exact diagonal identity

$$L_{jj} - F_{jj} = L_{jj} - 1 = -\frac{1}{2}\|GC(:, j)\|_2^2, \quad (12)$$

which proves the second and third lines of (6). The expansion $L - H = HC$ gives the fourth line of (6); the expansion in (10) gives the fifth line.

For (8), apply the direct diagonal/off-diagonal inequality (C1) to

$$Q - J = (L_{M'} - F_{M'}) \circ L_{M''} + F_{M'} \circ (L_{M''} - F_{M''}). \quad (13)$$

The first term is at most $1.01x(q + 1.01x) + x^2/2$, and the second is at most $(q + 2.01x)(1.01x) + x^2/2$. Their sum is at most $2.02qx + 4.051x^2$, hence the first bound in (8). Similarly, with $R_M = F_M - H_M$, whose diagonal is zero,

$$J - K = R_{M'} \circ F_{M''} + H_{M'} \circ R_{M''} \quad (14)$$

has norm at most $4.1qx + 4.1x^2$. The target Gram satisfies $\|K - I\|_\nu \leq q^2$, because its off-diagonal entries are products of the two target off-diagonal entries. This proves the second inequality in (8). Finally $q \leq q_* \leq \varrho/4$ and $x \leq \varrho$, so

$$q^2 + 4.1qx + 4.1x^2 \leq (1/16 + 4.1/4 + 4.1)\varrho^2 < 1/64. \quad (15)$$

The Neumann series in either induced norm gives (9); because J is the Gram of $P_{M'} \odot P_{M''}$, it is symmetric positive semidefinite, and the same bound gives its positive spectral floor. Thus the Moore–Penrose update is the ordinary update throughout the chart. $\square$

Proposition A.26 (one chronological block contraction). *On the event supplied by Proposition A.3, and under the hypotheses and conclusions of Lemmas A.18 and A.19, fix a mode M and let $x = d_Q(e, 0) \leq \varrho$ for an exact-span balanced positive-product state. Form the literal exact ALS update from the two held modes. The full-rank reserve below first proves positive solved columns; only then apply the product-preserving equal-norm representative. The landing conclusion of Proposition A.24 supplies one such in-chart input, but the estimate below is uniform over the entire displayed chart. The new mode direction and its refreshed product log satisfy, in both induced orientations and in the Euclidean column part of d_Q ,*

$$\|C_M^+\|_{\text{row},1} \vee \|C_M^+\|_{\text{col},1} \vee \max_j \|G_M C_M^+(\cdot, j)\|_2 \vee \|s_M^+\|_\infty \leq \ell x, \quad \ell := 8q_* + 32\varrho. \quad (16)$$

The ordinary full-rank output is in $\text{range}(G_M)$, has positive target orientation, and has positive active norm, so the canonical representative is defined.

Proof. In the balanced representative, write $D_M = D_\lambda^{1/3} E$ as in (2) and $D_{-M} = D_{M'} D_{M''} = D_\lambda^{2/3} E^2$. The unfolding identity and the full-rank conclusion of Lemma A.19 give the current normal-equation formula

$$\tilde{X}_M = G_M D_\lambda Q_M J_M^{-1} D_{-M}^{-1} = G_M \Xi_M D_\lambda^{1/3} E^{-2}, \quad \Xi_M := D_\lambda Q_M J_M^{-1} D_\lambda^{-1}. \quad (17)$$

Indeed, $T_{(M)} = G_M D_\lambda (G_{M'} \odot G_{M''})^\top$ and $(G_{M'} \odot G_{M''})^\top (P_{M'} \odot P_{M''}) = Q_M$. Using $Q_M J_M^{-1} - I = (Q_M - J_M) J_M^{-1}$, (8)–(9), and $\Gamma \leq 1.01$, diagonal similarity gives, for either ν ,

$$\|\Xi_M - I\|_\nu \leq 1.01 \frac{64}{63} (2.1qx + 4.1x^2) \leq 2.2qx + 4.3x^2. \quad (18)$$

Set $\beta := 2.2qx + 4.3x^2$. Since $2.2q_* + 4.3\varrho < 1/128$, we have $\beta < 1/128$. For each column $\xi_j = \Xi_M(\cdot, j)$, let $a_j = \|G_M \xi_j\|_2$. The column bound in (18) gives $|a_j - 1| \leq \beta$, hence $a_j > 0$. Moreover $\langle G_M(\cdot, j), G_M \xi_j \rangle \geq 1 - \beta > 0$, so the target orientation is preserved. Put

$$b_{M,j} := \lambda_j^{1/3} e^{-2s_j/3} > 0, \quad \tilde{x}_{M,j} = b_{M,j} G_M \xi_j, \quad \|\tilde{x}_{M,j}\|_2 = b_{M,j} a_j > 0. \quad (19a)$$

Thus the positive solved-column condition in Proposition A.25 is discharged before canonical normalization. The normalized coefficient field is

$$C_M^+ = \Xi_M \text{diag}(a_j^{-1}) - I. \quad (19)$$

Consequently, in either induced orientation,

$$\|C_M^+\|_\nu \leq \frac{2\beta}{1-\beta} \leq 4.5qx + 8.8x^2, \quad (20)$$

and the same bound holds for each Euclidean column error after applying G_M . To identify the refreshed product log, the two held balanced norms are $\lambda_j^{1/3} e^{s_j/3}$ each. Therefore, before the canonical rebalance,

$$\gamma_j^+ = (b_{M,j} a_j) (\lambda_j^{1/3} e^{s_j/3})^2 = \lambda_j a_j. \quad (20a)$$

The rebalance preserves this rank-one product, so its global product register satisfies the exact gauge cancellation

$$s_{M,j}^+ = \log \left(\frac{\gamma_j^+}{\lambda_j} \right) = \log a_j. \quad (20b)$$

The elementary logarithm bound on $[1 - \beta, 1 + \beta]$ now gives

$$|s_{M,j}^+| = |\log a_j| \leq \frac{\beta}{1 - \beta} \leq 2.3qx + 4.4x^2. \quad (21)$$

Since $q \leq q_*$ and $x \leq \varrho$, each right side in (20)–(21) is at most

$$(8q_* + 32\varrho)x = \ell x. \quad (22)$$

On the maintained chart, Lemma A.19 makes the current design Gram invertible, so the displayed ordinary formula (17) has a left factor G_M and proves the output is in $\text{range}(G_M)$. No singular/off-chart range assertion is used in this cyclic step. The positive normalizer just proved makes the normalized direction and canonical rebalance well-defined. This proves (16). $\square$

Proposition A.27 (chronological contraction and target-span invariant). *Under the event supplied by Proposition A.3, let Proposition A.24 supply the entry state e^{land} . Using Proposition A.26 and then Proposition A.25 (the latter only after the former has supplied positive solved columns), take any exact-span balanced positive-product state $e^0 = e$ in the fixed orientation chart with $x := d_Q(e, 0) \leq \varrho$, and define the setting's chronological states*

$$e_U = (C_U^+, C_V^0, C_W^0, s_U^+), \quad e_V = (C_U^+, C_V^+, C_W^0, s_V^+), \quad e_W = (C_U^+, C_V^+, C_W^+, s_W^+), \quad \Psi^q(e) := e_W. \quad (23)$$

Then all three blocks are defined, their outputs remain in the exact target-span chart with positive active products, and their refreshed records are quotient-equivalent to the literal ALS post-block states. Uniformly for every such in-chart input,

$$d_Q(\Psi^q(e), 0) \leq \ell d_Q(e, 0), \quad \ell = 8q_* + 32\rho_{\text{ALS}} = 17/512 < 1/16. \quad (24)$$

In particular, starting from e^{land} , the chronological orbit remains in the chart and defines the basin event; writing $e^{t+1} := \Psi^q(e^t)$ and $e^0 = e^{\text{land}}$, it has no additive forcing:

$$d_Q(e^t, 0) \leq \ell^t d_Q(e^{\text{land}}, 0), \quad d_Q(e^t, 0) - d_Q(e^{t+1}, 0) \geq (1 - \ell) d_Q(e^t, 0), \quad \sum_{t \geq 0} d_Q(e^t, 0) \leq \frac{d_Q(e^{\text{land}}, 0)}{1 - \ell}. \quad (25)$$

Proof. Fix an arbitrary input state in the domain of the proposition and write $x = d_Q(e, 0) \leq \varrho$. Lemma A.19 gives the full-rank held-design reserve for the U block. Proposition A.26 therefore gives

$$\|C_U^+\|_{\text{row},1} \vee \|C_U^+\|_{\text{col},1} \vee \max_j \|G_U C_U^+(\cdot, j)\|_2 \vee \|s_U^+\|_\infty \leq \ell x, \quad (26)$$

The positivity and target-span conclusions in the same proposition discharge the local condition of Proposition A.25; the full-rank reserve and equation (4) then identify the U record with the literal U post-block state. The unchanged V, W fields and their scale register are still bounded by x , while the newly written U fields/register are bounded by ℓx , so $d_Q(e_U, 0) \leq x$. Apply the same two propositions to V with the new U field and old W field. Its input is still in the chart, so the new V fields and s_V^+ are at most ℓx , and $d_Q(e_V, 0) \leq x$; the V record is again the literal post-block

state. Applying them to W gives the same bound for the new W fields and s_W^+ . At the end of the sweep, U, V, W and the common product register are all at most ℓx , proving the uniform map inequality (24), while equation (17) proves exact target-span membership before every subsequent block update.

For the landing state, Proposition A.24 gives $d_Q(e^{\text{land}}, 0) < \varrho/3$. Since $\ell < 1$, the uniform map inequality keeps every e^t in the chart and gives the first inequality in (25) by induction. Subtracting successive nonnegative terms and summing the geometric series gives the potential drop and finite budget in (25).

Finally,

$$8q_* + 32\rho_{\text{ALS}} = \frac{8}{4096} + \frac{32}{1024} = \frac{17}{512} < \frac{1}{16}, \quad (27)$$

so the contraction is strict with the fixed sketch constant. A singular or zero-normalizer tape outside the maintained chart receives no quantitative conclusion and follows the setting's observable unsuccessful/cap branch. $\square$

Proof. The landing conclusion of Proposition A.24, together with the event from Proposition A.3, supplies the legal positive exact-span entry. Lemma A.18 supplies the unique exact-span quotient coordinates and the balanced scale convention consumed by every block. Proposition A.26 first proves positive solved columns, target-span output, and the one-block bound on every in-chart input; only after that discharge does Proposition A.25 prove that positive product-one scaling is a genuine gauge symmetry of each full-rank literal normal-equation solve and that the setting's chronological Refresh_s records are the quotient representatives of the literal $U/V/W$ orbit. Lemma A.19 derives, rather than assumes, the dynamic pair-Gram reserve from the realized interference bound and the current radius; it also proves that every design used on the good trajectory is full rank. Proposition A.26 then retains the exact coefficient identity, displays the product/gauge scale cancellation $s_M^+ = \log a_j$, positive normalizers, target orientation, and all three parts of the one-block d_Q bound. Finally, Proposition A.27 composes the held input bounds in the actual $U/V/W$ order for every exact-span positive-product input with $d_Q \leq \rho_{\text{ALS}}$, proves target-span invariance before each block update, and exports the uniform map estimate (24) and the no-forcing recurrence (25). Therefore the generated basin event and Ψ^q satisfy the exact displayed claim, with no predecessor comparator, post-solve landing input, or hidden basin assumption.

At the exact orthogonal equal-weight certified state, $H_M = I$, $C_M = 0$, and $s = 0$. Hence $Q_M = J_M = I$, $\Xi_M = I$, $a_j = 1$, every Refresh_s register remains zero, and all three cyclic updates are stationary. The quotient error and the represented tensor residual are exactly zero, preserving the required baseline rather than replacing it by the finite-radius bound. $\square$

A.12 Stopping, runtime, and restarts

Proposition A.6 and Assumption 7 give, conditional on a fixed instance in the event of Proposition A.3,

$$\Pr(E_{\text{cov}} \mid \mathcal{F}_{\text{sm}}) \geq p_0, \quad p_0 := \frac{26}{27}. \quad (A1)$$

Proposition A.7 ensures that its witnesses are proof-only. Lemma A.18 and Proposition A.27 give, on the generated covered path,

$$x_0 := d_Q(e^{\text{land}}, 0) \leq 4096\omega < \rho_{\text{ALS}}/3, \quad x_{t+1} \leq \ell x_t, \quad \ell := \frac{17}{512} < \frac{1}{16}. \quad (A2)$$

The same proposition gives exact-span invariance, positive products, and literal ALS/refresh equivalence.

For the residual comparison we use $(A \odot B)^\top (A \odot B) = (A^\top A) \circ (B^\top B)$ and Schur order: if $H \succeq 0$ has unit diagonal and $aI \preceq K \preceq bI$, then $aI \preceq H \circ K \preceq bI$. We also use $\|A \otimes B\|_2 = \|A\|_2 \|B\|_2$, $\|C\|_2 \leq \sqrt{\|C\|_{\text{row},1} \|C\|_{\text{col},1}}$, and the isometry $\Delta(z) = \sum_j z_j e_j \otimes e_j \otimes e_j$. Every finite matrix has a Moore–Penrose pseudoinverse; this gives existence and dense cost on singular branches, not an inverse reserve. Conditional on the tensor, complete-run tapes are independent by Assumption 7, so the joint failure probability of completed runs is a product.

Lemma A.20 (balanced-chart residual transfer). *Under Assumption 1, the static-geometry interface of Proposition A.3, and the coordinate conclusion of Lemma A.18, let $e = (C_U, C_V, C_W, s)$ be a state supplied by Proposition A.27 represented in the exact-span, positively oriented, equal-norm chart, and suppose $x = d_Q(e, 0) \leq \rho_{\text{ALS}} = 1/1024$. If $\hat{T}(e)$ denotes the rank- r tensor represented by its active factors, with the inactive $k - r$ columns zero, then*

$$\frac{\|T - \hat{T}(e)\|_F}{\|T\|_F} \leq 5x \leq 8\kappa_0^2 x. \quad (1)$$

Proof. Write $G_M = U, V, W$ in the corresponding mode, $A_M = I + C_M$, and $\lambda = (\lambda_1, \dots, \lambda_r)^\top$. The fixed orientation chart and positive product register give

$$p_{M,j} = G_M A_M e_j, \quad \hat{T}(e) = \mathcal{G} \mathcal{A} \Delta(\lambda \odot e^s), \quad T = \mathcal{G} \Delta(\lambda), \quad (2)$$

where $\mathcal{G} = G_U \otimes G_V \otimes G_W$, $\mathcal{A} = A_U \otimes A_V \otimes A_W$, and e^s has entries e^{s_j} . This is the componentwise identity $\lambda_j e^{s_j} p_{U,j} \otimes p_{V,j} \otimes p_{W,j}$, so it does not add a factor or change the represented tensor.

By the definition of d_Q , both induced norms of each C_M are at most x , so $\|C_M\|_2 \leq x$ and

$$\|\mathcal{A}\|_2 \leq (1+x)^3, \quad \|\mathcal{A} - I\|_2 \leq (1+x)^3 - 1. \quad (3)$$

For $0 \leq x < 1$, the power-series comparison $e^x \leq 1/(1-x)$ gives

$$\|\lambda \odot e^s - \lambda\|_2 \leq (e^x - 1) \|\lambda\|_2 \leq \frac{x}{1-x} \|\lambda\|_2. \quad (4)$$

At $x \leq 1/1024$, $(1+x)^3 < 1.003$, $x/(1-x) < 1.001x$, and $(1+x)^3 - 1 < 3.004x$. The isometry of Δ therefore yields

$$\begin{aligned} \|\mathcal{A} \Delta(\lambda \odot e^s) - \Delta(\lambda)\|_F &\leq \|\mathcal{A}\|_2 \|\Delta(\lambda \odot e^s - \lambda)\|_F + \|\mathcal{A} - I\|_2 \|\Delta(\lambda)\|_F \\ &< (1.003)(1.001)x \|\lambda\|_2 + 3.004x \|\lambda\|_2 < 4.1x \|\lambda\|_2. \end{aligned} \quad (5)$$

The static geometry gives $\|G_M\|_2^2 = \lambda_{\max}(G_M^\top G_M) \leq 1 + q_*$, and hence

$$\|\mathcal{G}\|_2 \leq (1 + q_*)^{3/2}. \quad (6)$$

It remains to compare the denominator with the same coefficient norm used in (5). Put $H_U = U^\top U$, $H_V = V^\top V$, and $H_W = W^\top W$, and define the two-mode and full three-mode Khatri–Rao Grams

$$K_{VW} := (W \odot V)^\top (W \odot V) = H_W \circ H_V, \quad K_{UVW} := (W \odot V \odot U)^\top (W \odot V \odot U) = H_U \circ H_V \circ H_W,$$

where column j of $W \odot V \odot U$ is $w_j \otimes v_j \otimes u_j$. For each row j ,

$$\sum_{\ell \neq j} |(K_{VW})_{j\ell}| \leq \left(\max_{\ell \neq j} |(H_W)_{j\ell}| \right) \sum_{\ell \neq j} |(H_V)_{j\ell}| \leq q_*^2.$$

Thus symmetry and Gershgorin give $(1 - q_*^2)I \preceq K_{VW} \preceq (1 + q_*^2)I$. Since $H_U \succeq 0$ and $\text{diag}(H_U) = I$, the checked Schur-order fact in the displayed auxiliary fact gives the full sandwich

$$(1 - q_*^2)\|\lambda\|_2^2 \leq \lambda^\top K_{UVW}\lambda = \left\| \sum_{j=1}^r \lambda_j u_j \otimes v_j \otimes w_j \right\|_F^2 = \|T\|_F^2 \leq (1 + q_*^2)\|\lambda\|_2^2. \quad (7)$$

This is the required full three-mode Gram identity; the two-mode Gram is used only as the Schur-order input. Combining (5)–(7), direct substitution of $q_* = 1/4096$ gives

$$\frac{\|T - \hat{T}(e)\|_F}{\|T\|_F} \leq \frac{(1 + q_*)^{3/2}}{\sqrt{1 - q_*^2}} 4.1x < 5x. \quad (8)$$

Finally Assumption 1 gives $\kappa_0 \geq 1$, hence $5x \leq 8\kappa_0^2 x$. At $x = 0$, (2) is exactly the target tensor and the residual is zero. This is a conditional exact-entry statement; it does not assert that the frozen landing produces that entry in the baseline specialization. $\square$

Proposition A.28 (covered-path stopping before the cap). *Under Assumptions 1 and 8, Proposition A.27, and Lemma A.20, suppose a covered run has the generated E_{basin} entry $x_0 = d_Q(e^{\text{land}}, 0) \leq 4096\omega < 1$, and let x_t be its quotient error after t completed cyclic sweeps. With $\ell = 17/512$,*

$$\frac{\|T - \hat{T}(e^t)\|_F}{\|T\|_F} \leq 8\kappa_0^2 \ell^t. \quad (9)$$

Choose any universal C_{stop} with $C_{\text{stop}} \log(512/17) \geq 1$ (for example $C_{\text{stop}} = 1$), and set

$$m_{\text{cap}} = \left\lceil C_{\text{stop}} \log \frac{8\kappa_0^2}{\epsilon} \right\rceil. \quad (10)$$

Then the covered run has an original relative-Frobenius residual test at most ϵ no later than this counter.

Proof. Proposition A.27 gives $x_t \leq \ell^t x_0$. Since $x_0 < 1$, Lemma A.20 gives (9). Put $A = 8\kappa_0^2/\epsilon$. Assumptions 1 and 8 give $A > 1$. Since $\ell = 17/512$, (10) and the displayed choice of C_{stop} imply

$$\ell^{m_{\text{cap}}} \leq \exp\{-C_{\text{stop}} \log A \log(512/17)\} \leq A^{-1}. \quad (11)$$

The residual at the m_{cap} -th completed sweep is therefore at most $8\kappa_0^2 A^{-1} = \epsilon$. The setting tests the original residual after each completed sweep, so the first hit is no later than the cap. If an exact quotient entry $x_0 = 0$ is supplied, the contraction forces $x_t = 0$ for every completed sweep, and Lemma A.20 makes the first tested residual zero. This conditional statement does not prove the baseline landing trace. $\square$

Proposition A.29 (observable cap and Moore–Penrose branch separation). *Under Assumption 8, fix an instance in the event supplied by Proposition A.3; equation (7) gives $\|T\|_F > 0$. Define the algorithmically admissible raw domain of one complete run and of all J restarts by*

$$\Omega_{\text{raw}}^{(1),\circ} := \left\{ (\xi_i^{(M)})_{i \in [k], M \in \{U,V,W\}} : \|\xi_i^{(M)}\|_2 > 0 \text{ for every } i, M \right\}, \quad \Omega_{\text{raw}}^{(J),\circ} := (\Omega_{\text{raw}}^{(1),\circ})^J. \quad (12)$$

For every tape in $\Omega_{\text{raw}}^{(1),\circ}$, the setting’s procedure is initialized and has a finite observable outcome. A zero proposal contraction, empty certified pool, wrong cluster count, zero θ_a , or zero active norm at

the landing commit terminates that run as unsuccessful. Otherwise every prescribed Moore–Penrose solve is defined, including for a singular design. The cyclic phase executes no more than m_{cap} complete sweeps and returns either the first original-residual hit or an unsuccessful timeout. No singular or off-event tape receives the generated E_{basin} recurrence. Under the conditional Gaussian law, $\Pr(\Omega_{\text{raw}}^{(J),\circ} \mid \mathcal{F}_{\text{sm}}) = 1$.

Proof. The domain (12) is exactly where every initial normalization $\mathcal{N}(\xi_i^{(M)})$ in the written procedure is defined. A nondegenerate Gaussian vector in finite dimension is zero with probability zero; a finite union over $3kJ$ raw vectors proves the final probability-one assertion. No statement is made about a raw-zero array outside (12).

There are k proposal slots and each has at most $L_{\text{prop}} + 1$ old-state evaluations. A zero contraction is checked before normalization, so that gate has a finite failure branch. Certification, score filtering, graph construction, and connected components are finite operations on the k -slot list. The setting declares the other listed failures observable before a cyclic update.

For a surviving landing call, each frozen design is a finite matrix. The Moore–Penrose rule returns a matrix even when its Gram is singular; no ordinary inverse is silently substituted. The positive-norm landing gate either accepts all active columns or marks the run unsuccessful. After an accepted landing, the literal cyclic update also uses a Moore–Penrose solution at every mode. A singular cyclic design is thus a legal finite algebraic branch, but it is outside the hypotheses of the quotient Refresh_s argument; no d_Q , target-span, or contraction claim is made there. If a proof-only canonical refresh would require division by a zero output norm, that refresh is not invoked and the literal tape remains on the observable capped branch. The original residual is still evaluable for the literal factors because $\|T\|_F > 0$.

The counter is initialized before the first cyclic solve and ranges over the finite set $\{1, \dots, m_{\text{cap}}\}$. After each counted completed sweep, the original residual is tested; a hit returns success, and the final nonhit returns unsuccessful. Thus any admissible noncovered, singular, off-chart, or nonhitting tape cannot cause unbounded work or justify the good-path recurrence. This deterministic completion property is the event $E_{\text{run}}^{\text{cap}}$. □

Proposition A.30 (dense per-run runtime). *For the setting-defined finite dimensions and horizons, under the completion conclusion of Proposition A.29, put in the dense exact-arithmetic model*

$$B_s := n^3 s + n^2 s^2 + s^3, \quad L := L_{\text{prop}}, \quad m := m_{\text{cap}}. \quad (13)$$

There is a universal c_{cost} such that one complete run, including all unsuccessful branches, costs at most

$$W_{\text{run}} \leq c_{\text{cost}} [n^3 + kL(n^3 + n) + kn^3 + k^2n + (m + 1)B_k + kn]. \quad (14)$$

This bound is independent of every generated condition number.

Proof. Materializing the three dense unfoldings costs $O(n^3)$ once. A contraction of an $n \times n^2$ unfolding with one Khatri–Rao vector costs $O(n^3)$; three contractions per Jacobi commit, including certification look-ahead, give $O(kLn^3)$. Generating and normalizing raw vectors costs $O(kn)$. Scores use at most $O(kn^3)$ dense tensor contractions. The three modewise inner products for every retained pair and the connected graph cost $O(k^2n)$, covering filtering, clustering, and tie handling.

The frozen landing pass has three products of an $n \times n^2$ unfolding with an $n^2 \times k$ design, dense Gram formation, and three k -column Moore–Penrose factorizations. Their combined cost is $O(B_k)$; the one rebalance is lower order. A cyclic sweep has the same dense upper bound $O(B_k)$, including construction of the reconstructed tensor for the original residual test, which costs $O(n^3k)$. Inactive $k - r$ columns may be removed for a sharper B_r bound, but retaining the prescribed zero-padded

arrays is covered by B_k . Singular pseudoinverses use the same dimension-controlled factorization, so no inverse condition number enters. There are at most m sweeps by Proposition A.29, proving (14). $\square$

Proposition A.31 (conditional restart amplification). *Under Assumptions 7 and 8, condition on a fixed instance in the event supplied by Proposition A.3. Suppose complete runs use the cap in (10) and the conclusions of Propositions A.6, A.27, A.28, and A.29. Let*

$$J = \max\{1, \lceil C_{\text{rep}} \log(1/\delta_{\text{init}}) \rceil\}, \quad C_{\text{rep}} \geq 1/\log 27. \quad (15)$$

Then

$$\Pr(\text{all } J \text{ runs fail} \mid \mathcal{F}_{\text{sm}}) \leq (1 - p_0)^J = 27^{-J} \leq \delta_{\text{init}}. \quad (16)$$

On the complementary event, the smallest-residual successful output has at most k nonzero terms and original relative residual at most ϵ .

Proof. Proposition A.6 gives a one-run coverage probability at least $p_0 = 26/27$. On a fixed E_{sm} tensor, the deterministic chain through Proposition A.27 supplies the generated E_{basin} on that covered branch, and Proposition A.28 supplies a residual hit before the cap. Proposition A.29 makes every admissible tape a completed success/failure trial, and its domain has conditional probability one. Fresh complete-run tapes make these outcomes independent conditional on $\mathcal{F}_{\text{sm}}$, even though the tensor is reused. Therefore the failure product in (16) holds. Since

$$J \log 27 \geq C_{\text{rep}} \log(1/\delta_{\text{init}}) \log 27 \geq \log(1/\delta_{\text{init}}),$$

the final inequality in (16) follows. The clamp in (15) remains exact for δ_{init} near one.

Every successful run passed the original residual test, so minimum-residual selection cannot increase its residual. Only the active rank- r columns can be nonzero and $r \leq k$; all inactive columns remain exactly zero. $\square$

Proof. Fix a generated E_{sm} instance. For completed run $a \in [J]$, let $E_{\text{cov}}^{(a)}$ and $E_{\text{basin}}^{(a)}$ denote the generated interfaces on its fresh tape, and define

$$E_{\text{stop}}^{(a)} := E_{\text{cov}}^{(a)} \cap E_{\text{basin}}^{(a)} \cap \left\{ \exists t \leq m_{\text{cap}} : \frac{\|T - \hat{T}(e^{t,a})\|_F}{\|T\|_F} \leq \epsilon \right\}, \quad E_{\text{restart}} := \bigcup_{a=1}^J E_{\text{stop}}^{(a)}. \quad (17)$$

Let $E_{\text{run}}^{\text{cap}}$ be the event that every one of the J runs returns either success or an observable gate/timeout failure within its declared work. Proposition A.29 proves this for every tape in $\Omega_{\text{raw}}^{(J),\circ}$, a conditional probability-one domain. Its singular and off-event branches are never fed into the quotient recurrence.

Proposition A.6 gives $\Pr(E_{\text{cov}}^{(a)} \mid \mathcal{F}_{\text{sm}}) \geq 26/27$. On each covered branch, Proposition A.27 produces the exact-span chronology and contraction (A2). Lemma A.20 uses the full three-mode Gram sandwich (7) to transfer that quotient error to the original relative Frobenius residual, and Proposition A.28 proves $E_{\text{cov}}^{(a)} \subseteq E_{\text{stop}}^{(a)}$. Therefore Proposition A.31 gives the nested probabilities

$$\Pr(E_{\text{restart}} \mid \mathcal{F}_{\text{sm}}) \geq 1 - \delta_{\text{init}} \quad \text{on } E_{\text{sm}}, \quad \Pr(E_{\text{sm}} \cap E_{\text{restart}}) \geq (1 - \delta_{\text{sm}})(1 - \delta_{\text{init}}). \quad (18)$$

The second inequality is the tower calculation $\mathbb{E}[1_{E_{\text{sm}}} \Pr(E_{\text{restart}} \mid \mathcal{F}_{\text{sm}})]$; it does not move either confidence parameter into k . Every successful output passed the original residual test and has only its $r \leq k$ active columns nonzero.

It remains to charge all restarts rather than only one run. Define the proof-local ceiling bounds

$$\bar{k} := 1 + C_{\text{rank}} r^{5/3} (\log r)^{5/2}, \quad R_\epsilon := 2 + C_{\text{stop}} (\log 8 + 2 \log \kappa_0 + \log(1/\epsilon)).$$

From the setting definitions and $0 < \delta_{\text{init}} < 1$,

$$k \leq \bar{k}, \quad L_{\text{prop}} \leq 2 + (C_{\text{burn}} + C_{\text{cert}}) \log r, \quad m_{\text{cap}} + 1 \leq R_\epsilon, \quad J \leq 1 + C_{\text{rep}} \log(1/\delta_{\text{init}}). \quad (19)$$

Applying Proposition A.30 to each completed run and summing, including early failed runs, gives the displayed total-work bound

$$\begin{aligned} W_{\text{total}} &\leq J W_{\text{run}} \\ &\leq c_{\text{cost}} (1 + C_{\text{rep}} \log(1/\delta_{\text{init}})) \left[n^3 + \bar{k} \{ 2 + (C_{\text{burn}} + C_{\text{cert}}) \log r \} (n^3 + n) + \bar{k} n^3 + \bar{k}^2 n \right. \\ &\quad \left. + R_\epsilon (n^3 \bar{k} + n^2 \bar{k}^2 + \bar{k}^3) + \bar{k} n \right]. \end{aligned} \quad (20)$$

In particular, after exposing the prescribed rank powers, (20) is

$$\begin{aligned} W_{\text{total}} &= O \left((1 + \log(1/\delta_{\text{init}})) \left[n^3 + n^3 r^{5/3} (\log r)^{7/2} + n r^{10/3} (\log r)^5 \right. \right. \\ &\quad \left. \left. + (1 + \log \kappa_0 + \log(1/\epsilon)) \{ n^3 r^{5/3} (\log r)^{5/2} + n^2 r^{10/3} (\log r)^5 + r^5 (\log r)^{15/2} \} \right] \right). \end{aligned} \quad (21)$$

The hidden constant in (21) depends only on the fixed universal algorithmic constants. The bound is independent of ρ^{-1} , hence is in particular polynomial in the full list of variables allowed by the setting.

For the exact/noiseless baseline, the scope of this step is conditional: if the exact quotient entry $e^0 = 0$ is supplied, Proposition A.27 keeps every completed cyclic state at zero and Lemma A.20 gives an exactly zero first tested residual for every $\epsilon > 0$. Neither the cap nor the restart rule weakens that conclusion. This step does not prove that the frozen landing produces $e^0 = 0$; the complete landing specialization is proved separately in Proposition A.33. $\square$

Proposition A.32 (Rate Specialization Bridge). *Under Assumptions 1–8, let E_{sm} be the event supplied by Proposition A.3, and condition on a fixed instance in that event. Choose*

$$k = \left\lceil C_{\text{rank}} r^{5/3} (\log r)^{5/2} \right\rceil, \quad m_{\text{cap}} = \left\lceil C_{\text{stop}} \log \frac{8\kappa_0^2}{\epsilon} \right\rceil, \quad J = \max \left\{ 1, \left\lceil C_{\text{rep}} \log \frac{1}{\delta_{\text{init}}} \right\rceil \right\}. \quad (\text{B.1})$$

Let

$$K_r := r^{5/3} (\log r)^{5/2}, \quad \bar{k} := 1 + C_{\text{rank}} K_r, \quad L_r := 2 + (C_{\text{burn}} + C_{\text{cert}}) \log r, \quad R_\epsilon := 2 + C_{\text{stop}} (\log 8 + 2 \log \kappa_0 + \log(1/\epsilon)). \quad (\text{B.2})$$

Then

$$k \leq \bar{k}, \quad L_{\text{prop}} \leq L_r, \quad m_{\text{cap}} + 1 \leq R_\epsilon, \quad J \leq 1 + C_{\text{rep}} \log(1/\delta_{\text{init}}). \quad (\text{B.3})$$

If $p_0 \geq 26/27$ is the one-run coverage probability from Proposition A.6, then

$$(1 - p_0)^J \leq 27^{-J} \leq \delta_{\text{init}} \quad \text{whenever } C_{\text{rep}} \geq 1/\log 27. \quad (\text{B.4})$$

Thus

$$\Pr(E_{\text{restart}} \mid \mathcal{F}_{\text{sm}}) \geq 1 - \delta_{\text{init}} \quad \text{on } E_{\text{sm}}, \quad \Pr(E_{\text{sm}} \cap E_{\text{restart}}) \geq (1 - \delta_{\text{sm}})(1 - \delta_{\text{init}}). \quad (\text{B.5})$$

Moreover, the dense per-run bound of Proposition A.30 gives

$$W_{\text{total}} \leq c_{\text{cost}}(1 + C_{\text{rep}} \log(1/\delta_{\text{init}})) \left[n^3 + \bar{k} L_r (n^3 + n) + \bar{k} n^3 + \bar{k}^2 n + R_\epsilon (n^3 \bar{k} + n^2 \bar{k}^2 + \bar{k}^3) + \bar{k} n \right]. \quad (\text{B.6})$$

For $r \geq 3$ and $k \leq n$, set $C_K := 1 + C_{\text{rank}}$, $C_L := 2 + C_{\text{burn}} + C_{\text{cert}}$, and $C_R := 2 + C_{\text{stop}}(\log 8 + 2)$. Then

$$K_r \geq 1, \quad \bar{k} \leq C_K K_r, \quad L_r \leq C_L \log r, \quad R_\epsilon \leq C_R (1 + \log \kappa_0 + \log(1/\epsilon)), \quad (\text{B.7})$$

and every term suppressed from (B.6) is dominated through

$$\begin{aligned} \bar{k} L_r (n^3 + n) + \bar{k} n^3 + \bar{k} n &\leq 4C_K (C_L + 1) n^3 K_r \log r, \\ \bar{k}^2 n &\leq C_K^2 n K_r^2, \\ R_\epsilon (n^3 \bar{k} + n^2 \bar{k}^2 + \bar{k}^3) &\leq C_R C_K^3 (1 + \log \kappa_0 + \log(1/\epsilon)) (n^3 K_r + n^2 K_r^2 + K_r^3). \end{aligned} \quad (\text{B.8})$$

Consequently,

$$W_{\text{total}} = O\left((1 + \log(1/\delta_{\text{init}})) \left[n^3 + n^3 r^{5/3} (\log r)^{7/2} + n r^{10/3} (\log r)^5 + (1 + \log \kappa_0 + \log(1/\epsilon)) \{ n^3 r^{5/3} (\log r)^{5/2} + n^2 r^{10/3} (\log r)^5 + r^5 (\log r)^{15/2} \} \right] \right). \quad (\text{B.9})$$

The hidden constant depends only on the fixed universal algorithmic constants. Assumptions 1 and 4 give $\kappa_0 \leq r^{d_\kappa}$ and $\rho^{-1} \leq r^{d_\rho}$; the cost bound is independent of ρ^{-1} and hence is polynomial in the full declared list.

Proof. The ceiling in the definition of k gives the first inequality in (B.3), and the definitions of L_{burn} and L_{cert} give the second. Since $0 < \epsilon < 1$ and $\kappa_0 \geq 1$,

$$\log \frac{8\kappa_0^2}{\epsilon} = \log 8 + 2 \log \kappa_0 + \log(1/\epsilon),$$

which proves the third inequality in (B.3); the last follows from the ceiling and $0 < \delta_{\text{init}} < 1$. The conditional one-run bound is Proposition A.6; independence of complete runs from Assumption 7 gives the first inequality in (B.4), and the choice of C_{rep} gives the second. The deterministic covered-run chain through Propositions A.27 and A.28 converts coverage into success, so (B.4) gives the conditional statement in (B.5). Proposition A.3 gives $\Pr(E_{\text{sm}}) \geq 1 - \delta_{\text{sm}}$; the tower property gives the joint bound in (B.5).

Substituting (B.3) into the per-run expression of Proposition A.30 and multiplying by the bound on J gives (B.6). Since $\log r \geq \log 3 > 1$, (B.7) follows directly. Also $n^3 + n \leq 2n^3$ and $n \leq n^3$, which proves the first line of (B.8); the other two lines follow by substituting $\bar{k} \leq C_K K_r$. Finally, $K_r^2 = r^{10/3} (\log r)^5$ and $K_r^3 = r^5 (\log r)^{15/2}$, so (B.8) gives (B.9) with no omitted term. The two envelope inequalities are primitive parts of the cited assumptions. No confidence parameter is absorbed into k , and the probability conversion remains conditional until the displayed tower calculation. $\square$

A.13 Exact orthogonal baseline

Assume the deterministic scope condition `ExactCertifiedSeed` from the preliminaries. The proof below checks the frozen-input solves, the product-preserving rebalance, all chronological mode orderings, and the original residual directly. We use only the elementary identities $(A \odot B)^\top (A \odot B) =$

$(A^\top A) \circ (B^\top B)$ and the Moore–Penrose formula for a positive scalar multiple of an orthonormal matrix; their three mode-wise instances are displayed explicitly.

Set

$$G_U := U, \quad G_V := V, \quad G_W := W, \quad K_U := W \odot V, \quad K_V := W \odot U, \quad K_W := V \odot U. \quad (1)$$

The Khatri–Rao column $K_M(:, j)$ is the tensor product of the two held target directions in the order used by the corresponding unfolding. Under `ExactCertifiedSeed`,

$$K_U^\top K_U = K_V^\top K_V = K_W^\top K_W = I_r, \quad (2)$$

because each factor Gram is I_r . The target tensor and its unfoldings are

$$T = \lambda \sum_{j=1}^r u_j \otimes v_j \otimes w_j, \quad T_{(1)} = \lambda U K_U^\top, \quad T_{(2)} = \lambda V K_V^\top, \quad T_{(3)} = \lambda W K_W^\top. \quad (3)$$

Lemma A.21 (exact orthogonal equal-weight seed and frozen Grams). *Under the local specialization `ExactCertifiedSeed` and the seed convention from Proposition A.16, the observable scores satisfy $\theta_j = \lambda$, the balanced frozen seed is*

$$X^0 = \lambda^{1/3} U, \quad Y^0 = \lambda^{1/3} V, \quad Z^0 = \lambda^{1/3} W, \quad (4)$$

and, for every $M \in \{U, V, W\}$,

$$P_M^0 = G_M, \quad D_M^0 = \lambda^{1/3} I_r, \quad C_M^0 = 0, \quad N_M^0 = 0, \quad s^0 = 0, \quad (5)$$

$$F_M^0 = (P_M^0)^\top P_M^0 = I_r, \quad G_M^\top P_M^0 = I_r. \quad (6)$$

Consequently, in all three held-mode orderings,

M	$J_M^0 = F_{M'}^0 \circ F_{M''}^0$	$Q_M^0 = (G_{M'}^\top P_{M'}^0) \circ (G_{M''}^\top P_{M''}^0)$
U	$I \circ I = I$	$I \circ I = I$
V	$I \circ I = I$	$I \circ I = I$
W	$I \circ I = I$	$I \circ I = I$

(7)

and every actual frozen pair Gram is $\lambda^{4/3} I_r$.

Proof. For an exact representative,

$$\theta_j = \left\langle \lambda \sum_{\ell=1}^r u_\ell \otimes v_\ell \otimes w_\ell, u_j \otimes v_j \otimes w_j \right\rangle = \lambda \quad (8)$$

by the orthonormality identities. If a product-one sign chart is used, the three signs multiply to one and the same inner product and represented rank-one term result. The setting’s observable seed therefore gives (4), and normalizing its columns gives (5). Equations (6)–(7) follow from the orthonormality identities and the Khatri–Rao identity. Finally, for each mode M ,

$$H_M^0 = (P_{M'}^0 D_{M'}^0) \odot (P_{M''}^0 D_{M''}^0) = \lambda^{2/3} K_M, \quad (9)$$

so (2) gives $(H_M^0)^\top H_M^0 = \lambda^{4/3} I_r$. This also proves that the exact seed has no perpendicular or product-log defect. The line $\lambda^{4/3} I_r$ is valid for each of the U, V, and W orderings in (7). Thus the actual left-hand defects in the bridge (the differences bounded by $\delta_L, \delta_F, \delta_{FL}$, and $Q_M^0 - J_M^0$) are zero, while the conservative budget constants $\eta_J, \eta_{QJ}, \eta_A$ may remain positive at the main-theorem threshold values. All these bounds are therefore sharp upper bounds at the baseline, and the budget constants become zero in the optional formal $q_*, \tau_r \downarrow 0$ no-floor limit; no budget is used to weaken the equalities above.

□

Proposition A.33 (three exact frozen landing solves). *Under ExactCertifiedSeed, with the synchronized frozen designs of the setting, the Moore–Penrose outputs are well-defined and, mode by mode,*

$$\begin{array}{c|c|c|c}
M & H_M^0 & T_{(M)}H_M^0 & T_{(M)}H_M^0((H_M^0)^\top H_M^0)^\dagger \\
\hline
U & \lambda^{2/3}K_U & \lambda^{5/3}U & \lambda^{1/3}U \\
V & \lambda^{2/3}K_V & \lambda^{5/3}V & \lambda^{1/3}V \\
W & \lambda^{2/3}K_W & \lambda^{5/3}W & \lambda^{1/3}W
\end{array} \tag{10}$$

Thus $\tilde{X}_U = \lambda^{1/3}U$, $\tilde{X}_V = \lambda^{1/3}V$, and $\tilde{X}_W = \lambda^{1/3}W$, with every active output norm equal to $\lambda^{1/3} > 0$.

Proof. The U row follows from (2)–(3) and (9):

$$T_{(1)}H_U^0 = \lambda U K_U^\top (\lambda^{2/3}K_U) = \lambda^{5/3}U, \quad ((H_U^0)^\top H_U^0)^\dagger = \lambda^{-4/3}I_r. \tag{11}$$

Their product is $\lambda^{1/3}U$. Replacing (U, K_U) by (V, K_V) and (W, K_W) gives, respectively,

$$T_{(2)}H_V^0 = \lambda^{5/3}V, \quad T_{(3)}H_W^0 = \lambda^{5/3}W, \tag{12}$$

with the same Gram inverse and the same output scale. These are three independent evaluations from the one frozen seed; no output in one row is an input to another row. Since $\lambda^{4/3}I_r$ is non-singular, the Moore–Penrose rule agrees with the displayed ordinary inverse, while the calculation remains valid at the exact $\rho_{\text{sm}} = 0$ specialization. $\square$

Proposition A.34 (joint rebalance and product-one gauge). *Under ExactCertifiedSeed and the outputs of Proposition A.33, the setting’s single joint rebalance returns*

$$X_U^{\text{land}} = \lambda^{1/3}U, \quad X_V^{\text{land}} = \lambda^{1/3}V, \quad X_W^{\text{land}} = \lambda^{1/3}W, \tag{13}$$

preserves every represented rank-one product, and has $\tilde{\gamma}_j^{\text{land}} = \lambda$ and $s_j^{\text{land}} = 0$ for every j . The same represented tensor and quotient state result after any positive product-one scale gauge or proof-only product-one sign gauge.

Proof. Proposition A.33 gives $\|\tilde{x}_j\|_2 = \|\tilde{y}_j\|_2 = \|\tilde{z}_j\|_2 = \lambda^{1/3}$. Hence

$$\tilde{\gamma}_j = \|\tilde{x}_j\|_2 \|\tilde{y}_j\|_2 \|\tilde{z}_j\|_2 = \lambda, \tag{14}$$

and the rebalance multiplier is $\tilde{\gamma}_j^{1/3} = \lambda^{1/3}$. Dividing each output by its norm therefore gives (13). More generally, for nonzero factors and positive scalars $R_{U,j}, R_{V,j}, R_{W,j}$ with $R_{U,j}R_{V,j}R_{W,j} = 1$,

$$(R_{U,j}x_j) \otimes (R_{V,j}y_j) \otimes (R_{W,j}z_j) = x_j \otimes y_j \otimes z_j. \tag{15}$$

The rebalance uses positive norms and is exactly such a product-one rescaling. A product-one sign gauge has the same identity with signs in place of positive scalars. Thus neither gauge changes the represented term, and the common product remains λ , giving $s_j = \log(\lambda/\lambda) = 0$. $\square$

Proposition A.35 (all cyclic normal equations are stationary). *Under ExactCertifiedSeed, start a cyclic active ALS sweep from the balanced state (13), with inactive columns zero. For each mode $M \in \{U, V, W\}$, regardless of which mode is first, the held design is $\lambda^{2/3}K_M$, its Gram is $\lambda^{4/3}I_r$, and the exact normal equation returns $\lambda^{1/3}G_M$. After the product-preserving canonical refresh, the state is unchanged and the refreshed common-product register is $s = 0$. Consequently every chronological Refresh_s state and every full cyclic sweep equal the same exact quotient state; this holds for all six permutations of the U/V/W mode order.*

Proof. For the displayed U/V/W order, the U block uses the held factors $Y = \lambda^{1/3}V$, $Z = \lambda^{1/3}W$, hence

$$H_U = \lambda^{2/3}K_U, \quad H_U^\top H_U = \lambda^{4/3}I_r, \quad T_{(1)}H_U = \lambda^{5/3}U, \quad \tilde{X}_U = \lambda^{1/3}U. \quad (16)$$

Its solved product with the held factors is λ in each component, so Refresh_s writes $s_U^+ = 0$ and the canonical representative is again the state (13). The V block then has the unchanged held U and W factors and similarly obeys

$$H_V = \lambda^{2/3}K_V, \quad T_{(2)}H_V = \lambda^{5/3}V, \quad \tilde{X}_V = \lambda^{1/3}V, \quad s_V^+ = 0. \quad (17)$$

The W block gives

$$H_W = \lambda^{2/3}K_W, \quad T_{(3)}H_W = \lambda^{5/3}W, \quad \tilde{X}_W = \lambda^{1/3}W, \quad s_W^+ = 0. \quad (18)$$

Thus the U/V/W chronological intermediate states are all exactly (13). For any other first mode, use the same generic identity

$$T_{(M)} = \lambda G_M K_M^\top, \quad H_M = \lambda^{2/3}K_M, \quad K_M^\top K_M = I_r, \quad (19)$$

and then apply the remaining two identities in their chosen order. This proves all six mode permutations, not only the displayed cyclic order.

For completeness, the scale-equivariance interface can be checked directly. If a literal state is changed by positive diagonal $R_U R_V R_W = I$, then the mode-M design changes to $H_M R_M^{-1}$, and

$$T_{(M)} H_M R_M^{-1} (R_M^{-1} H_M^\top H_M R_M^{-1})^{-1} = T_{(M)} H_M (H_M^\top H_M)^{-1} R_M. \quad (20)$$

Therefore the solved factors differ only by the same gauge, and the canonical Refresh_s representative is (13). The sign gauge is handled by the proof-only orientation and preserves each product by (15). Induction over sweeps proves stationarity forever, with no accumulated forcing term. $\square$

Proposition A.36 (exact residual and no-floor stopping). *Under ExactCertifiedSeed and Assumption 8, after the joint landing and after every cyclic refresh, the active factors represent exactly*

$$\hat{T} = \lambda \sum_{j=1}^r u_j \otimes v_j \otimes w_j = T, \quad (21)$$

and the inactive $k - r$ columns contribute zero. Hence

$$\frac{\|T - \hat{T}\|_F}{\|T\|_F} = 0 \leq \epsilon, \quad \|T\|_F = \lambda\sqrt{r} > 0. \quad (22)$$

The first original residual test is therefore successful (whether it is performed immediately after landing or after the first completed cyclic sweep), for every $\epsilon > 0$; no ω -, τ -, or positive-radius error floor is present.

Proof. Equation (13) gives, componentwise,

$$x_j^{\text{land}} \otimes y_j^{\text{land}} \otimes z_j^{\text{land}} = \lambda u_j \otimes v_j \otimes w_j. \quad (23)$$

Proposition A.35 shows that the same equality is preserved by every block and every refresh. The tensors $u_j \otimes v_j \otimes w_j$ are mutually orthonormal in the Frobenius inner product under the orthonormality identities, so $\|T\|_F^2 = r\lambda^2$. This proves (21)–(22) and the stopping assertion. In the

deterministic $\rho_{\text{sm}} = 0$ limit there is no perturbation term to dominate, and in the optional formal no-floor limit $q_*, \tau_r \downarrow 0$ the same exact calculation remains valid. The positive ρ_{ALS} chart radius is only a neighborhood for the main theorem; the baseline state is its exact origin, so no radius term enters (22). $\square$

Proof. Lemma A.21 derives the exact observable score, balanced seed, zero coefficient/perpendicular/product-log fields, and all three identity pair/cross Grams. Proposition A.33 then evaluates the three synchronized frozen-input pseudoinverse calls from that single seed. It explicitly gives the U, V, and W outputs and their positive normalizers, so no mode-specific predecessor or post-solve design is used. Proposition A.34 applies the one joint positive rebalance, proves product preservation, and handles both positive scale and product-one sign gauges. Proposition A.35 checks the chronological U/V/W normal equations and their generic permutation, and proves that every Refresh_s register remains zero. Finally, Proposition A.36 identifies the represented tensor with T , computes its strictly positive Frobenius norm, and applies the residual-test conclusion of Proposition A.28. Thus all three frozen landing outputs, every cyclic refresh, the common product error, and the original residual have the exact values required by the stated result. The argument is a deterministic specialization and leaves the positive- ρ main-theorem assumptions untouched. $\square$

A.14 Proof of the Main Theorem

Proof of the main theorem. Proposition A.3 constructs E_{sm} , proves $\Pr(E_{\text{sm}}) \geq 1 - \delta_{\text{sm}}$, and supplies the realized norm, Gram, weight-ratio, and Khatri–Rao conclusions. Conditional on a fixed instance in that event, Proposition A.5 gives the two-sided window mass and Proposition A.6 gives coverage probability at least 26/27.

Lemma A.8 proves the displayed R, S recurrence; Proposition A.8 proves denominator persistence and finite chart entry, and Proposition A.9 gives the stored certificate. Proposition A.11 supplies the score and coordinate ledger. Proposition A.13 applies the public observable graph rule, while Lemma A.11 gives the target permutation and product-one gauge. Propositions A.14 and A.15 close both coefficient orientations, and Proposition A.16 gives the balanced-seed bounds.

Proposition A.20 collects the complete pre-solve Gram, inverse, congruence, coefficient, and normalizer interface. It is consumed by Propositions A.21, A.22, and A.23, whose composition in Proposition A.24 produces the exact-span landing state without a cross-mode landing dependency. Proposition A.27 then gives the literal cyclic orbit identification, chart invariance, and contraction.

Lemma A.20 converts quotient error to the original Frobenius residual. Propositions A.28 and A.29 prove covered-path stopping and finite work on every tape. Proposition A.30 supplies the dense per-run cost, Proposition A.31 gives conditional restart confidence, and Proposition A.32 proves the displayed total-work rate and the nested probability conversion. Combining these exact producer statements proves every conclusion of Theorem 3.1 without an additional assumption or rate loss. $\square$

References

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